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SCH 209 MODULE BY SIR OTONDE MICHAEL , THE INCOMING DELEGATE -0741050074





KENYATTA UNIVERSITY

**DIGITAL SCHOOL OF VIRTUAL AND OPEN LEARNING
IN COLLABORATION WITH
SCHOOL OF PURE & APPLIED SCIENCES
DEPARTMENT OF CHEMISTRY**

SCH 209 _ ATOMIC STRUCTURE AND CHEMICAL BONDING

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SCH 209 ATOMIC STRUCTURE AND CHEMICAL BONDING

Pre-requisite-SCH 100



INTRODUCTION

Welcome to SCH 209: Atomic Structure and Chemical Bonding. This course explores the basic theories of atomic structure and bonding. It provides a detailed but simplified theoretical background toward the understanding of various theories used to explain the atomic structure and bonding. It is assumed that you have taken SCH 109: Fundamentals of Inorganic Chemistry.

Each lecture opens with a brief introduction, followed by detailed discussions and examples. There are also some interspersed questions with suggested answers.



COURSE DETAILS Purpose

This course is intended to equip the students with the basic quantum concepts of the structure of the atom and the theories of chemical bonding between atoms.

Learning Outcomes

At the end of this unit you should be able to:

- i). Describe the experimental foundation of quantum theory ii). Explain the dual nature of light and matter iii). Appreciate the Schrödinger equation and significance of its solutions
 - Solve Schrödinger equation for a particle in-a-box
 - Appreciate how Schrödinger equation for hydrogen – like atom is solved to get quantum numbers.
- iv). Use Aufbau principle and Pauli's Exclusion Principle to write electronic configurations.
- v). Describe the periodicity of the periodic table
- vi). Describe various bonding models; Lewis structures, Resonance structures, Valence Bonding Theory (VBT), Valence Shell Electron Pair Repulsion (VSEPR) Model, Hybridization and Molecular Orbital Theory (MOT).
- vii). Appreciate Lattice energies of ionic solids and its calculation from thermochemical data viii). Describe hydrogen bonding and its significance.

Course Description

This course covers the experimental foundations of quantum theory; particles and waves; uncertainty principles; waves and the wave equation; properties of a well-behaved wavefunction; the Schrödinger equation and its solutions; the physical significance of the wavefunction; solutions of the Schrödinger equation for hydrogen-like atoms: quantum numbers, radial angular and functions, charge clouds and orbital; energy levels, electron configuration and the periodic table; ionic, covalent, hydrogen and metallic bonds, valence bond theory and molecular orbital theory for simple molecules and ions; complex ions; definition and examples of Lewis acids, Lewis bases, co-ordinate bond, ligands and coordination compounds; ionic structures and calculation of lattice energies.

Practicals include proper handling of the analytical balance, use of standardized solutions, acid-base titrations, redox titrations and EDTA titrations and identification of cations and anions in solution.

Course Outline

- Experimental foundations of Quantum Theory.
 - Wave-particle duality of matter
 - Uncertainty principle
 - Waves and the wave equation
 - Properties of well-behaved wave functions
 - Schrödinger equation and its solutions for a particle in a box
 - The physical significance of the wave function
 - Solutions of the Schrödinger equation for hydrogen-like atoms: quantum numbers, radial functions, radial distribution functions, charge clouds and orbitals.
 - Energy levels electron configurations, and the Periodic Table.
 - Ionic, covalent, hydrogen and metallic bonds
 - Valence bond theory and molecular orbital theory for simple molecules and ions
 - Complex ions: Lewis acids and bases coordinate bond, ligands and coordination compounds
 - Ionic structures and calculation of lattice energies, ionic sizes and salvation energies
- Assessment
- CATs and Practicals 40%,
 - End of Semester Examination 60%



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1.1 Objectives of the Course

At the end of this unit you should be able to;

- (a) Describe the experimental foundation of quantum theory
- (b) Explain dual nature of light and matter
- (b) Explain the Schrödinger equation and significance of its solutions
 - Solve Schrödinger equation for a particle in-a-box
 - Appreciate how Schrödinger equation for hydrogen – like atom is solved to get quantum numbers.
- (c) Use Aufbau principle and Pauli's Exclusion Principle to write electronic configurations.
- (d) Describe the periodicity of the periodic table
- (f) Describe various bonding models; Lewis structures, Resonance structures, Valence Bonding Theory (VBT), Valence Shell Electron Pair Repulsion (VSEPR) Model, Hybridization and Molecular Orbital Theory (MOT).
- h) Calculations of Lattice energies
- i) Describe hydrogen bonding and its significance.



1.2 Some Useful References and Recommended Reading:

- 1 Manas Chanda, "Atomic Structure and the Chemical Bond- Including Molecular Spectroscopy". 4th edition. Tata McGraw-Hill Publishing Company Limited, New Delhi (2000).
- 2 E. Cartmell and G. W. A. Fowles, "Valency and Molecular Structure". 4th edition. Butterworth and Co (Publishers) Ltd (1977).
- 3 Shriver D. F., Atkins P. W., Longford C. H. (1994), "Inorganic Chemistry", 2nd edition Oxford University Press: Oxford.
- 4 Huhe J. E. *et.al.* (1993), "Inorganic Chemistry", Principles of Structure and Reactivity", 5th edition, New York, Harper Collins Colleges Publishers. 5 Coulson C. A. (1952). "Valence". London, Oxford University Press.
- 6 Petrucci R. H. and Harwood W. S., "General Chemistry: Principles and Modern Applications". 7th edition. Prentice Hall. New Jersey.
- 7 McMurry J. and Fay R. C., "Chemistry". Prentice Hall. New Jersey.
- 8 Raymond Chang, "Essential Chemistry". McGraw-Hill Companies, Inc. Toronto
- 9 Hill G. C. and Holman J. S., "Chemistry in Context" 4th edition 1995, Thomas Nelson and Sons Ltd Surrey UK.
- 10 P.W. Atkins, Physical Chemistry, ELBS and Oxford University press
- 11 Any other college inorganic chemistry text book.
- 12 Internet Search

1 LECTURE 1

ATOMIC STRUCTURE



Introduction

In this lecture we will review the classical theory of atomic structure. You will also learn the experimental foundation that led to the development of the concept of quantum theory of atomic structure. Hydrogen spectra were well explained by Bohr's model of the atom.



Lecture 1 Objectives

At the end of this lecture, you should be able to:

- Explain the classical view of atoms
- Describe the experimental foundations of the quantum view of atoms
- Explain the hydrogen spectrum
- Describe the Bohr model of atoms



1.3 Classical View of Atomic Structure

The classical view of atomic structure was constructed with the body of knowledge accumulated in physics over several centuries preceding the 20th century. This body of knowledge is called *classical physics*. Through experience, physicists have identified a long list of particles which make up the atoms. The most fundamental particles are:

- Electrons
- Protons (also made up of 3 quarks)
- Neutrons (also made up of 3 quarks)

Table 0.1: Properties of some fundamental particles and the hydrogen atom (e is the elementary unit of electronic charge = $1.60217733 \times 10^{-19}$ coulombs)

Particle	Symbol	Mass/ 10^{-31} kg	RAM	Charge	Spin
Proton	p	16726.231	1.0072765	+e	$\frac{1}{2}$
Neutron	n	16749.286	1.0086649	zero	$\frac{1}{2}$
Electron	e	9.1093897	0.0005486	-e	$\frac{1}{2}$
Hydrogen	H	16735.339	1.007825	zero	-

^aThe atomic unit of mass is given by $m_u = m(^{12}\text{C})/12$ kg. The relative atomic mass (RAM) of an element is given by m_{element}/m_u where m_{element} is the mass of one atom of the element in kg.

From the table we can see that:

- The mass of proton and neutron are nearly identical.
- The nucleus (protons plus neutrons) contain virtually all the mass of the atom.
- The electron while equal and opposite in charge to the proton has a mass equivalent to only 0.05% of the mass of the proton.



1.4 Beginning of Quantum¹ Theory

The laws of motion put forward by Newton form the pillars of classical mechanics. Until early 1900, it was thought that these classical concepts and laws hold good both for macroscopic bodies like planets and microscopic or sub-atomic objects like electrons. The important assumptions of classical mechanics include the following:

- There are no restrictions on the value that a dynamic variable (e.g. energy, momentum etc.) can have.
- There is no limit to the accuracy with which one or more of the dynamic variables of a system can be measured except the limit imposed by the precision of the measuring instruments.
- There is no restriction on the number of dynamic variables that can be accurately measured at the same time.

When these and other ideas of classical physics (e.g. thermodynamics, optics and electromagnetic theory) developed by Galileo, Kepler and Newton were applied to atomic and molecular systems, they were found to be inadequate. A number of experiments done in the latter half of nineteenth century and the first two decades of the twentieth century gave results totally at variance with the predictions of classical physics. There was, therefore, need for a theory to describe, correlate and predict the behaviour of the microscopic or sub-atomic particles and this is how the quantum theory of Max Planck came into being. Some of the specific cases which defied classical physics included:

- Black-body radiation
- Compton effect
- Photoelectric effect
- Atomic spectra
- Heat capacity variation

The quantum theory, proposed by Max Planck and applied by Einstein and Bohr to explain different aspects of behaviour of matter, is an important milestone in the formulation of the modern concept of the atom.

¹ In physics, **classical mechanics** is one of the two major sub-fields of study in the science of mechanics, which is concerned with the set of physical laws governing and mathematically describing the motions of bodies and aggregates of bodies. The other sub-field is quantum mechanics. Quantum mechanics is concerned with the set of physical laws governing and mathematically describing the motion of microscopic bodies.

In this unit, we will study how black body radiation, photoelectric effect, the Compton Effect, and the atomic hydrogen spectrum can be explained on the basis of theories proposed by Max

Planck, Einstein and Bohr. They based their theories on the postulate that all interactions between matter and radiation occur in terms of definite packets of energy, known as quanta. Their ideas, when extended further, led to the evolution of wave mechanics, which shows the dual nature of matter and energy.

The importance of the quantum theory in the study of chemical problems cannot be overemphasized. The classical theories, although they have rendered great service and continue to be very useful in modern chemistry, are however, not always sufficient in the study of chemical problems. For example, the spectra, the structure of the periodic table, the nature of chemical bonding, the structures of molecules and their properties cannot be properly understood without recourse to quantum theory. The quantum theory itself had its origin in the branch of physics dealing with radiation. It was the study of the radiation problem, which had been defying solution on the basis of classical theories that lead Max Planck in 1901 to introduce the concept of quanta, making the foundation of the quantum theory. It is, therefore, pertinent to start with a discussion of this historic problem.

Statement of Planck's Quantum Theory: *Energy is emitted or absorbed only in discrete amounts called quanta and not continuously.*



1.4.1 Black body radiation²

Up until the late 1800's the wave picture of light was the prevalent theory, as it could explain most of the experiments done on light. However, there were a few notable exceptions. One such exception was that associated with *blackbody radiation*, i.e. the characteristic radiation that a body emits when heated.

When a solid body is heated, the constituent particles gain energy and vibrate at a rate dependent on the temperature. In a state of rapid vibration the particles emit radiant energy in the form of waves which are identical with electromagnetic waves (Maxwell's electromagnetic theory). The total radiation emitted by such a hot body is composed of electromagnetic waves of a varying range of wavelengths constituting a spectrum. The total intensity of radiation as well as its distribution among the various wavelengths of the spectrum is different for different objects even at the same temperature. If the radiation emerging through a small hole in the wall

² As bodies become hotter, their maximum brightness shifts towards higher frequencies. This is why an object, heated to 300-400 °C, emits mostly infra-red or heat waves. As the temperature is increased, the object appears to be red, then orange, and finally white or even blue. The brightness did increase initially, but only up to a limit. Then, actually, it began to fall. Classical theories totally failed to explain this discrepancy between the known facts and the observations.

of an isothermal hollow body is measured, it is found that the total intensity of radiation and its distribution among wavelengths are independent of the material and of the size and shape of the enclosure. The radiation so emitted is called *Black-body radiation* and experimentally the enclosure approximates a black-body.

A *black-body* is “a hypothetical body that absorbs without reflection all of the electromagnetic radiation incident on its surface.” It absorbs all the radiation incident upon it and is heated by the same radiation to a higher temperature than any other body. A heated black body radiates more intensely than other bodies at the same temperature. In the small hole in a wall approximation of a black-body, almost all the radiation entering the hole will be absorbed by successive reflections inside the enclosure. In this way the hole almost exactly fulfills the definition of a black-body.

Black-body radiation was investigated by O. Lummer and E. Pringshein through a complete series of experiments during the period 1897-1899. The typical curves from black body at different temperatures are shown in Fig. 1 below. The task of theory was to fit the curves of Fig. 1 and to lead to the two fundamental laws of thermal radiation. Various attempts were made to fit experimental observation to mathematical laws:

The following observations can be made from the figure:

- (i) At shorter wavelength region that is at higher frequency region intensity of radiation is low.
- (ii) At every temperature, there is a wavelength at which energy radiated is maximum. This wavelength is called λ_{max} value of that temperature.
- (iii) At higher temperatures, there is increased intensity of radiation in the shorter wavelength region. The maxima of the curves shift to shorter wavelengths with increasing temperature. These facts are pictorially represented in Fig. 1.

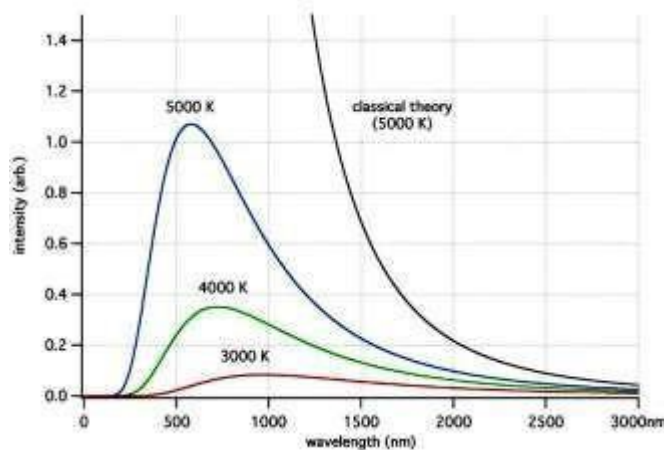


Figure 1.0.1: Emission of radiation from a black-body at different temperatures. As the temperature decreases, the peak of the black-body radiation curve moves to lower intensities and longer wavelengths. The black-body radiation graph is also compared with the classical model of Rayleigh and Jeans. http://en.wikipedia.org/wiki/Black_body

A number of mathematical laws were suggested to try and fit experimental observations to theory (physical laws):

(a) *The Stefan-Boltzmann law*: states that the total energy radiated per unit surface area of a black body per unit time U , is directly proportional to the fourth power of the black body's thermodynamic temperature T (also called absolute temperature):

$$U = \sigma T^4$$

Where σ = Stefan's constant = $5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

This was discovered experimentally by Stefan in 1879 and derived using the principles of thermodynamics by Boltzmann in 1884.

(b) *Wien's law*: The wavelength of maximum intensity is proportional to the absolute temperature of the black-body i.e. the product of the absolute temperature (T) at which a particular curve of Fig. 1 and the wavelength (λ_{max}) at which the curve has a maximum is a constant: $\lambda_{\text{max}} T = \text{a constant} = 2.8978 \times 10^{-3} \text{ mK}$.

This is a special case of Wien's displacement law of 1894 formulated from theoretical considerations. This law expresses the intensity of the radiation E_λ as;

$$E_\lambda \propto \frac{a}{\lambda^5 c^2} e^{-\frac{b}{\lambda T}} \text{ where } a \text{ and } b \text{ are constants, and } c \text{ is speed of light.}$$

This equation holds true for short wavelengths if a and b are properly selected but it is ineffective in the longer wavelength range.

(c) *Rayleigh-Jeans Law*: Attempted to describe the spectral radiance of electromagnetic radiation at all wavelengths from a black body at a given temperature through classical arguments. For wavelength λ , it is;

$$B_\lambda(T) \propto \frac{2ckT}{\lambda^4}$$

where c is the speed of light, k is Boltzmann's constant and T is the temperature in kelvins

The Rayleigh–Jeans expression agrees with experimental results at large wavelengths (or, equivalently, low frequencies) but strongly disagrees at short wavelengths (or high frequencies). As λ decreases this expression increases without limit and without going through a maximum. It suggests, for example, that oscillators of extremely short wavelengths (corresponding to ultraviolet light, X-rays and even γ -rays) are strongly excited even at room temperature. This inconsistency is commonly known as the *ultraviolet catastrophe*. It suggested that the radiation intensity would shoot toward infinity at the ultraviolet region of the electromagnetic scale but experimentally it was observed to be highest toward the middle of the visible range of the spectrum, which seemed entirely counter-intuitive. (d) *Max Planck's Law*

In 1900 Max Planck empirically derived an expression for Blackbody Radiation expressed in terms of wavelength $\lambda = c / \nu$;

$$I(\lambda, T) = \frac{2\pi^5 k^4 T^4}{15 h^3 c^2} \frac{1}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1}$$

where h is Planck's constant.

The Planck law does not suffer from an ultraviolet catastrophe, and agrees well with the experimental data, but its full significance (which ultimately led to quantum theory) was only appreciated several years later. In the limit of very high temperatures or long wavelengths, the term in the exponential becomes small, and so the denominator becomes approximately $hc/\lambda kT$ by power series expansion.

In 1900 while trying to explain this law, Planck came up with a bold hypothesis. He proposed that *emission and absorption of energy can occur only in discrete amounts called quanta*. - The energy associated with each quantum is given by the formula

$$E = nh\nu,$$

where n is an integer ($n = 1, 2, \dots$), ν is the frequency of radiation emitted and h is a constant, now called *Planck's constant* ($h = 6.626176 \times 10^{-34} \text{ Js}$).

This was against the entire world-view that had been built from the time of Newton onward. In the physics that had been built up since the time of Newton, and indeed in the minds of most thinkers before Newton, matter and energy were thought to be smooth and continuous.

Furthermore, he postulated that when a molecule went from a higher energy state to a lower one it emitted a quantum (packet) of radiation, or *photon*, which carried away the excess energy. With this quantization picture, Planck was able to successfully explain the blackbody radiation curves, both at long and at short wavelengths. However, it was a radical departure from the conventional picture of light emitted from a blackbody, and at the time many people felt it was simply a calculational trick invented to obtain the right answer.



1.4.2 The Photoelectric Effect

This is the emission of electrons when certain metals surfaces are irradiated with high energy electromagnetic radiation like ultraviolet light. It was first noted by Heinrich Hertz in 1887. The electrons, so emitted, are known as photoelectrons, to differentiate them from the electrons remaining inside the metal atoms. The characteristic features of this phenomenon are:

- No electrons are ejected, regardless of the intensity of the light, unless its frequency exceeds a threshold value, ν_0 , characteristic of each metal. Increasing the frequency of the light beyond threshold frequency value, only increases the velocity of the photoelectrons.
- Even at low intensities of light, electrons are ejected immediately if the frequency is above the threshold.
- The maximum energy of the emitted electrons is linearly dependent on the frequency of the incident light but is independent of the intensity of the incident light (Philipp Eduard Anton von Lenard 1902)

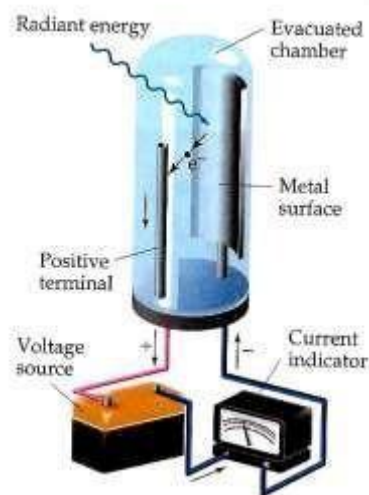


Figure 0.2: The set up to observe the photoelectric effect

These observations could not be explained by the nineteenth century view of light as a wave phenomenon. They however, strongly suggest that the photoelectric effect depends on an electron suffering a collision with a localized projectile carrying enough energy to knock it out of the metal

In 1905 Albert Einstein used Planck's quantization theory to explain these observations. He postulated that radiation consists of a stream of particles or corpuscles (now called photons) each with one quantum of energy, $h\nu$. Each of these gave up its energy to an electron on the

surface of the metal such that part of the energy is used just to remove the electron from the surface while the remainder is used to accelerate the electron from the surface (i.e. it appears as the kinetic energy of the electron).

Based on this assumption Einstein proposed the following relation for the kinetic energy T of the emitted electrons:

$$\frac{mv^2}{2} = T = h\nu - W_0 \quad 1.2$$

where T is the kinetic energy of the electron, h is the Planck's constant, ν is the frequency of the incident light and W_0 is the work function of the metal being used. Work function is the minimum amount of work required to free an electron from the metal surface.

An interesting consequence of Eq. (1.2) is that for a given metal the kinetic energy of the emitted electrons is independent of light intensity. It depends only on the frequency of the incident light.

Would you have guessed that the kinetic energy would be independent of light intensity?

The wave model of light would have predicted that the kinetic energy depends on light intensity. The simple relation of Eq. (1.2) completely explains the photoelectric effect and suggests a way to determine Planck's constant h and the work function W_0 . To experimentally investigate the Einstein relation we note that one must determine the kinetic energy T of the electrons as a function of light frequency ν .

Consider the setup shown in Figure 2. Suppose initially the applied potential V is zero. Then when light strikes the metal surface, electrons are emitted with kinetic energy T . They proceed through the vacuum until they strike the other metal plate and are registered as an electrical current by the ammeter. Now consider what happens as the potential V is increased. The electrons now experience a retarding force and when V is high enough all the electrons emitted will be just brought to rest and no current will be registered. When this happens we know the kinetic energy of the electrons.

$$eV = \frac{mv^2}{2} = T = h\nu - W_0 \quad 1.3$$

Thus we have a means of determining one of the parameters in the Einstein relation. Light frequency ν is selected by using an appropriate color filter which allows only the light of a certain frequency to pass through. For example, a red filter transmits light quanta of frequencies corresponding to the red color and absorbs quanta of all other frequencies. Another approach is to use a grating to separate the incident light into its component frequencies. Different frequencies then can be selected from the spectrum with the help of a slit that allows only a certain part of the spectrum to pass through. The rest of the spectrum is blocked. Thus by using the light of different frequencies and determining the stopping voltage we can determine both h and W_0 in the Einstein relation.

In 1916, while using alkali metals Li, Na, and K which display photoelectric effect very strongly with visible and UV light, Millikan tested the validity of equation 1.2. He found that all of his results agreed exactly with Einstein's predictions about photons, not with the wave theory. In 1921 Einstein actually won the Nobel Prize for his work on the photoelectric effect, not for his more famous theory of relativity. Using specially prepared apparatus he measured the maximum energy of escaping electrons for a range of frequencies ν , and plotted his results in a straight line graph.

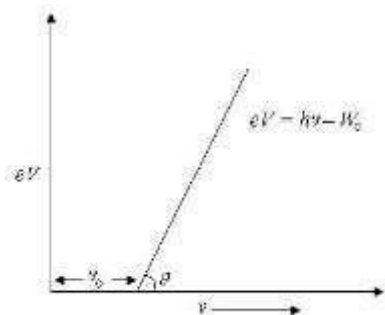


Figure 0.3: Variation of maximum kinetic energy of electrons with frequency in photoelectric effect

Millikan found the slope of the straight line to be equal to $\tan \theta$
 $= h \approx 6.55 \times 10^{-37} \text{ ergs sec}$

This value was in good agreement with what was obtained from black-body radiation. The intercept of the horizontal axis was also found to be equal to W/h . Thus the threshold frequency ν_0 is given by

$$\nu_0 = \frac{W}{h}$$



1.4.3 The Compton Scattering

Another experiment which illustrated the particulate nature of light was performed in 1923 by Compton, and is called *Compton scattering*.

When a photon of high energy radiation (e.g. X-rays, γ -rays) collides with an electron it gives part of its energy to it, as a result the radiation is scattered and its wavelength is increased. Here photons behave as if they are particles since they give momentum to the particle, and if this is true, it should be possible to treat collisions between a photon and a material particle such as an electron in the same manner as, say billiard ball collisions are treated in accordance with the usual laws of mechanics.

The scattered light has a different wavelength from that of the incident light, an effect that again cannot be explained using the wave picture of light. Compton analyzed this problem by considering light as a photon with energy given by Planck's hypothesis, as well as a momentum given by;

$$\frac{h p}{m_e c}$$

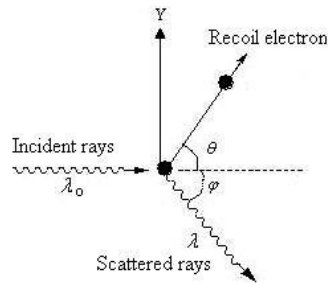


Figure 1.0.4: The Compton Effect

which was motivated in part by Einstein's theory of Special Relativity. Using the energy and momentum conservation for such a collision, he found the following relation between the incident and scattered photon: where $m_e = 9.11 \times 10^{-31}$ kg is the mass of the electron. Such a small difference in wavelengths between the incident and scattered light is very hard to detect using visible light, with wavelengths of the order of 500 nm. However, it is a comparatively large effect for X-rays, which have wavelengths of the order of 0.1 nm, and so X-rays were used to demonstrate that such scattering does indeed take place.

Using the cosine rule on the diagram above, the energy equation, with some special relativity (or by approximation) one can derive the change in wavelength as a function of scattering angle:

$$\lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos \theta)$$

Here m_e is the electron mass and the term $h/m_e c$ is called the "Compton wavelength". The Compton effect supports the quantum theory in that we have conservation of momentum. The energy difference $h\nu - h\nu'$ is proportional to the kinetic energy of the ejected electron.



1 State Planck's Quantization theory".

Suggested answer

Planck's quantum theory states that energy is emitted or absorbed only in discrete amount called quanta.

or

$E = h\nu$ or $E = hf$ where h = Planck's constant and ν = frequency

2 Describe very briefly two experiments that show the dual nature of electromagnetic radiation and at the same time confirm the reliability of Planck's quantum theory.

Suggested answer

Any brief description of photoelectric effect or Compton effects



1.4.4 The Atomic Hydrogen Emission Spectrum

Each element displays a characteristic spectrum as first pointed out by Kirchhoff in 1859. The law governing the distribution of lines in a spectrum was discovered by Balmer and Rydberg. The interpretation of this law in terms of atomic structure began with the work of Bohr in 1913. It revealed that a study of spectra could provide a wealth of information about the atomic structure. The study of these spectra has provided a key to the interpretation of many of the spectra of more complex atoms.

In the visible region, the spectrum of hydrogen was known to consist of 4 main lines:



Red 656.3 nm
Blue 486.1 nm
Violet 434.0 and 410.2 nm

Denoted as H_{α} , H_{β} , H_{γ} and H_{δ} .

Near the ultra violet there were other lines getting closer together converging to a limit of 364.6 nm

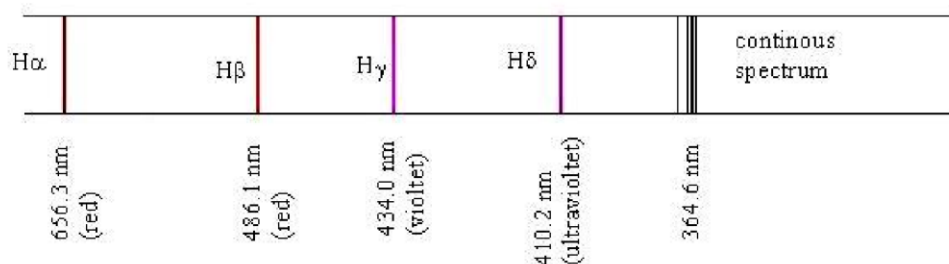


Figure 0.5: The Balmer series of atomic hydrogen excited by electric discharge. The dark lines correspond to the bright emission lines of the spectrum.

It was discovered by Balmer as early as 1885, that the wavelength (λ) of the nine lines, then known in this spectrum, could be accurately expressed by a single formula;

$$\frac{1}{\lambda} = \frac{k n^2}{n^2 - 2^2} \quad (1)$$

Where k is a constant and n is an integer. The different wavelength corresponding to the lines H α , H β , H γ and H δ etc. are obtained by putting the integers 3, 4, 5, 6 etc respectively in place of n . This set of line constitutes a spectral series and is known as the **Balmer series**.

Expressing in terms of wave numbers, equation (1) is rearranged as

$$\frac{1}{\lambda} = \frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n^2} \right] \quad (2)$$

Where R is a constant called the Rydberg constant $R = 109677.8 \text{ cm}^{-1}$.
 n was an integer which can take all values greater than 2, $n = 3, 4, 5, \dots \infty$.

$\frac{1}{\lambda}$ - the wave number, i.e. the number of wavelength in unit length.

The hydrogen spectrum was investigated in the ultra violet (UV), infrared (IR) regions and revealed that there were few other series such as Lyman (UV region), Paschen and Brackett and Pfund. All these series were found to be in excellent agreement with the general equation

$$\frac{1}{\lambda} = \frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad (3)$$

This is the universal formula for the atomic hydrogen spectrum. Where n_1 and n_2 are integers somehow defining the energy stages of the atom. The atomic spectrum of hydrogen atom consists of the following series of lines.

Series	Region	n_1	n_2
Lyman	Ultraviolet	1	2, 3, 4, 5..... ∞
Balmer	Visible	2	3, 4, 5, 6..... ∞
Paschen	Near infrared	3	4, 5, 6, 7..... ∞
Brackett	Far infrared	4	5, 6, 7, 8..... ∞
Pfund	Far infrared	5	6, 7, 8, 9..... ∞

Though equation (3) represented with great precision the entire known spectrum of atomic hydrogen, it was however an empirical formula described from experiment. The interpretation of this correct but underived formula was first given by theory of Niels Bohr (1913).



1.5 Bohr Model of the Atom (1913)

In order to explain the hydrogen spectrum, Bohr made the following assumptions governing the behaviour of electrons:

- Electrons revolve in orbits of specific radius around the nucleus without emitting radiation.
- Within each orbit, each electron has a fixed amount of energy; electrons in orbits further from the nucleus have greater energies.
- An electron may 'jump' from one orbit of high energy to another of lower energy causing the energy difference to be emitted as a photon of electromagnetic radiation such as light.
- An electron may absorb a photon of radiation and jump from a lower-energy orbit to a higher-energy one.

The energy emitted or absorbed corresponds to the difference in the energy for the initial and final state of the system.

$$E_{\text{final}} - E_{\text{initial}} = E = h\nu$$

$$r \propto \frac{n^2 h^2}{m e^2}$$

The radius of each orbit in Bohr's model is given by $\frac{n^2 h^2}{m e^2}$

Derivation

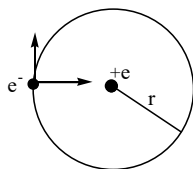


Figure 1.0.6: An electron moving in a circular orbit about a proton

The

coulombic electrostatic force upon the electron is given by $F_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$

mv^2

The centripetal force F_c is given by $F_c = \frac{mv^2}{r}$

Equating the two forces

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\text{therefore, } mv^2 = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \quad \text{and}$$

We know that kinetic energy is given by $K.E = \frac{1}{2}mv^2$. We can therefore write the above equation as follows:

$$\frac{1}{2}mv^2 = \frac{1}{2} \left(\frac{1}{4\pi\epsilon_0} \frac{e^2}{r} \right)$$

The potential energy of the electron is given by

$$U = - \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} dr = - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

Total energy (kinetic plus potential) is given by

$$\frac{h^2}{2(4\pi^2 m_0 r)} \quad \frac{h^2}{4\pi^2 m_0 r} \quad \frac{h^2}{2(4\pi^2 m_0 r)} \quad \begin{matrix} e^2 \\ E \end{matrix} \quad \begin{matrix} e^2 \\ E \end{matrix} \quad \begin{matrix} e^2 \\ E \end{matrix}$$

The wavelength of the electron was given by the deBroglie relation

$$h = \lambda p = mv\lambda$$

The stable electron orbits for the hydrogen atom are restricted to those with an integer number of wavelengths. i.e.

$$2\pi r = n\lambda$$

These two combined lead to

$$\frac{h}{mvr} = n\lambda \quad \text{or} \quad \frac{h}{mvr} = \frac{2\pi r}{n}$$

Eliminating v from the equation

$$\frac{mv^2}{2} = \frac{e^2}{2(4\pi\epsilon_0 r)}$$

and

$$mvr = \frac{h}{n}$$

$$r = \frac{n^2 m_e h^2}{4\pi^2 m_e e^2}$$

Substitution of all the values of the various constants gives $r = 0.529 \text{ \AA}$ when $n = 1$ The total energy

$$E = \frac{e^2}{2(4\pi\epsilon_0 r)} - \frac{2\pi^2 m_e e^4}{n^2 h^2 (4\pi\epsilon_0 r)^2}$$

$$E$$

0

Using the second postulate, the frequency was given by

$$h\nu = E_2 - E_1 = \frac{h^2}{4\pi^2 m e^2 r^4} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

The wave number $\left(\frac{1}{\lambda} = \frac{\nu}{c} \right)$ become

$$\frac{1}{\lambda} = \frac{2\pi^2 m e^4}{h^3 R} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where R was the Rydberg constant

Of the entire possible orbit only certain ones were acceptable – namely which had a specified angular momentum. He assumed circular electron orbit with quantized electronic angular momentum.

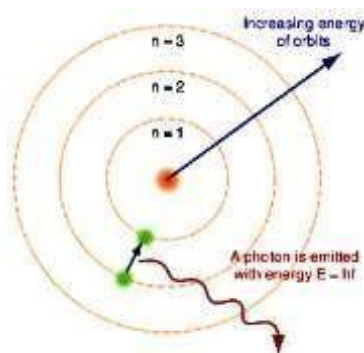


Figure 1.0.7: Bohr atomic model (Source: www.geocities.com/arhie_hend/images/bohr.bmp)

One of the implication of these quantized energy states is that only certain photon energy are allowed when electrons jump down from high to lower levels producing the hydrogen spectrum.

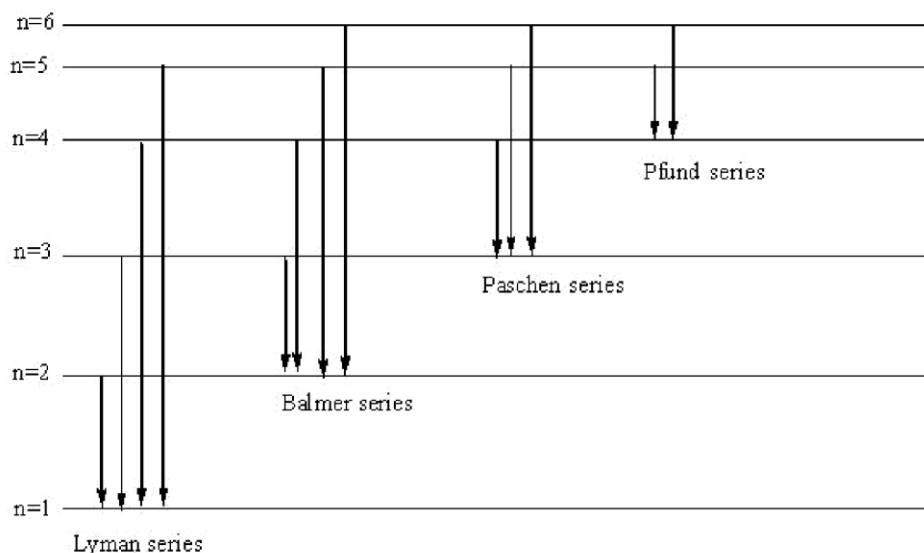


Figure 0.8: Various series in hydrogen spectrum

For hydrogen the energy of electron in a n energy level is given by:

$$E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2} = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

or

$$E_n = -\frac{Z^2 m e^4}{8 h^2 \epsilon_0^2} \frac{1}{n^2} = -13.6 \frac{Z^2}{n^2} \text{ eV, because } 1 \text{ eV} = 1.6022 \times 10^{-19} \text{ J.}$$

Where

m = mass of electron $9.1079 \times 10^{-31} \text{ kg}$
 e = electronic charge $1.602 \times 10^{-19} \text{ C}$
 h = Plank's constant $6.626 \times 10^{-34} \text{ J s}$
 ϵ_0 = permittivity of free space³ = $8.854188 \times 10^{-12} \text{ C}^2 \text{ s}^2 \text{ kg}^{-1} \text{ m}^{-3}$
 Z = nuclear charge

$$\Delta E = E_{n_2} - E_{n_1}$$

³The permittivity of free space (sometimes called the vacuum permittivity) is a constant of nature that specifies how strong the electric force between electric charges is in vacuum, or put another way it tells us how strongly the electric field reacts to the presence of charges in empty space. This quantity is normally represented with the symbol ϵ_0 (read "epsilon zero" or "epsilon naught"). In SI units, the vacuum permittivity is defined to have an exact value of $\epsilon_0 = 1/(\mu_0 c^2)$, where μ_0 is the magnetic permeability of free space and c stands for the speed of light in vacuum. This has the approximate numerical value $\epsilon_0 \cong (8.854 \ 187 \ 817) \times 10^{-12} \text{ F/m}$. In more familiar units, $\text{F/m} = \text{C}^2/(\text{Nm}^2)$. The permittivity of free space tells us how strong the electric force is, because it determines the constant that appears in Coulomb's law: $k = 1/(4\pi\epsilon_0)$

$$E_n = 2.18 \times 10^{-18} Z^2 \left(\frac{1}{n^2} - \frac{1}{n'^2} \right) \text{ J}$$

or

$$E_n = 13.6 Z^2 \left(\frac{1}{n^2} - \frac{1}{n'^2} \right) \text{ eV, since } 1 \text{ eV} = 1.6022 \text{ J.}$$



1.5.1 Failures of the Bohr Model

While the model was a major step towards understanding the quantum theory of the atom, it is not in fact a correct description of the nature of electron orbit. Some of the shortcomings of the models are:-

- It fails to provide any understanding why some spectral lines are brighter than others.
- There is no mechanism for the calculation of transition probability.
- The Bohr model treats electrons as if they were miniature planet with definite radius and momentum. This is a direct violation of the Heisenberg uncertainty principle which dictates that position and momentum cannot be simultaneously determined.

The Bohr model gives us a basic conceptual model of electron orbits and energies. The precise details of spectra and charge distribution must be left to quantum mechanical calculations as with the Schrödinger equation.

2 LECTURE 2

THE QUANTUM MODEL OF AN ATOM



2.1 About this Lecture

In this lecture you will learn the current understanding of the electronic structure of an atom expressed in terms of quantum mechanics. The fundamental concept of quantum mechanics is that all matter has wave and particle-like properties. You will also review, qualitatively, the quantum numbers that describe an electron in an atom.



2.2 Lecture 2 Objectives

At the end of this lecture, you should be able to:

- i) Describe the dual nature of matter and light
- ii) State Heisenberg uncertainty principle
- iii) Write Schrödinger equation
- iv) Qualitatively explain the quantum numbers



2.3 Quantum Mechanics

Our current understanding of the electronic structure of an atom has come from quantum mechanics. A fundamental concept of quantum mechanics is that all matter has wave and particle-like properties. In this section we shall discuss the wave particle duality and the Schrodinger wave equation and its solutions for the hydrogen atom.



2.4 The Wave-Particle Duality

The de Broglie Hypothesis

It has been shown that light can behave both as a wave and as particles (photons). Following these observations Louis Victor de Broglie (1924) extended the idea of wave particle duality and claimed that all matter, not just light, has a wave-like nature. He suggested that the wavelength and momentum of such matter are related by the equation

$$\lambda = \frac{h}{p}, \text{ where } p \text{ is the momentum and } h \text{ is Planck's constant.}$$

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This can be arrived at from Einstein's special relativity theory and Planck's quantization theory. According to Einstein's theory, the fundamental law relating energy E , and the mass m of a particle is

$$E = mc^2$$

From Planck's quantum theory, $E = h\nu$. Therefore, $E = h\nu = mc^2$. But $\nu = \frac{c}{\lambda}$

$$h\nu = mc\lambda$$

This equation is immediately applicable to photons but not atomic particles which cannot have a velocity equal to the speed of light. Thus de Broglie modified this equation by replacing the speed of light c with the velocity of the atomic particle v . Thus,

$$h\nu = mv\lambda$$

λ = wavelength of the moving particle
 h = Planck's constant = 6.626×10^{-34} J.s
 m = mass of the moving particle.

since $mv = p$, the momentum of the particle, then $\lambda = \frac{h}{p}$

That is, all particles in motion have a wavelength that depends on the particle's momentum. The larger the momentum, the smaller the wavelength. For this hypothesis de Broglie received the Nobel Prize in Physics in 1929.

de Broglie's equation was confirmed experimentally three years later. In 1927, Davisson and Germer at Bell Laboratories showed that when a beam of electrons was directed at a nickel crystal, a diffraction pattern was observed. The pattern obtained was similar to that obtained when light is allowed through a thin slit and allowed to fall on a screen. Thus electrons could be diffracted just like light waves and so they too possessed wave properties. For this work they received the Nobel Prize in Physics in 1932.

The wave-particle duality strikes at the heart of classical physics where particles and waves are treated as entirely distinct entities. Clearly a new type of mechanics had to be devised – *Quantum Mechanics*.



2.5 Heisenberg Uncertainty Principle

The dual nature of matter, both particles and wavelike properties, raised questions regarding the applicability of the classical physics to the motion of small bodies the size of atoms and

subatomic particles like electrons. It was known that a wave extends in space and its location is not precisely defined.

From the foregoing considerations the German Physicist Werner Heisenberg (1901-1961) concluded that the dual nature of matter places a fundamental limitation on how precisely we [2608.2020@ students.Ku.ac.ke](mailto:2608.2020@students.Ku.ac.ke) can know both the location and the momentum of any object. The uncertainty principle states that *it is impossible to simultaneously know both the position and momentum of an object as small as an electron*. Mathematically it can be stated as follow:

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Δx = error (or uncertainty) in position

Δp = error(or uncertainty) in momentum

The principle implies that the more accurately we know the position of the electron, the less accurately we know the momentum and vice versa.



2.5.1 Consequences of the Heisenberg Uncertainty Principle

- 1 It did away with Bohr's ideas of electrons being in definite orbits where the position and velocity are known exactly. These were replaced by the probability of finding an electron in a particular position or volume of space. Thus orbits were replaced with orbitals and electron particle replaced with electron cloud. Thus an orbital is a volume in space where 99% of the electron cloud is expected to be found.
- 2 It demonstrated the need to find a microscopic/atomic equation equivalent to the Newton's second law of motion which would have to include the Planck postulate $E = h\nu$ as well as the de Broglie wave postulate.



2.6 The Schrödinger Wave Equation

After the confirmation that electrons too could possess wave properties, it was time to find a wave equation that would describe the motion of the electron in an atom. It is one of the postulates of quantum mechanics that for a physical system consisting of a particle there is an associated wavefunction. This wavefunction determines everything that can be known about the system.

In 1926 and 1927 Schrodinger and Heisenberg published papers on wave mechanics (descriptions of the wave properties of electrons in atoms) that used very different mathematical techniques. What was interesting is that as much as their approaches were different, it was shown that their theories were equivalent. It is however, the Schrodinger's differential equation that is commonly used to introduce the theory.

In developing a model for the hydrogen atom, Erwin Schrodinger (1887 – 1961) devised an equation that enabled him to calculate the shape of the wave associated with any particle.

He then solved the equation for electron in the hydrogen atom.

So what were his findings?

He found out that waves would fit into the hydrogen atom only for certain energy values i.e only certain waves with their associated energies were allowed. This confirmed quantization of energy postulated by Planck. Then he also worked out the mathematical expressions for the shapes of the allowed waves which are now called wavefunctions and denoted by the Greek letter ψ (psi).

In its simplest form the Schrödinger equation is written;

$$H\psi = E\psi$$

where H = the Hamiltonian operator³ ψ = is the wavefunction

E = the energy associated with

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the wave

The equation implies that if the operations represented by H are performed on the function ψ , the result will contain information about ψ and its associated permitted energies.

The other way of presenting the Schrödinger equation is

$$\nabla^2 \psi + \frac{2m(E - V)}{\hbar^2} \psi = 0$$

where E = total energy of the particle

V = potential energy of the particle
particle/electron

m = mass of the

$$\nabla^2 (\text{del squared}) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

E-V = Kinetic energy of the particle ψ = is the wavefunction. Thus, the equation in full is,

$$\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) + (E - V)\psi = 0$$

This is the celebrated time independent Schrodinger equation. It describes the wave properties of an electron in terms of its position, mass, total energy, and potential energy.

It is applicable only in the case where the state of the system does not change with time. For example in atoms electrons remain in a given energy level (called orbit by Bohr) except

³ An operator is an instruction or set of instructions that states what to do with the function that follows it. It may be a simple instruction like “multiply the following function with 6,” or it may be much more complicated than the Hamiltonian.

when excited. This suggests that the motion of the electron in an atom is analogous to a system of standing waves.

Some common examples of standing waves are: a string fixed at both ends plucked to produce a musical tone in a guitar or violin.

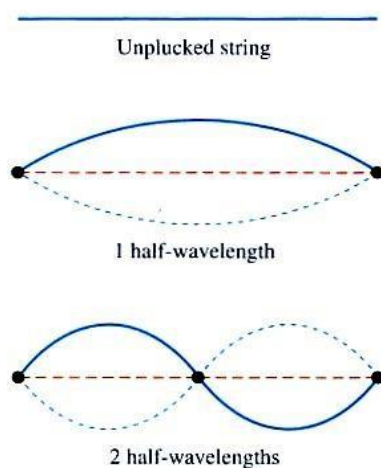


Figure 2.0.1: Standing wave



2.6.1 Solutions to the Schrodinger Equation

<https://www.youtube.com/watch?v=UjaAxUO6-Uw>

The main task of applying the Schrodinger equation ~~is~~ to a given problem is to obtain a suitable expression of ψ showing how the wave amplitude varies with distance along the x, y and z axes and then derive the solutions to the differential equation. In this course you are not required to solve the Schrodinger Equation or to derive it. Acceptable solutions (wavefunctions) are called *Eigen*⁴ functions and the energies associated with the acceptable Eigen functions are called *Eigen* values.

⁴ Eigenvalue" comes from the German "Eigenwert" which means proper or characteristic value. "Eigenfunction" is from "Eigenfunktion" meaning "proper or characteristic function".

The Schrodinger equation being a differential equation of the second order has innumerable many solutions but not all of these can be accepted as physically meaningful solutions. An acceptable solution must meet the following conditions:

- a) ψ must be a solution of the Schrodinger equation.
- b) ψ must be finite and single valued for all values of the coordinates

c) ψ must be continuous functions of the coordinates

ψ , $\frac{\partial \psi}{\partial x}$, $\frac{\partial \psi}{\partial y}$ and $\frac{\partial \psi}{\partial z}$ must be continuous functions of x , y , and z respectively

- e) The probability of finding the electron over all space from $+\infty$ to $-\infty$ must be equal to one. The probability of finding the electron at any point in space is given by the square of the wave function i.e. ψ^2 . Thus the sum of all probabilities

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \psi^2 dx dy dz = 1$$

Notes

- ψ has positive and negative amplitudes, like any wave.
- Due to the wave-like behavior of an electron, we cannot pin down its position and momentum at the same time (Heisenberg Uncertainty Principle). In other words, we do not know the exact location of an electron and how it moves from one spot to another in an atom.
- ψ has no physical significance but depends on x, y and z coordinates. Each ψ describes the wave properties of a given electron in a particular orbital. Every ψ matches an atomic orbital but it is not an atomic orbital.
- ψ^2 at any point around the nucleus is measure of the electronic charge density at that point. It may therefore be regarded as the probability density function for finding the electron at that point in space.



2.6.2 Types of functions (ψ) usually chosen

- Exponential functions; $e^{\pm kx}$
- Sine functions; $\sin x$ and
- Cosine functions; $\cos x$

These are chosen because their second derivatives give the same function multiplied by a constant i.e.

$$\frac{\hbar^2 k^2}{2m} = E$$



2.7 The quantum mechanics of some simple systems

Wave-mechanical treatment of physical problems is highly complicated mathematically. The mathematics entailed in the complete solution of the wave equation even for the simplest atomic systems, namely, the hydrogen and hydrogen-like atoms, is extremely complicated. For this reason, the study of wave mechanics is reserved for those who have the required proficiency in mathematics. Nevertheless, even a relatively limited mathematical background is sufficient for one to grasp the basic procedures involved in the wave mechanical treatment. To show this, we consider here a very simple problem, namely, the particle-in-a box problem, which, though hypothetical, bears some relevance to real molecular problems.

Broadly, our objectives in considering this problem are:

- To illustrate quantum mechanical principles
- To illustrate how Schrödinger equation is solved satisfying the given boundary conditions
- To show how discrete energy levels arise when a small particle is confined to a region of space.
- To illustrate how Schrödinger equation can predict the absorption spectrum of some linear conjugated molecules by treating the π electrons as free particles in a 1dimensional box with infinite walls.



2.7.1 The quantum mechanics of a particle in a one dimensional box

Consider a particle of mass m , bouncing back and forth between the walls of a box

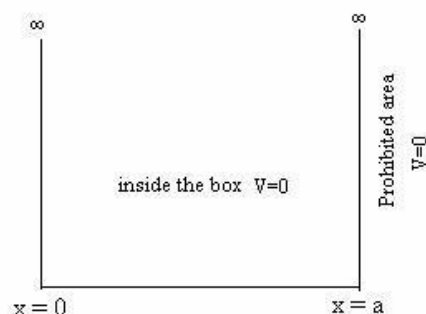


Figure 2.0.2:One-dimensional potential box particle box walls of infinite height at $x=0$ and $x=a$. Suppose that the particle travels only along the x -axis and is confined only to the area between

$x=0$ and $x=a$ by two infinitely hard walls so that the particle has no chance of penetrating them. Suppose that the particle does not lose energy when it collides with such walls, so that its total energy remains constant. The Schrödinger equation for the motion of a particle in one dimension is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + [E - V(x)]\psi(x) = 0$$

Since the potential outside the box is infinitely high, the probability of finding the particle outside must be zero, that is to say, ψ^2 and hence ψ must be zero when $0 > x > a$. To avoid a discontinuity at $x=0$ and $x=a$, the wavefunction must be zero at these points also and so the boundary conditions to be satisfied in the present problems are $\psi = 0$ at $x=0$ and $\psi = 0$ at $x=a$. Our task is now to find what is inside the box, in the region $0 < x < a$.

Since $V=0$ inside the box, the one-dimensional wave equation reduces to

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} - E\psi(x) = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{8\pi^2 m}{h^2} E \psi$$

Since E is positive, the coefficient of ψ is a negative constant at any value of E, and the differential equation can be written

$$\frac{\partial^2 \psi}{\partial x^2} = -k^2 E \psi \quad \text{Where } k^2 = -\frac{8\pi^2 m}{h^2} E$$

The solution to this (a well-known general equation) is

$$\psi = A \sin kx + B \cos kx$$

where A, B and k remain to be determined

That is,

$$\psi = A \sin \left(\frac{8\pi^2 m}{h^2} E \right)^{1/2} x + B \cos \left(\frac{8\pi^2 m}{h^2} E \right)^{1/2} x$$

Where A and B are integration constants.

For $x=0$, $\sin kx=0$

$\cos kx=1$, so $\psi=B$ at $x=0$.

However, we have already shown that ψ must be zero at $x=0$, and therefore $B=0$.

Consequently, the wave function

$\psi = A \sin kx + B \cos kx$ Reduces to

$$\psi = A \sin kx$$

or

$$\psi = A \sin \left(\frac{8\pi^2 m}{h^2} E \right)^{1/2} x$$

Applying the condition $\psi=0$ at $x=a$

$$\psi = A \sin kx = 0$$

Since $A \neq 0$, $A \sin kx = 0$, it follows that $ka = n\pi$. Therefore,

$$\psi = A \sin \frac{n\pi}{a} x$$

$$\left(\frac{8\pi^2 m}{h^2} E \right)^{1/2} = \frac{n\pi}{a}$$

Where $n=1, 2, 3, \dots$

$$E = \frac{n^2 h^2}{8ma^2}$$

NB: $n=1, 2, 3, \dots$ (n cannot be zero, since this would make ψ^2 zero between $x=0$ and $x=a$ i.e. the particle would not be in the box. That is,

$$E_n = \frac{n^2 h^2}{8ma^2}$$

Where $n=1, 2, 3$ and is the quantum number determining the energy of a particle of mass m confined within a well of length a . So we see that the limitations placed on the value of ψ lead to quantized energy levels whose spacing is determined by m and a . The corresponding wavefunction is given by

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$$\psi_n = A \sin \frac{n\pi x}{a}$$

Therefore, only certain values of E , namely, E_n , are possible, and these are the only ones allowed. The energy value E_n are usually called *eigenvalues* and ψ_n *eigen functions*; the integer n may be termed a *quantum number*, and it arises here merely as a consequence of the boundary conditions imposed. This quantum number can be equal to 1, 2, 3....but not zero because ψ , and hence ψ^2 or probability of finding the particle, would be zero everywhere within the box.

Example

Calculate the permitted energies of an electron confined in a box 1 Å wide.

Solution

The permitted energies are given by

$$E_n = \frac{n^2 h^2}{8ma^2} \quad ; \quad 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$E_n = \frac{n^2 (6.623 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} (1 \times 10^{-10})^2}$$

$$= 6.038 \times 10^{-18} n^2 \text{ J}$$

$$= 37.69 n^2 \text{ eV}$$

$$\approx 38 n^2 \text{ eV}$$

The minimum energy that said electron can have is $38 n^2 \text{ eV}$, corresponding to $n=1$. The other permitted energy levels are:

$$E_2 = (2)^2 \times 38 = 152 \text{ eV}$$

$$E_3 = (3)^2 \times 38 = 342 \text{ eV}$$

$$E_4 = (4)^2 \times 38 = 608 \text{ eV, etc}$$

$$E_n = \frac{n^2 h^2}{8ma^2}$$

Since n in $E_n = \frac{n^2 h^2}{8ma^2}$ cannot be zero, the minimum energy of E_1 corresponds to $n = 1$. It is the zero point energy (ZPE) of the system. This implies that matter cannot exist without motion.

The value of A is determined from the normalization condition. This involves setting to unity the integral of the probability between two fixed limits. Thus, since the particle must be

somewhere between $x=0$ and $x=a$, the integral of the probability between these two limits must, obviously, be unity, that is

$$\int_0^a P_x dx = \int_0^a \psi_n^2 dx = 1$$

Substituting $\psi_n = A \sin \frac{n\pi x}{a}$

$$\int_0^a A \sin \frac{n\pi}{a} x dx = 1$$

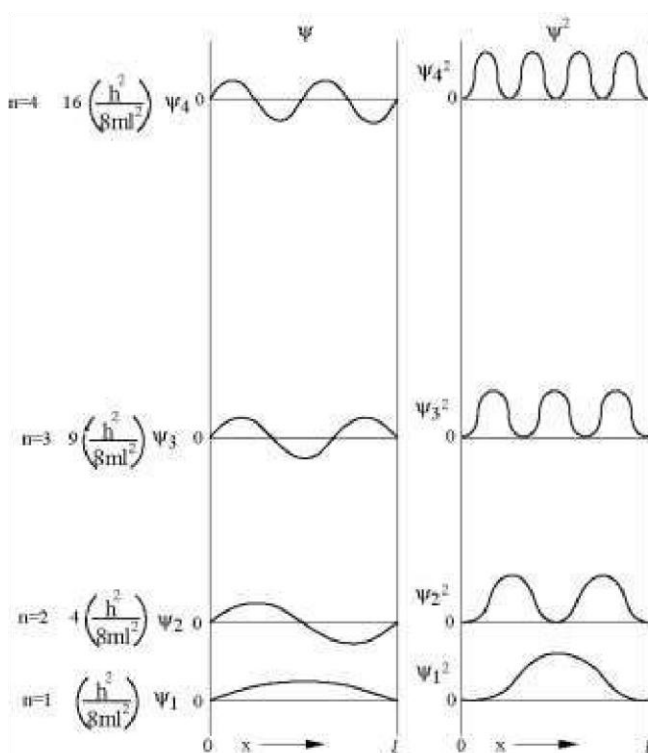
$$A^2 \int_0^a \frac{1}{2} \cos \frac{2n\pi}{a} x dx = 1$$

$$A^2 \left[x \right]_0^a \frac{c}{2n\pi} \sin \frac{2n\pi}{a} x \Big|_0^a = 1$$

Hence

$$A = \sqrt{\frac{2}{a}}$$

Therefore, the normalized wavefunction is given by $\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$. The plots of ψ and ψ^2 for a particle in a 1-dimensional potential well appear as follows:



- As n increases E increases and ψ becomes more wiggly. The λ becomes shorter and shorter.
- The probability finding the particle at the various points of the well along the allowed length l (or a) is not the same. For values of $n > 1$, ψ^2 at certain points = 0 which is classically impossible.

Figure 0.3: Variation of ψ and ψ^2 with x for $n=1, 2, 3$, and 4 for a particle in one dimensional box



Study questions

- What happens if the walls are removed?
- What happens if finite walls are used?
 - Case 1: $E < V_0$



- b) Case 1: $E > V_0$
- 3 a) What is quantum mechanical tunneling?
- b) State four examples of quantum mechanical tunneling
4. For a particle in a one dimension box, $E_n = \frac{n^2 h^2}{8ma^2}$ m = mass of the particle, and a = length of the box
- a) Calculate the energy difference between $n=1$ and $n=2$ levels for an electron ($m=9.109 \times 10^{-31} \text{kg}$) confined to a one dimension box having length of $4 \times 10^{-10} \text{m}$. (in Joules.
- b) the normalized wave function for a particle in a box is given by,

$$\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$$

Sketch curves for the variation of ψ with x for $n=1, 2$ and 3 for a particle in a one dimension box for $0 < x < a$

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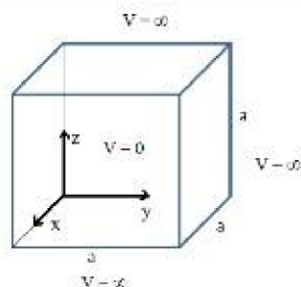
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2.8 The quantum mechanics of a particle in a three dimensional box

In place of one dimensional box, we now consider a three dimensional square box having sides a_x , a_y and a_z , and with potential zero everywhere inside the box but infinite outside. The wave function $\psi(x, y, z)$ in this case depends on three coordinates x , y and z . The Schrödinger equation now is



$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

$$E = E_x + E_y + E_z$$

Solution to these equations leads to

$$\psi_{n_x, n_y, n_z} = \sqrt{\frac{8}{a^3}} \sin \frac{n_x x}{a} \sin \frac{n_y y}{a} \sin \frac{n_z z}{a}$$

$$E_{n_x, n_y, n_z} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$

The energy thus depends on the sum of the squares of the three quantum numbers (n_x , n_y , and n_z) and evidently, it is possible for groups of different states, each specified by unique set of the three quantum numbers, to possess the *same energy*. In such a case, the energy level and the corresponding independent states are said to be *degenerate*. The zero point energy is given by:

$$E_{(1,1,1)} = \frac{3h^2}{8ma^2}, \text{ corresponding to the wavefunction } \psi_{(1,1,1)}.$$

$$6h^2$$

For the level for which the energy is $\frac{6h^2}{8ma^2}$, there are three independent states having quantum number (2,1,1), (1,2,1) and (1,1,2) for (n_x , n_y , n_z) corresponding to the wavefunction

$\psi_{(2,1,1)}$, $\psi_{(1,2,1)}$, and $\psi_{(1,1,2)}$. Such an energy level is said to be triply degenerate. The next four

$\frac{9h^2}{8ma^2}$	$\frac{11h^2}{8ma^2}$	$\frac{12h^2}{8ma^2}$	$\frac{14h^2}{8ma^2}$
energy levels	energy levels	energy levels	energy levels

respectively.

The degeneracy breaks down on applying even a small modification to the system. This has relevance to what is often observed in practice, such as the splitting of spectral lines in a magnetic or electric field. As a result of the application of a magnetic or electric field the degenerate levels break up into separate levels, and consequently, spectral lines which were single become multiplets.

3 LECTURE 3

THE HYDROGEN-LIKE ATOMS



3.1 Purpose

In this lecture you will learn how the Schrödinger equation for hydrogen-like atoms is solved. You will learn how the quantum numbers arise. You will also learn the shapes of s, p and d orbitals.



3.2 Lecture 3 objectives

At the end of this lecture, you should be able to:

- Explain the wave functions of hydrogen like atoms
- Solve the Schrödinger equation of hydrogen like atom
- Explain the four quantum numbers that describe an electron
- Draw *s*, *p*, and *d* orbitals



3.3 The Schrodinger Equation for hydrogen-like atoms

Hydrogen-like atoms are cations that have one electron moving in the field of force of the nucleus. The problem of the structure of the hydrogen atom is the most important problem in the field of atomic and molecular structure, not only because the theoretical treatment of this atom is simpler than that of other atoms and of molecules, but also because it forms the basis for the discussion of more complex atomic systems.

The wave mechanical treatment of polyelectronic atoms and of molecules is usually closely related in the procedure to that of the hydrogen atom, often being based on the use of hydrogen – like or closely related wavefunctions. Moreover, almost without exception the applications of qualitative and semi –quantitative wave-mechanical arguments to chemistry involve the functions which occur in the treatment of the hydrogen atom.

The hydrogen atom has held a prominent place in the development of physical theory. The first spectral series expressed by a simple formula was the Balmer series of hydrogen. Bohr's treatment of the hydrogen atom marked the beginning of the old quantum theory of atomic structure, and wave mechanics had its inception in Schrödinger's first paper, in which he gave the solution of the wave equation for the hydrogen atom. Only in Heisenberg's quantum mechanics was there extensive development of the theory before the treatment of the hydrogen atom, characterized by its difficulty, was finally given by Pauli.

The nucleus has a charge of $+Ze$, where Z =atomic number. The electron has a charge of $-e$. The potential $V(r)$ is given by

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r} \quad (5.1)$$

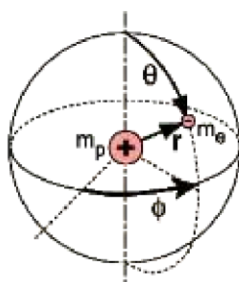
For the proton – electron Coulombic attraction, kinetic energy is given by

$$T = \frac{p^2}{2m} \quad (5.2)$$

Where $m = \frac{m_e m_p}{m_e + m_p}$ the reduced mass m_e

The Hamiltonian depends on the kinetic energy of the particle with the reduced mass of the system, as well as the potential energy (in Cartesian coordinates).

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \quad (5.3)$$



□

Figure 3.0.1: Positions of coordinates of electrons.

The electron-nuclear distance r which appears in the potential energy term can be expressed in terms of Cartesian coordinates as $r^2 = x^2 + y^2 + z^2$. However, owing to the symmetry of the physical situation, it will be more convenient to express equation (5.1) in terms of the spherical polar coordinates (r, θ, ϕ) (Fig 3.1), where r is the length of the radius vector, θ the angle between the axis of the sphere and the radius vector (or, in other words, it corresponds to 2° - 'latitude') and ϕ the angle between a fixed meridian plane and the meridian plane in which r is measured (i.e. it corresponds to the 'longitude').

A particle situated at (x, y, z) in Cartesian coordinates can be transformed to the (r, θ, ϕ) , system by the transformation

$$x = r \sin \theta \cos \phi; y = r \sin \theta \sin \phi \text{ and } z = r \cos \theta$$

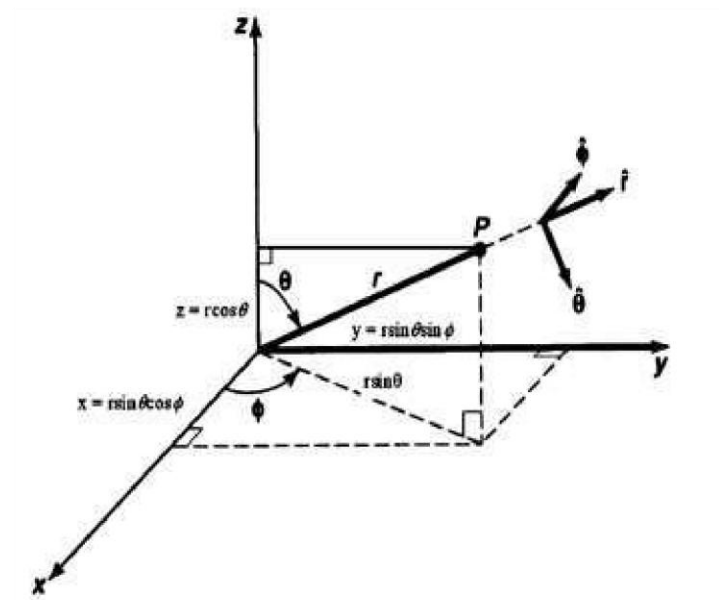


Figure 3.2: The relationship between Cartesian and polar coordinates $0 \leq r \leq \infty$, $0 \leq \theta \leq 180$, $0 \leq \phi \leq 360$



3.4 Atomic Wavefunctions

A wavefunction ψ describes an atomic orbital. For most calculations, it is simpler to solve the wave equation if ψ is converted from the Cartesian coordinates (x, y, z) to spherical ψ coordinates (r, θ, ϕ) . In spherical coordinates, ψ can be factored into a radial wavefunction R , and an angular wavefunction A :

$$\psi(x, y, z) = R(r)A(\theta, \phi)$$

The effect of this transformation from (x, y, z) to (r, θ, ϕ) is that the three-dimensional

Schrödinger equation when written in polar coordinates becomes:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = E \psi \quad (5.4)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = -\frac{2mE}{\hbar^2} \psi$$

The advantage of writing the Schrödinger equation in terms of spherical polar coordinates for the problem of one – electron atom is that in this form it may be readily separated into three

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independent equations each involving only one coordinate. By separation of the variables, (r, θ, ϕ) can be written as

$$\psi(r, \theta, \phi) = R(r) \Theta(\theta) \Phi(\phi) \quad (5.5)$$

Where Radial function $R(r)$ is a wavefunction dependent on r only, $\Theta(\theta)$ is a wavefunction dependent on θ only and $\Phi(\phi)$ is wavefunction dependent on ϕ only i.e. the three variables can be written in terms of their own wavefunctions, independent of other variable. $R(r)$ is called the radial function and $\Theta(\theta) \Phi(\phi)$ is called the angular function and may be denoted $Y(\theta, \phi)$.

Notes

- The radial function $R(r)$ describes how ψ of the electron varies with the radial distance from the nucleus while θ and ϕ are kept constant. It determines the spatial extent of the orbital.
- $\Theta(\theta)$ describes how ψ varies with the zenith angle θ along a constant meridian on a sphere centred at the nucleus while r and ϕ are kept constant.
- $\Phi(\phi)$ describes how ψ varies with the azimuthal angle ϕ along a constant meridian on a sphere centred at the nucleus while r and θ are kept constant.
- The product $\Theta(\theta) \Phi(\phi)$ is known as the angular wavefunction and is denoted $Y(\theta, \phi)$. It describes the shape of an atomic orbital and its orientation in space.

The radial and angular components of the Schrodinger equation can now be solved separately after separation of variables. In its most general form the solution to radial part of Schrodinger equation for the hydrogen atom may be expressed as a function of the quantum numbers n and l . The solution to the angular component on the other hand can be written as a function of the quantum numbers l and m .

The integer n is known as the **principal quantum number** and is related to the distance between the nucleus and electron. The integer l is known as the **azimuthal quantum number** and describes the general shape of the wavefunction for the atomic orbital. The integer m is known as the **magnetic quantum number** because in the presence of a magnetic field, states with the same values of m will have different energies (Zeeman effect). It determines the orientation of a given orbital in space.

The quantum numbers n , l and m are related as follows:

Principal quantum number $n = 1, 2, 3, \dots, \infty$

Azimuthal quantum number $l = 0, 1, 2, \dots, (n - 1)$

Magnetic quantum number $m = 0, \pm 1, \pm 2, \dots, \pm l$

The three quantum numbers n , l and m characterize each solution of the Schrodinger equation. Of these three quantum numbers only n enters the energy formula. Therefore all the solutions having the same values of n but different values of l and m will be energetically degenerate.

For the lowest energy wavefunction $n=1$ and $l=0$ and $m=0$. Since for this function no other choices are available this level is said to be non-degenerate.

By convention (from spectroscopy) a solution to the Schrodinger equation in which $l=0$ is denoted as an s orbital (or wavefunction). The other notations are as follows:

l	0	1	2	3	4	5	etc	Notation	s	p	d	f
g	h											

The orbital (wave function) labels are worked out as follows:

- For the energy level characterized by $n=1$,
When $l=0$, $m=0$. This gives an s orbital with only one orientation in space.
Thus the $n=1$ energy level consists of no sublevels. It is only one orbital denoted $1s$.
The 1 stands for the principal quantum number and the s shows the shape of the orbital i.e. $l=0$
- For the energy level characterized by $n=2$, $l=0, 1$ When $l=0$, $m=0$ as well and is denoted $2s$ orbital.
When $l=1$, $m=-1, 0, +1$. Thus we have a p orbital with three orientations in space. Thus, we have $2p_x$, $2p_y$ and $2p_z$ orientations.

Thus the $n=2$ energy level consists of two sublevels (orbitals) $2s$ and $2p$. The $2p$ sublevel (also called subshell) consists of three orbitals in three different orientation in space. iii) For the energy level characterized by $n=3$, $l=0, 1, 2$ When $l=0$, $m=0$ as well and is denoted $3s$ orbital.

When $l=1$, $m=-1, 0, +1$. Thus we have a p orbital with three orientations in space. Thus, we have $3p_x$, $3p_y$ and $3p_z$ orientations.
When $l=2$, $m=-2, -1, 0, +1, +2$. Thus we have a d orbital with five orientations in space. Thus, we have $3d_{xy}$, $3d_{xz}$, $3d_{yz}$, $3d_{x^2-y^2}$ and $3d_z^2$ orientations.

Thus the $n=3$ energy level consists of three sublevels (orbitals) $3s$ and $3p$ and $3d$. The $3p$ sublevel (also called subshell) consists of three orbitals in three different orientations in space while the $3d$ sublevel (or subshell) consists of five orbitals in five different orientations in space. iv) For the energy level characterized by $n=4$, $l=0, 1, 2, 3$ Thus, this level consists of the sublevels $4s$, $4p$, $4d$ and $4f$. The $4f$ has seven orientations ($m=-3, -2, -1, 0, 1, 2, 3$).

We shall use this knowledge to explain why each energy level can only contain a specific number of electrons when we discuss electron configurations.



3.4.1 Radial Wavefunctions and Radial Distribution Functions

The method of describing the shape of an orbital in terms of its projection of its wavefunction along an axis, as in the p_x orbital seen earlier, is a way of describing the orientation dependent

part of the wavefunction. That the wavefunction of the p_x orbital is orientationally dependent means that its projection is not the same along each of the Cartesian axes.

The electronic wavefunction, as we have seen above, describes the distribution of the electron positions in terms of the distance of the electron from the nucleus, r , and the orientation of the electron relative to the nucleus. We can separate the wavefunction into an orientationally dependent part, known as the angular wavefunction, and an orientationally independent part, which is known as the *radial wavefunction*.

$$\psi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) Y_{n,l,m_l}(\theta, \phi)$$

The Radial Wavefunction, $R_{n,l}(r)$, depends only on the distance of the electron from the nucleus, and is characterized by the values of the principal and orbital angular momentum quantum numbers, whereas the angular wavefunction, Y , depends on the angles of the electron from the nucleus, and is characterized by the values of the orbital angular momentum and magnetic quantum numbers.

The s orbitals consist only of a radial part to the wavefunction, and $Y = 1$. The angular wavefunctions for other types of orbitals are complicated mathematical expressions, so generally only the shapes of the orbitals, as shown above, are important.

Another important function we need to consider is the *Radial Distribution Function*, $P_{n,l}(r)$. This is defined as the probability that an electron in the orbital with quantum numbers n and l will be found at a distance r from the nucleus. It is related to the radial wavefunction by the following relationship:

$$P_{n,l}(r) = 4\pi r^2 R_{n,l}^2(r); \text{normalized by } \int_0^\infty P_{n,l}(r) dr = 1$$

The factor $4\pi r^2$ arises because the *radial distribution function* refers to the probability of finding an electron not at a specific point in space (which equals ψ^2), but on a spherical shell of area $4\pi r^2$, at a distance r from the nucleus. The integral results from the fact that the total probability of finding the electron is one, as it must be found somewhere around the nucleus.

The number of radial nodes, where the sign of the $R_{n,l}(r)$ changes, in the radial wavefunction,

g



is equal to $n - l - 1$. The number of maxima in the radial distribution function is equal to $n - l$.

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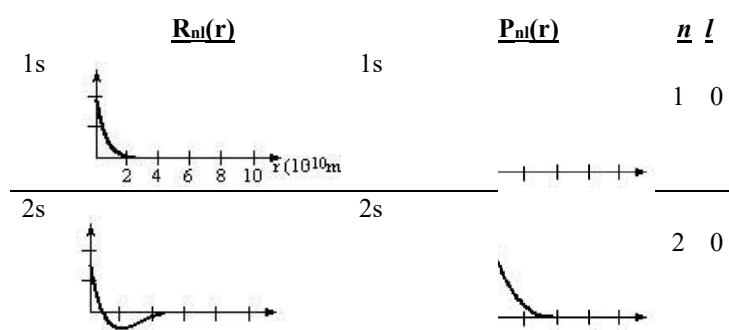
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3.4.2 Radial wavefunction $R(r)$

Describes how the electron density changes with distance from the nucleus. s orbitals have finite electron density [$R(r) \neq 0$] at the nucleus but this drops off exponentially as the distance from the nucleus increases. Other orbitals (e.g., p and d orbitals) have zero electron density [$R(r) = 0$] at the nucleus.

Notes

1. The point at which the radial wavefunction $R(r) = 0$ (except the origin) is called a radial node.
2. The sign of the radial wavefunction $R(r)$ changes (from +ve to -ve or vice versa) after passing the radial node
3. The number of radial nodes for a given orbital is given by $[n - (l + 1)]$. For example,
 - a) ns orbitals have $(n - 1)$ radial nodes, e.g. $2s$ has 1 radial node.
 - b) np orbitals have $(n - 2)$ radial nodes, e.g. $2p$ has 0 radial node, $3p$ has 1 radial node.
 - c) nd orbitals have $(n - 3)$ radial nodes, e.g. $3d$ has 0 radial node, $5d$ has 2 radial nodes.
 - d) nf orbitals have $(n - 4)$ radial nodes, e.g. $4f$ has 0 radial node.
 - e) $1s$ orbital has no radial node.
 - f) $2s$ orbital has two regions of maximum probability, separated by a spherical surface of zero probability (1 radial node)
 - g) $3s$ orbital has three regions of maximum probability, separated by two radial nodes.



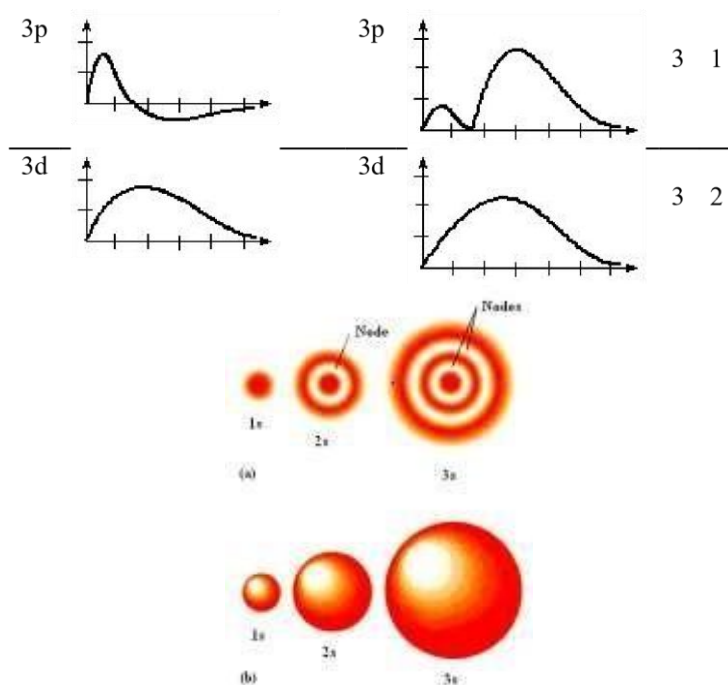
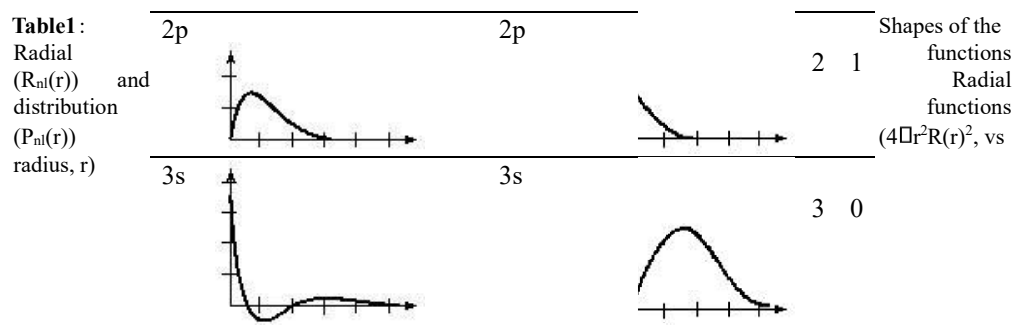


Figure 3.34: Boundary surface diagram of s orbitals and electron density plots showing the presence of radial nodes (diagram from Zumdahl)



3.4.3 Radial distribution function $4\pi r^2 R(r)$

Tells us where an electron will be most likely found at a given distance from the nucleus. Plots of radial distribution functions for $1s$, $2s$ and $3s$ orbitals show that there is at least one maximum and the fact that an electron tends to be further from the nucleus as its principal quantum number n increases. *Therefore the size of an atomic orbital increases as the principal quantum number n increases.*

Plots of radial distribution functions for $3s$, $3p$ and $3d$ orbitals show that s orbital has electron density closest to the nucleus than p and d orbitals. We say that an electron in an s orbital is more penetrating than that in a p or d orbital.



3.4.4 Electron position

The probability of locating an electron within a volume v is given by

$$\int_V |\psi|^2 d\tau$$

Energy values

The energy eigen values for a hydrogen like atom are specified by

$$E_n = -\frac{2\pi^2 m_e Z^2 e^4}{n^2 h^2}, \text{ where } n=1, 2, 3, \dots$$

This result is identical to that of the Bohr theory. We have thus, obtained three quantum numbers which we may list together with their permissible values as follows

$$\begin{aligned} n &= 1, 2, 3, \dots \\ l &= 0, 1, 2, \dots, (n-1) \\ m &= 0, \pm 1, \pm 2, \pm 3, \dots, \pm l \end{aligned}$$

In summary, wavefunctions of the one-electron atom may be written in the general form

$$\psi_{n,l,m}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m}(\theta, \phi)$$

Since each of the component functions making up ψ is separately orthonormal, it follows

$$\int \psi_{n,l,m}^*(r, \theta, \phi) \psi_{n',l',m'}(r, \theta, \phi) d\tau = \delta_{nn'} \delta_{ll'} \delta_{mm'}$$



3.5 Hydrogen-like Wavefunctions

Table 0.1: Real forms of one-electron wavefunction

Wavefunction symbol	Orbital symbol	Wavefunction equation
Ψ_{100}	1s	$\psi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a_0} \right)^{3/2} e^{-z/a_0}$ $\psi_{2s} = \frac{1}{4\sqrt{2\pi}} \left(\frac{z}{a_0} \right)^{3/2} (2 - \frac{z}{a_0}) e^{-z/2a_0}$ $\psi_{2px} = \frac{1}{4\sqrt{2\pi}} \left(\frac{z}{a_0} \right)^{3/2} \frac{z}{a_0} \sin\theta \cos\phi$ $\psi_{2py} = \frac{1}{4\sqrt{2\pi}} \left(\frac{z}{a_0} \right)^{3/2} \frac{z}{a_0} \sin\theta \sin\phi$ $\psi_{2pz} = \frac{1}{4\sqrt{2\pi}} \left(\frac{z}{a_0} \right)^{3/2} \frac{z}{a_0} \cos\theta$ $\psi_{3s} = \frac{1}{81\sqrt{\pi}} \left(\frac{z}{a_0} \right)^{3/2} (27 - 18\frac{z}{a_0} + 2\frac{z^2}{a_0^2}) e^{-z/3a_0}$ $\psi_{3px} = \frac{1}{81\sqrt{\pi}} \left(\frac{z}{a_0} \right)^{3/2} (6 - \frac{z}{a_0}) \frac{z}{a_0} \sin\theta \cos\phi$
ψ_{200}	2s	
ψ_{210}	2px	
ψ_{211}	2py	
ψ_{210}	2pz	
ψ_{300}	3s	

$\Psi_{3l m_l}$	Orbital	Wavefunction
$\Psi_{31 \pm 1}$	3px	$\Psi_{3px} = \frac{(z/a_0)^{3/2}}{81\sqrt{\pi}} (6-\rho) \rho e^{-\rho/3} \sin\theta \sin\phi$
Ψ_{310}	3py	$\Psi_{3py} = \frac{(z/a_0)^{3/2}}{81\sqrt{\pi}} (6-\rho) \rho e^{-\rho/3} \sin\theta \cos\theta$
Ψ_{320}	3pz	$\Psi_{3pz} = \frac{\sqrt{2}(z/a_0)^{3/2}}{81\sqrt{\pi}} (6-\rho) \rho e^{-\rho/3} \cos\theta$
$\Psi_{32 \pm 1}$	3dz	$\Psi_{3dz} = \frac{\sqrt{2}(z/a_0)^{3/2}}{81\sqrt{6\pi}} \rho^2 e^{-\rho/3} (3\cos^2\theta - 1)$
$\Psi_{32 \pm 1}$	3dx+z	$\Psi_{3dx+z} = \frac{\sqrt{2}(z/a_0)^{3/2}}{81\sqrt{6\pi}} \rho^2 e^{-\rho/3} \sin\theta \cos\theta \cos\phi$
$\Psi_{32 \pm 1}$	3dy+z	$\Psi_{3dy+z} = \frac{\sqrt{2}(z/a_0)^{3/2}}{81\sqrt{6\pi}} \rho^2 e^{-\rho/3} \sin\theta \cos\theta \sin\phi$
$\Psi_{32 \pm 2}$	3dxy	$\Psi_{3dxy} = \frac{(z/a_0)^{3/2}}{81\sqrt{6\pi}} \rho^2 e^{-\rho/3} \sin^2\theta \cos 2\phi$
$\Psi_{32 \pm 2}$	3dx+y	$\Psi_{3dx+y} = \frac{(z/a_0)^{3/2}}{81\sqrt{6\pi}} \rho^2 e^{-\rho/3} \sin^2\theta \sin 2\phi$

With $\rho = \frac{z}{a_0} r$



3.6 Pictorial Representation of Atomic Orbitals

1. An atomic orbital is defined as a boundary surface which encloses 95 % of the electron density for a particular ψ .
2. Atomic orbitals can be represented pictorially by boundary surface diagrams (H&S Fig. 1.9, 1.10, 1.11) or electron density contour plots.

s orbitals

- $l = 0, m_l = 0$, the angular wavefunction $A(\theta, \phi)$ is independent of the angle θ and ϕ . Therefore, an s orbital is spherically symmetrical about the nucleus.

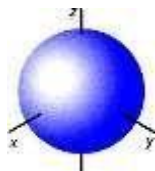
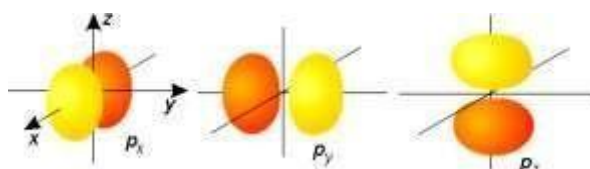


Figure 3.0.47: Boundary surface diagram of an s orbital. (Figure taken from Shriver & Atkins) *p*

orbitals

- $l = 1$; $m_l = -1, 0, +1$. 3 possible orientation of p orbitals: along the x, y and z axes.
- Dumb-bell shaped; electron density concentrated in identical lobes on either side of the nucleus.
- p_z orbital has an angular node (point where $A(\theta, \phi) = 0$) along the xy plane; p_x orbital has an angular node along the yz plane; p_y orbital has an angular node along the xz plane.
- The electron density contour map shows that there is no radial node for the $2p$ orbitals but 1 radial node for the $3p$ orbitals. [see example 1]



OR

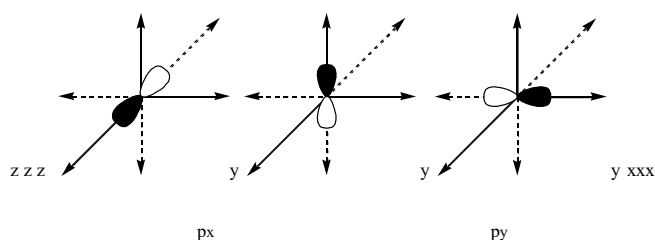


Figure 3.0.58 Boundary surface diagram of p orbitals. The lightly shaded lobe has a positive amplitude; the more darkly shaded lobe is negative (Figure taken from Shriver & Atkins)

d orbitals $\ell = 2$, $m_\ell = -2, -1, 0, +1, +2$. There are five d orbitals.

- d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$ have clover leaf shape and have 4 lobes of maximum electron probability centered in the plane indicated in the orbital label. The four lobes are separated by two nodal planes through the nucleus.
- d_{xy} , and $d_{x^2-y^2}$ are both centered in the xy plane while the lobes of d_{xy} lie between the axes.
- d_{z^2} has two lobes along the z axis and a "belt" centered in the xy plane.

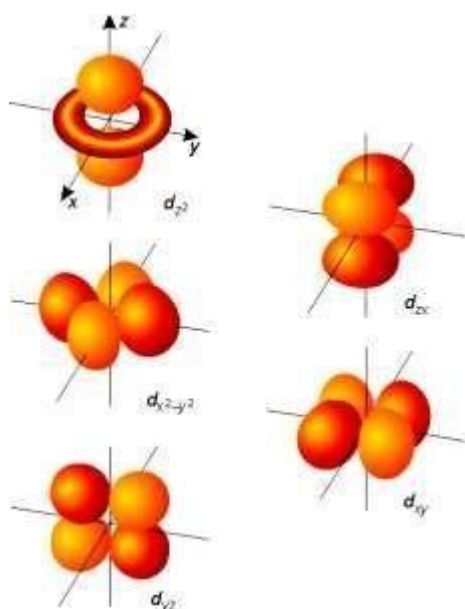


Figure 3.0.69: Boundary surface diagram of d orbitals. The lightly shaded lobe has a positive amplitude; the more darkly shaded lobe is negative (Figure taken from Shriver & Atkins).



3.7 Nodal Surfaces and Electron Density Contour Plots of Atomic Orbitals

Electron density contour maps are the most informative way of representing the ψ^2 function. The contour maps are obtained by computing values of electron density at a large number of points of various coordinates x , y , and z .

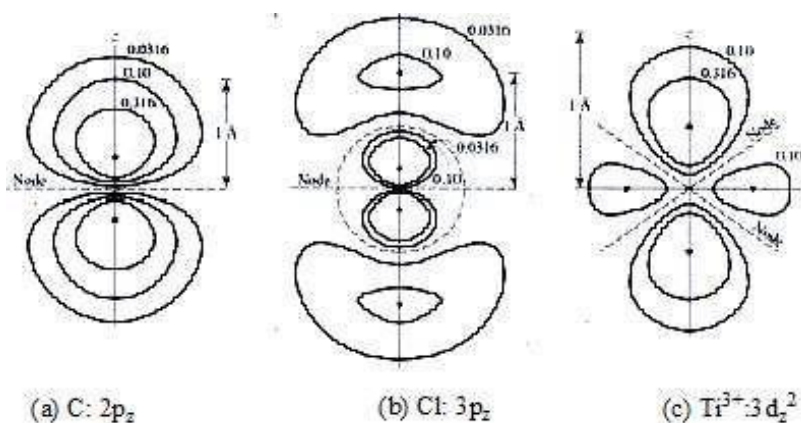


Figure 0.740: Electron density contour plots for selected atomic orbitals showing radial or angular nodes (diagram taken from Miessler&Tarr).



3.8 The Multi-electron atoms

In principle The Schrödinger equation can be applied to multi-electron atoms however, it can only be solved approximately. The orbitals derived from the Schrödinger equation for the hydrogen atom are used for multi-electron atoms with a correction to take into account the different effective nuclear charge.

Recall

- H has got the simplest structure; 1-electron moving in the field of force of the nucleus. The potential energy function V in the S.E is given by

$$V = -\frac{e^2}{r}$$

- For hydrogen-like ions e.g. Li^{2+} and multi-electron atoms the charge in the nucleus is Ze . Thus;

$$V = -\frac{Ze^2}{r}$$

In multi-electron atoms, a given electron moves under the influence of the nucleus and repulsion of other electrons. The influence of electrons on each other in an atom can be envisaged in two interrelated phenomena:

- The screening effect
- Penetration effect

Questions to ponder

- How do the electrons in a many-electron atom populate the available orbitals?
- What determines the orbitals in which electrons reside?



3.8.1 The Screening effect

Consists of a reduction of the nuclear attraction on an electron due to the presence of other electrons nearer the nucleus than the electron under observation. The other electrons lower the full charge of the nucleus from Z to Z^* - the effective nuclear charge. Thus, the electron is said to be screened and experiences less nuclear attraction (Z^*) given by

$$Z^* = Z - S$$

Where Z = nuclear charge, S = screening/shielding constant

Screening increases with number of e^- shells. The calculation of the screening constant will be discussed in the next lecture.



3.8.2 Penetration effect

Penetration refers to how effectively electrons can get close to the nucleus. It describes the proximity of electrons in an orbital to the nucleus. It is actually just how well the electrons feel the nucleus. Arises from the wave nature of the electron – quantum mechanics. Some of the time even outermost electrons are in the region near the nucleus where screening is minimum. Thus, they can be said to penetrate through the inner electron shells to the nucleus. For a given n value, penetration varies as follows: $s > p > d > f$. The energy of the subshells in a given energy level decrease in the orders $< p < d < f$. And for different values of shell (n) and subshell (l), penetrating power of an electron follows the trend:

$$1s > 2s > 2p > 3s > 3p > 4s > 3d > 4p > 5s > 4d > 5p > 6s > 4f \dots$$

and the energy of an electron for each shell and subshell goes as follows... $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p \dots$

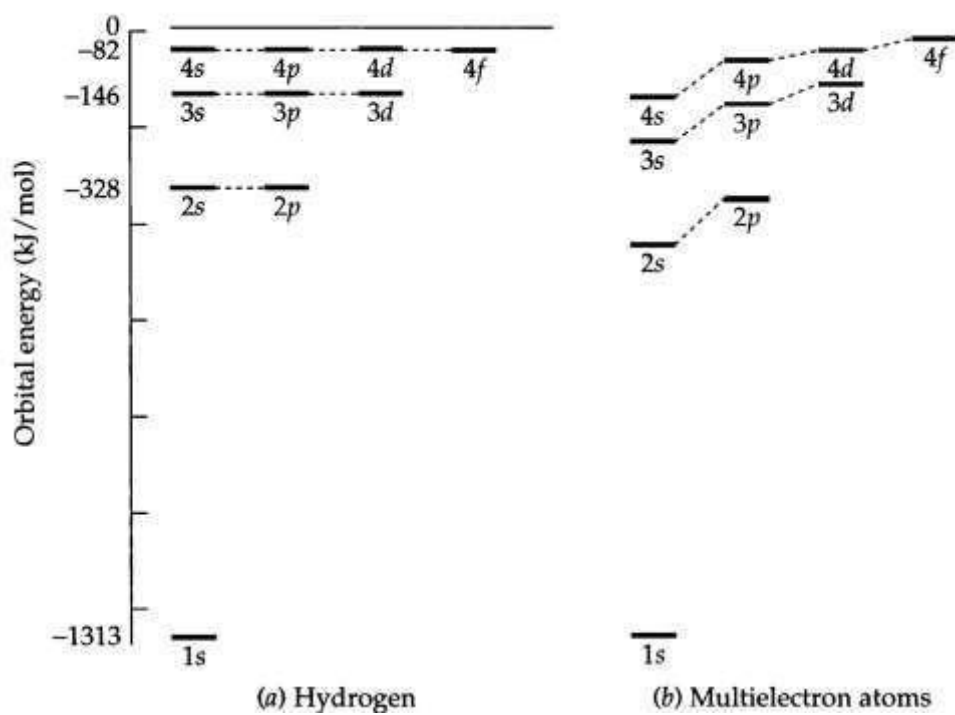


Figure 3.0.811: Orbital energy level diagrams for a hydrogen atom and multi-electron atoms (diagram taken from McMurry & Fay)

4 LECTURE 4

ELECTRONIC CONFIGURATIONS



4.1 About this Lecture

In this lecture, you will learn energy levels of orbitals and how to use *Aufbau* principle to write the electronic configurations of atoms. You will practice how write electron configuration of all the elements in the periodic table.



4.2 Lecture 4 Objectives

At the end of this lecture, you should be able to:

- Describe energy levels of orbitals in an atom
- Use *Aufbau* principle to write electronic configurations of the first 30 elements of the periodic Table.



4.3 Energy Levels in Hydrogen and Other Atoms

For H ~~and hydrogen-like~~ atoms, the energy of the electron only depends on n , principal quantum number. Since they have only one electron, it only experiences only nuclear attraction and no screening. Thus the subshells s, p, d, f have the same energy in each energy level characterized by the quantum number n . In the ground state, the electron in the H atom resides on the $1s$ orbital.

For other atoms with more than one electron, electron-electron repulsion and shielding play a role in deciding the energies of the subshells. Thus, the **energy of a given subshell depends on angular momentum quantum number l as well as the principal quantum number n .**

They vary in the order, $ns < np < nd < nf$. This ordering can be explained by considering the ability of having electron density closest to the positively charged nucleus in different atomic orbitals. The plot of radial distribution functions, $4\pi r^2 R(r)^2$, for $3s, 3p, 3d$ orbitals versus radius (r) (seen in Lecture 3) shows the first maximum for the s orbital is closest to the nucleus than the p or d orbitals, i.e., the s orbital has electron density closest to the nucleus. We say that electrons in an s orbital can penetrate closer to the nucleus than an electron in a p, d , or f orbital can. Since orbitals with electron density closer to the nucleus are more stable, therefore, for a given n , the energy of the orbital increases in the order: $s < p < d < f$.

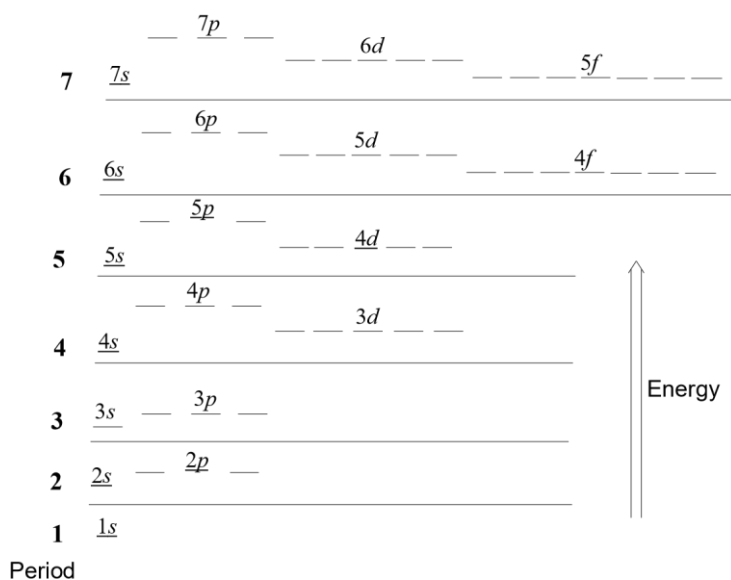


Figure 4.0.1: Relative energy level of various atomic orbitals in a multielectron atom (diagram from RaynerCanham).

Please note for the value of ns and $(n - 1) d$ orbitals, there are crossover points (e.g., $4s < 3d$; $5s < 4d$; $6s < 5d$; $7s < 6d$), this becomes important when filling electrons in multi-electron atoms. The effect is so pronounced for d and f electrons that they lie at higher energy than the s or p of the $(n + 1)$ shell in some cases. E.g. Energy level of $3d > 4s$, $4d > 5s$; $4f > 5s$, $5p$.



4.4 Electronic Configurations

These describe how electrons are distributed in various orbitals in an atom. The filling of orbitals with electrons is governed by the following principles:



4.4.1 The Aufbau Principle

1. Lowest energy orbitals are filled with electrons first.
2. Only two electrons with opposite spins per orbital (because Pauli exclusion principle [says](#) [state](#) that no two electrons in an atom can have the same four quantum numbers).
3. If two or more degenerate orbitals (i.e. orbitals with the same energy) are available, electrons go into each degenerate orbital with parallel spins until the orbitals are half-filled (Hund's rule).

Order of orbital filling in polyelectronic atoms is as follows:

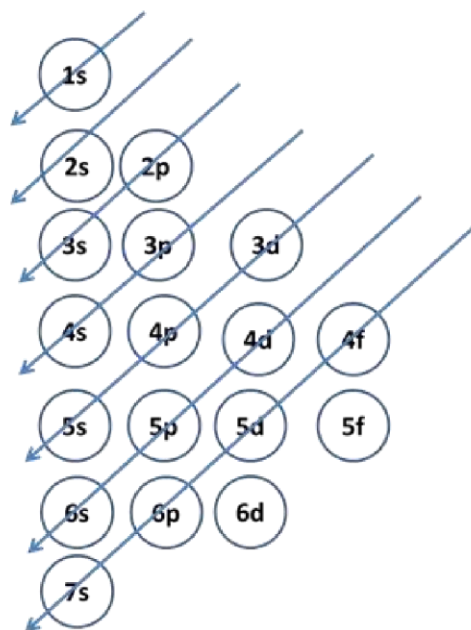


Figure 4.0.2: Order of orbital filling in poly-electronic atoms

Table 0.1: Filling of electrons for the first 30 elements

Atom	Atomic number	Electronic configuration	Atom	Atomic number	Electronic configuration
H	1	$1s^1$	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
He	2	$1s^2$	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Li	3	$1s^2 2s^1$	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Be	4	$1s^2 2s^2$	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
B	5	$1s^2 2s^2 2p^1$	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
C	6	$1s^2 2s^2 2p^2$	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
N	7	$1s^2 2s^2 2p^3$	Ti	22	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
O	8	$1s^2 2s^2 2p^4$	V	23	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$

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F	9	$1s^2 2s^2 2p^5$	Cr	24	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d_5$
He	10	$1s^2 2s^2 2p^6$	Mn	25	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d_5$
Na	11	$1s^2 2s^2 2p^6 3s^1$	Fe	26	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d_6$
Mg	12	$1s^2 2s^2 2p^6 3s^2$	Co	27	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d_7$
Al	13	$1s^2 2s^2 2p^6 3s^2 3p_1$	Ni	28	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d_8$
Si	14	$1s^2 2s^2 2p^6 3s^2 3p_2$	Cu	29	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d_{10}$
P	15	$1s^2 2s^2 2p^6 3s^2 3p_3$	Zn	30	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d_{10}$

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4.4.2 Calculation of Shielding Constant

In atoms with more than one electrons, the effect of electron-electron repulsion depends on where the various electrons are located in the atom. Electrons in the outer shell (higher n) are shielded by electrons in the inner shells (lower n). As a result, the net nuclear charge (or effective nuclear charge, Z_{eff}) felt by an outer electron is substantially lower than the actual nuclear charge (Z). We say that the outer electrons are **shielded** from the full charge of the nucleus by the inner electrons.

Electrons in the same shell (same n) have an immediate effect in shielding the nuclear charge. Electrons in the outer shell (higher n) do not shield the nuclear charge from the inner electrons (lower n).

In 1930, Slater formulated a set of rules for the **effective nuclear charge** felt by electrons in different atomic orbitals based on experimental data. The effective nuclear charge Z_{eff} can be calculated by the following equation:

$$Z_{\text{eff}} = Z_{\text{actual}} - S$$

where, Z : atomic number (number of protons); S : Slater screening constant;

Slater Rules

The values of S for various electrons are estimated as follows:

- i). Write out the electron configuration of an element in groups of: $(1s)$, $(2s, 2p)$, $(3s, 3p)$, $(3d)$, $(4s, 4p)$, $(4d)$, $(4f)$, $(5s, 5p)$ etc. Electrons with higher n do not shield those in lower n ; $S = 0$.
- ii). If the electron in question resides in an s or p orbitals, then,
 - Each of the other electrons in the same ns, np group, $S = 0.35$.
 - Each electrons in the $(n - 1)$ shell, $S = 0.85$.
 - Each electrons in the $(n - 2)$ and lower shell, $S = 1.00$.
- iii). If the electron in question resides in a d or f orbitals, then,
 - Each of the other electron in the same nd, nf subshell contributes $S = 0.35$.
 - Each electrons in the $(n - 1)$ shell, $S = 1.00$.



Study Questions

1. Calculate the effective nuclear charge for the outermost electron in Oxygen, given the electron configuration of O is $1s^2 2s^2 2p^4$.
2. Calculate the effective nuclear charge on a 3d electron in a Nickel atom, given the electron configuration of Ni: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$.

Solution

$$(1s^2)(2s^2 2p^6)(3s^2 3p^6) (3d^8)(4s^2)$$

$$Z^* = 28 - \{(2 \times 1) + (8 \times 1) + (8 \times 0.85) + (7 \times 0.35)\} =$$

5 LECTURE 5 THE PERIODIC TABLE



5.1 About this lecture

In this lecture you will learn about the quantum basis of the periodic table and how to place an atom in the periodic table. You will also learn periodic trends of some atomic properties.



5.2 Lecture 5 Objectives

At the end of this lecture, you should be able to: i) Describe the modern periodic table ii) To place an element in the periodic table given the electron configuration iii) Explain various trends in periodic table



5.3 Structure of the Periodic Table

Elements are arranged with increasing atomic number in rows (called periods) and columns (called groups). The periods correspond to the occupied outermost energy level, n , while rows correspond to the outermost occupied subshell denoted by angular quantum number l . It is divided into four blocks based on the orbitals filling with electrons (Fig5.1):

- i). Elements in Groups 1 and 2 are also called s block elements; valence electrons fill in s orbitals.
- ii). Elements in Groups 13 to 18 are also called p -block elements; valence electrons fill in p orbitals.
- iii). Elements in Groups 1, 2 and 13 - 18 are called main group elements. iv). Elements in Groups 3 - 12 are called transition metal because they have partially filled d -orbitals. They are also called d -block elements because valence electrons are contained in d orbitals.
- v). f -block elements (also called Lanthanides and Actinides) have valence electrons filling in f orbitals.
- vi). Elements in each group have similar valence electron configuration (electrons in the outermost shell) and therefore similar properties.

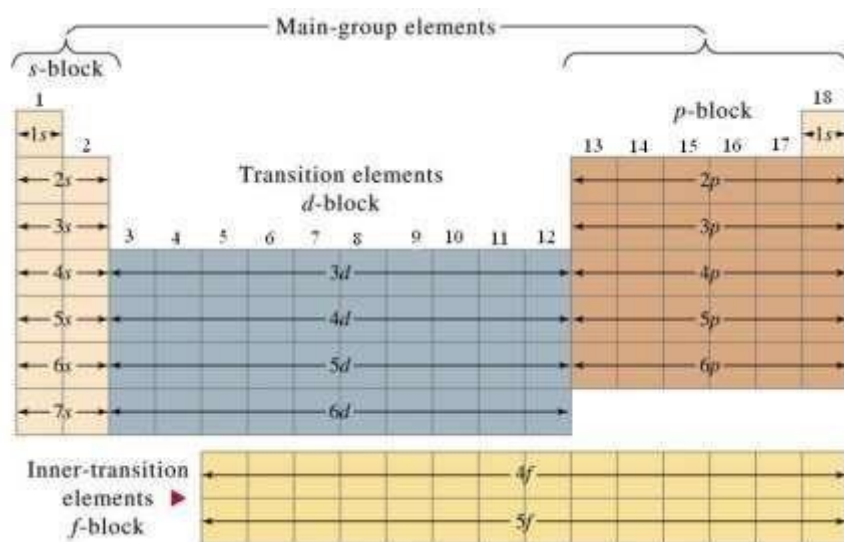


Figure 5.0.1: Sections of the Periodic Table(http://en.wikipedia.org/wiki/Periodic_table)



5.4 Periodic Trends of Atomic Parameters

Atomic properties such as effective nuclear charge, atomic radii, ionization energies, electron affinity and electronegativity are important in accounting for the chemical properties of an element.



5.4.1 Periodic Trend in Effective Nuclear Charge

Below is a table of the values of effective nuclear charge (Z_{eff}) for s and p electrons in elements Li - Ne:

Element	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
1s	2.69	3.68	4.68	5.67	6.66	7.66	8.65	9.64
2s	1.28	1.91	2.58	3.22	3.85	4.49	5.13	5.76
2p			2.42	3.14	3.83	4.45	5.10	5.76

In the same atom, each electron with different n and l values can have a different Z_{eff} because the repulsive effects due to the other electrons is different. E.g., the Z_{eff} for the B atom: $1s \gg 2s > 2p$ and the size of orbitals is: $1s \ll 2s < 2p$

For the same n and l value, Z_{eff} increase across a period (e.g., from Li to Ne) as Z increases. As a result, the size of the orbital decreases (electrons gets closer to the nucleus) and the size

of the atoms decreases. E.g. the Z_{eff} for the $2p$ orbitals: $B < C < N < O < F < \text{Ne}$ and the size of the $2p$ orbitals is: $B > C > N > O > F > \text{Ne}$



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Periodic Table of the Elements
and ground-state electronic configurations
(see opposite for Table of Standard Atomic Weights)

Period ↓	1	2											13	14	15	16	17	18
	1	2																
	3	4											5	6	7	8	9	10
2	Li [He]2s ¹	Be [He]2s ²											B 2s ² 2p ¹	C 2s ² 2p ²	N 2s ² 2p ³	O 2s ² 2p ⁴	F 2s ² 2p ⁵	Ne 2s ² 2p ⁶
3	11	12											13	14	15	16	17	18
	Na [Ne]3s ¹	Mg [Ne]3s ²	3	4	5	6	7	8	9	10	11	12	Al 3s ² 3p ¹	Si 3s ² 3p ²	P 3s ² 3p ³	S 3s ² 3p ⁴	Cl 3s ² 3p ⁵	Ar 3s ² 3p ⁶
4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
	K [Ar]4s ¹	Ca [Ar]4s ²	Sc 3d ¹ 4s ²	Ti 3d ² 4s ²	V 3d ³ 4s ²	Cr 3d ⁵ 4s ¹	Mn 3d ⁵ 4s ²	Fe 3d ⁶ 4s ²	Co 3d ⁷ 4s ²	Ni 3d ⁸ 4s ²	Cu 3d ¹⁰ 4s ¹	Zn 3d ¹⁰ 4s ²	Ga 4s ² 4p ¹	Ge 4s ² 4p ²	As 4s ² 4p ³	Se 4s ² 4p ⁴	Br 4s ² 4p ⁵	Kr 4s ² 4p ⁶
5	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
	Rb [Kr]5s ¹	Sr [Kr]5s ²	Y 4d ¹ 5s ²	Zr 4d ² 5s ²	Nb 4d ⁴ 5s ¹	Mo 4d ⁵ 5s ¹	Tc 4d ⁵ 5s ²	Ru 4d ⁷ 5s ¹	Rh 4d ⁸ 5s ¹	Pd 4d ¹⁰	Ag 4d ¹⁰ 5s ¹	Cd 4d ¹⁰ 5s ²	In 5s ² 5p ¹	Sn 5s ² 5p ²	Sb 5s ² 5p ³	Te 5s ² 5p ⁴	I 5s ² 5p ⁵	Xe 5s ² 5p ⁶
6	55	56	57 *	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
	Cs [Xe]6s ¹	Ba [Xe]6s ²	La 5d ¹ 6s ²	Hf 5d ² 6s ²	Ta 5d ³ 6s ²	W 5d ⁴ 6s ²	Re 5d ⁵ 6s ²	Os 5d ⁶ 6s ²	Ir 5d ⁷ 6s ²	Pt 5d ⁹ 6s ¹	Au 5d ¹⁰ 6s ¹	Hg 5d ¹⁰ 6s ²	Tl 6s ² 6p ¹	Pb 6s ² 6p ²	Bi 6s ² 6p ³	Po 6s ² 6p ⁴	At 6s ² 6p ⁵	Rn 6s ² 6p ⁶
7	87	88	89 †	104	105	106	107	108	109	110	111	112						
	Fr [Ra]7s ¹	Ra [Ra]7s ²	Ac 6d ¹ 7s ²	Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub						



Figure 5.0.2: The standard periodic Table (From: http://en.wikipedia.org/wiki/Periodic_table, slightly modified)





5.4.2 Sizes of Atoms and Ions

Since electrons can only be located by probability, there is no real boundary to an atom. Radii of atoms and ions are obtained experimentally from measuring bond lengths in molecules, metals and ionic crystals by X-ray diffraction.

Covalent radii r_{cov}

Defined as half the distance between the nuclei of two atoms of the same element joined by a covalent bond. E.g., Cl_2 bond distance = 198 pm, atomic radius of Cl = $(198)/2 = 99$ pm.

Van der Waals radius r_{vdw}

This is defined as half the distance between the nuclei of two non-bonded atoms or molecules. It is obtained from the closest approach between two non-bonded atoms or molecules in the solid state. E.g., van der Waals radius of Cl = 180 pm. In general, $r_{vdw} > r_{cov}$.

Ionic radius

It is defined as the distance between the nuclei of adjacent cations and anions in a purely ionic lattice. In general, $r_{anions} > r_{cov} > r_{cation}$ for any given element. For example ionic radius of sodium ion, $r_{\text{Na}^+} = 116$ pm while covalent radius of sodium atom, $r_{\text{Na}} = 154$ pm.

Explanation of the difference

From Na ($1s^2 2s^2 2p^6 3s^1$) to Na^+ ($1s^2 2s^2 2p^6$) Z_{eff} increases since the number of electrons decreases.

The covalent radius of chlorine, $r_{\text{Cl}} = 97$ pm while the ionic radius of the chloride ion $r_{\text{Cl}^-} = 167$ pm;

Explanation of the difference

From Cl ($1s^2 2s^2 2p^6 3s^2 3p^5$) to Cl^- ($1s^2 2s^2 2p^6 3s^2 3p^6$): Z_{eff} decreases since no. of electrons increases.



5.4.3 Periodic Trends in Atomic Radii

Across a row, the atomic radius decreases because the n value for the outer shell remains the same but Z_{eff} is increasing. *Example*

For the row started by Na the outer shell is $3s^1, 3s^2, 3s^23p^1, 3s^23p^2, \dots, 3s^23p^6$. The atomic radius decreases (from Na to Ar) because the Z value is increasing and the added electrons do not completely shield the outer electrons from the increased Z .

Down a group, the increase in Z_{eff} does not completely counteract the fact that the outer electrons are in orbitals with higher n , so radius increase down a group. There is an increase in the number of occupied orbitals.

□ E.g. For group headed by Be outer shell is $2s^2, 3s^2, 4s^2, 5s^2, 6s^2, 7s^2$, atomic size increases from Be to Ra.

Table 0.1: Covalent radii (pm) of some selected elements

Li	Be	B	C	N	O
132	89	82	77	77	73
Na	Mg	Al	Si	P	S
154	136	118	111	106	102
K	Ca	Ga	Ge	As	Se
203	174	126	122	120	117
Rb	Sr	In	Sn	Sb	Te
216	191	144	140	140	136

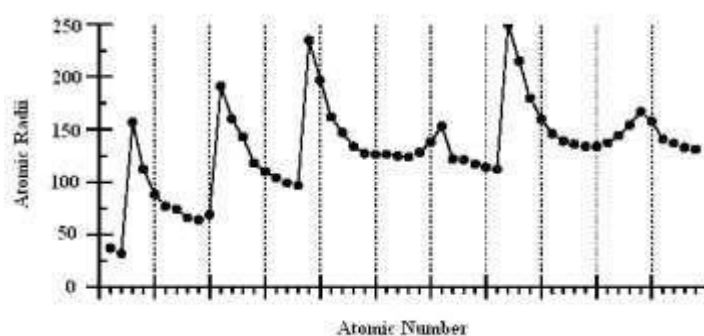


Figure 5.0.3: Variation of atomic radii with atomic number (across the periods)



5.4.4 Periodic Trends in Ionic Radii

Down a group in cations and anions- sizes increase, e.g.

- $\text{Mg}^{2+}(2s^22p^6) > \text{Ca}^{2+}(3s^23p^6) > \text{Sr}^{2+}(4s^24p^6) > \text{Ba}^{2+}(5s^25p^6)$
- $\text{F}^{-}(2s^22p^6) > \text{Cl}^{-}(3s^23p^6) > \text{Br}^{-}(4s^24p^6) > \text{I}^{-}(5s^25p^6)$
- n value of outer orbital increase but is not counteracted by the increasing Z_{eff} .

Size of isoelectronic series (ions/atoms containing the same number of electrons):

- $O^{2-}(2s^2 2p^6) > F^-(2s^2 2p^6) > Ne(2s^2 2p^6) > Na^+(2s^2 2p^6) > Mg^{2+}(2s^2 2p^6)$
- Z_{eff} increasing > Since same no. of electrons and Z increasing.
- $r(O^{2-}) > r(F^-) > r(Ne) > r(Na^+) > r(Mg^{2+})$

Table 0.2: Ionic radii (pm) of some selected main group ions

Li⁺	Be²⁺			O²⁻	F⁻
60	31			140	136
Na⁺	Mg²⁺	Al³⁺		S²⁻	Cl⁻
95	65	50		184	181
K⁺	Ca²⁺	Ga³⁺		Se²⁻	Br⁻
133	99	62		198	
Rb⁺	Sr²⁺	In³⁺	Sn⁴⁺	Te²⁻	I⁻
248	113	81	71	221	216



5.4.5 Periodic Trends in the First Ionization Energy

The ionization energy of an atom is the minimum energy required to remove an electron from a gaseous atom in its ground state. The magnitude of ionization energy is a measure of the effort required to force an atom to give up an electron, the higher the ionization energy, the more difficult it is to remove the electron.

For many-electron atom, the amount of energy required to remove the first electron from the atom in its ground state is called the **first ionization energy (I₁)**. The second ionization energy (I₂) and the third ionization energy (I₃) are shown in the following equations:



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Higher ionization potentials labeled I², I³, I⁴ and so on will correspond to the successive removal of additional electrons. The table below gives the first ionization energies of the first 20 elements.

Table 0.3: Ionization Potential (Energy) for the First 20 Elements

Element	1st	2nd	3 rd	4th	5th	6th	Element	1st	2nd	3rd	4th	5th	6th
¹ H	1312						¹¹ Na	495.9	4560	6900	9540	23400	16600
² He	2373	5248					¹² Mg	738.1	1450	7730	10500	13600	18000
³ Li	520	7300	11808				¹³ Al	577.9	1820	2750	11600	14800	18400
⁴ Be	899	1757	14850	20992			¹⁴ Si	786.3	1580	3230	4360	16000	20000
⁵ B	801	2430	3660	25000	32800		¹⁵ P	1012	1904	2910	4960	6240	21000
⁶ C	1086	2350	4620	6220	38000	47232	¹⁶ S	999.5	2250	3360	3660	6990	8500
⁷ N	1400	2860	4580	7500	9400	53000	¹⁷ Cl	1251	2297	3820	5160	6540	9300
⁸ O	1314	3390	5300	7470	11000	13000	¹⁸ Ar	1521	2666	3900	5770	7240	8800
⁹ F	1680	3370	6050	8400	11000	15200	¹⁹ K	418.7	3052	4410	5900	8000	9600
¹⁰ Ne	2080	3950	6120	9370	12200	15000	²⁰ Ca	589.5	1145	4900	6500	8100	11000

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Ionization energies are positive quantities. If the ionization energy was plotted against the atomic number, the periodicity will be clearly evident. The 1st Ionization energy generally increases across the period. Group I elements (alkali metals) have the lowest ionization energies. This is because the elements have one valence electron that is effectively shielded by the completely filled inner shells. Consequently, it is energetically easy to remove an electron from the atom of an alkali metal to form unipositive ion (Li^+ , Na^+ , K^+ ...).

On the other hand non-metals have much higher ionization energies. The ionization energies of the metalloids usually fall between those of metals and non-metals. This explains why metals readily form cations while non-metals form anions. However, there are irregularities in the increase of the ionization energy across the periods. For example going across the period from group 2 to 3, thus from Be to B and from Mg to Al. The group 3 elements have a single electron in the outermost sub level ($ns^2 np^1$), which is well shielded by the inner electrons and the ns^2 electrons. Less energy is therefore needed to remove a paired s electron from the same principle energy level. This brings about the lower ionization energies in group 3 elements compared with those in group 2 in the same period.

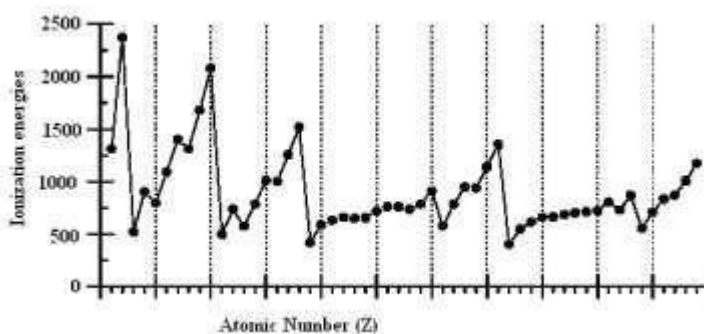


Figure 5.0.4: Ionization energies of the first 55 elements

The following observations are clear from the figure above:

- Ionization energy increases across each row, e.g. from Li to Ne; Na to Ar; K to Kr and so on (because Z_{eff} increases and the electrons in a given orbital are held more tightly, therefore, to remove the electron from the atom requires more energy).
- Group 1 (Alkali) metals (i.e. Li, Na, K, Rb, Cs and Fr) have minimum I.E. This is because they have only a single loosely held electron in the valence shell ns^1 .
- Group 18 (noble gases) (i.e. He, Ne, Ar, Kr, Xe and Rn) have the highest I.E. This is because they have a filled valence shell, $ns^2 np^6$ and high effective nuclear charge, Z_{eff} . Thus valence electrons are tightly held.
- Minor irregularities can be seen; for example, e.g. I.E. of Be = 899.4 kJ/mol while I.E. of B = 800.6 kJ/mol. This can be attributed to there being a completely filled ns^2 configuration ($2s^2$) in Be while in B the arrangement is $2s^2 2p^1$ in B. The unpaired

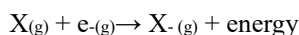
electron in B is easily lost leading to lower I.E. ○ I.E of N = 1402.3 kJ/mol while the I.E. of O = 1313.9 kJ/mol. This attributed to the stability of the half filled p orbital in N ($2s^2 2p^3$). In O the valence electron arrangement is $2s^2 2p^4$ in which one p electron is paired and hence easily lost to reduce electron-electron repulsion.

- I.E decreases down a group, e.g.
Li > Na > K > Rb > Cs > Fr. Similarly
He > Ne > Ar > Kr > Xe > Rn. Due to electrons filling in larger shell (larger n), further away from the nucleus. Therefore, valence electrons are well shielded from the nucleus by inner shell electrons and are loosely held.
- The ionization energy increases only slightly across a row in the transition series (including lanthanides and actinides series).



5.4.6 Periodic Trend in Electron Affinity

The electron affinity of an atom is defined as the energy change obtained when a neutral atom in the gaseous state captures an electron. Thus, the energy is released by the reaction represented below



It is the reverse of the 1st ionization potential and may be looked at as the ability of an atom to accept one or more electrons. The largest electron affinities are those of the halogens. This is as expected since the addition of one electron yields the stable octet configuration of the 18th group elements—the noble gases.

We assign a negative value to the affinity when energy is released. The more negative the electron affinity, the greater the tendency of the atom to accept an electron. The tendency to accept electrons increases as we move from left to right across the periodic table. Thus Electron Affinity becomes more negative. The E.A of metals are generally more positive (or less negative) than those of non-metals. The values differ little within a group, but the halogens have the most negative E.A values, while the noble gases that have filled outer s and p sub shells have no tendency to accept electrons. The E.A of oxygen has a negative value, which means that the process $O_{(g)} + e^{-} \rightarrow O^{-}_{(g)}$ is favourable, while that of the O²⁻ i.e. $O^{-}_{(g)} + e^{-} \rightarrow O^{2-}_{(g)}$ is positive (780 kJ/mol) meaning that this process is not favourable in the gas phase.

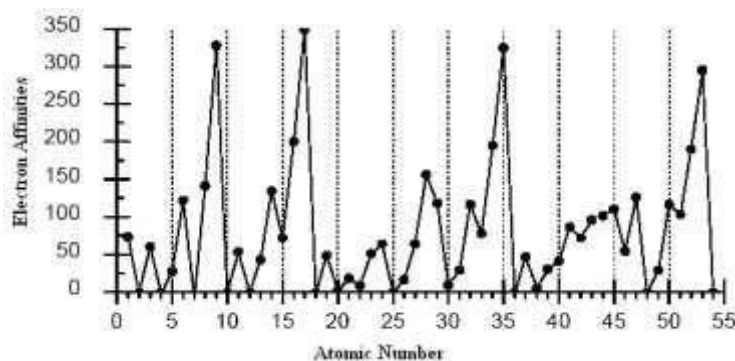


Figure 5.0.5: Variation of electron affinity with atomic number



5.4.7 Periodic Trends in Electronegativity, χ

The tendency of an atom to attract electrons to itself in a chemical bond is referred to as **electronegativity**. The greater the electronegativity of an atom, the more strongly the atom attracts the electrons of a bond. This concept was proposed by Linus Pauling in 1937. Electronegativity cannot be calculated accurately or measured directly. However, we expect it to depend on the magnitude of the charge and on the distance of the bonding pair of electrons from the nucleus. There are two important trends in electronegativity within the periodic table: -

- electronegativity increases across a period as the charge increases
- electronegativity generally decreases from top to bottom in a group; because with each successive shell, the bonding electrons are further from the nucleus.

Because the electronegativity of an atom cannot be defined quantitatively, it cannot be given a precise value but approximate values. Table shows the electronegativity values

Table 0.4: Electronegativity χ of some elements

Group →	1	2		13	14	15	16	17	18
Period ↓									
1	H								He
2	Li	Be	B	C	N	O	F		Ne
		1.0	1.5	2.0	2.5	3.1	3.5	4.1	-
3	Na	Mg	Al	Si	P	S	Cl		Ar
		1.0	1.2	1.3	1.7	2.1	2.4	2.8	-
4	K	Ca	Ga	Ge	As	Se		Br	Kr
		0.9	1.0	1.8	2.0	2.2	2.5	2.7	3.1
5	Rb	Sr	In	Sn	Sb	Te		I	Xe
		0.9	1.0	1.5	1.7	1.8	2.0	2.2	

Electronegativity may be used to estimate bond polarity. e.g.

2.4

- Atoms with similar electronegativity (i.e. $\Delta\chi \leq 0.4$) form non-polar bonds.

- Atoms whose electronegativities differ by more than 2 form ionic bonds.
- Atoms whose electronegativities differ by less than 2 form covalent bonds.



5.5 Common Oxidation States of the Elements

Oxidation number is the apparent charge assigned to an atom in a molecule or in a compound. Some oxidation numbers are fixed:

Elements	Oxidation number
O	-2 except in peroxide
H	+1 except in hydride
Group 1 metals	+1
Group 2 metals	+2
Halogens	-1 except in oxygen compounds



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6 LECTURE

ELECTRONIC STRUCTURE AND CHEMICAL BONDING



1.1 Introduction

In this lecture you will learn simple representation of chemical bonding. You will start with Lewis structure, Valence Shell Electron Pair Repulsion Theory (VSEPR) to explain the geometry of simple molecules. Finally you will learn Valence Bond Theory (VBT) and hybridization of atomic orbitals.



Lecture 6 Objectives

At the end of this lecture, you should be able to:

- i) Explain Lewis dot structure.
- ii) Describe and use Valence Shell Electron Pair Repulsion theory (VSEPR).
- iii) Describe Valence bond theory (VBT) and hybridization of atomic orbital.
- iv) Use the various bonding model to explain bonding



6.1 Bonding

Some important questions running in the mind of Linus Pauling while doing undergraduate studies which later worked on leading to description of the nature of the chemical bond:

- What forces held atoms together to form molecules?
- How did those forces give the molecules particular shapes and qualities?

We can also ponder the following in this section include:

- What is a bond?
- How many types of bonds can atoms form?
- Why do atoms form bonds?
- How do atoms bond or how do atoms combine to form molecules?

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6.1.1 A bond and bond types

A **bond** may be described as a linkage between a particular pair of atoms e.g. H and Cl in HCl, N and H in NH₃ etc. A bond may be described as being strong, weak, long, short, polar or non polar, single, double, triple and even quadruple.

The three major types of bonds we shall meet in this study include *ionic bonds*, *covalent bonds* and *metallic bonds*. The *dative* or *coordinate bond* is a special type of covalent bond we shall

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also discuss later in the sections below. These are in most cases two-centre-two electron bonds. A fifth type of bond which is of great importance is hydrogen bond. It will soon become clear that few bonds are purely ionic, covalent or metallic. Most are intermediates between the three main types and show some properties of at least two and sometimes of all three types.



6.1.2 Why atoms combine

It may safely be said that atoms form molecules as a way of attaining chemical stability. A molecule (combination of two or more atoms) only forms if it is chemically more stable and has a lower energy than individual atoms. The stability manifested as a lack of reactivity, is because the atoms already have a low energy which may not be lowered further by forming compounds.

From the electronic point of view, the most stable electronic arrangement is a noble gas electronic structure and most molecules have this arrangement. Thus, because it is only electrons in the outermost shell of an atom that are involved in forming bonds, each atom acquires a stable electron configuration by forming one or more bonds.



6.2 Theories of Bonding

There are four different but related approaches to explain how atoms combine to form covalent bonds

- Lewis Theory
- Valence Shell Electron Pair Repulsion theory (VSEPR)
- Valence bond theory (VBT) and hybridization of atomic orbitals
- Molecular Orbital (MO) theory.

In this lesson we will describe the interactions between atoms called chemical bonds. Most of our discussion will centre on the simplest method of representing chemical bonding, known as the Lewis theory. We will explore, however, another relatively simple theory, one for prediction of probable shapes – Valence – Shell Electron – Pair Repulsion (VSEPR) theory.



6.2.1 Lewis Theory

In the early 1920s, Lewis (left) published an idea about the bonds between atoms that he had developed with General Electric researcher **Irving Langmuir**. They theorized that an element's valence arose naturally from its atomic structure. Atoms, it was known, consisted of a positively charged nucleus surrounded by negatively charged electrons. Lewis and Langmuir hypothesized that atoms were most stable when the electrons orbited the nucleus in shells containing eight at a time (except for the atom's innermost shell, which contained two electrons).

According to the Lewis and Langmuir model, if an atom had seven electrons in an outer shell, it would tend toward collecting an eighth for maximum stability. One way to do this was to combine with another atom that had one extra electron in its outer shell. The two atoms would "share" an electron, creating a more stable product. The resulting "shared electron bond" tied the atoms together into a molecule.

(Source: <http://osulibrary.oregonstate.edu/specialcollections/coll/pauling/bond/narrative/page5.html>) Some

fundamental ideas in Lewis' theory are:



- Electrons, especially those of the outermost (valence) electronic shell, play a fundamental role in chemical bonding.
- In some cases electrons are *transferred* from one atom to another. Positive and negative ions are formed and attract each other through electrostatic forces called **ionic bonds**.
- In other cases one or more pairs of electrons are shared between atoms; this *sharing* of electrons is called a **covalent bond**.
- Electrons are transferred, or shared, in such a way that each atom acquires an especially stable electron configuration. Usually this is a noble gas configuration, one with eight outer shell electrons, or an **octet**. This octet rule is broken in a significant number of cases:
 - In atoms such as Be or B which have less than four outer electrons. Even if all the outer electrons are used to form bonds an octet cannot be attained.
 - Where atoms have an extra energy level which is close in energy to the *p* level, which may accept electrons and be used in bonding. PF₃ obey the octet rule but PF₅ does not because the phosphorous uses one 3*s*, three 3*p* and one 3*d* orbitals. These violations become increasingly common in elements after the first two periods of eight elements in the periodic table.
 - The octet rule does not work in molecules with an odd number of electrons such as NO and ClO₂.



6.2.2 Lewis Symbols and Lewis Structures

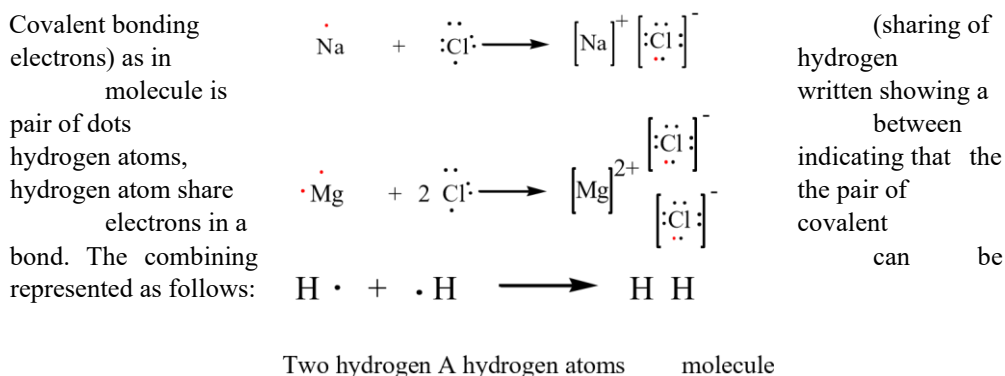
A **Lewis symbol** consists of a chemical symbol to represent the nucleus and *core* (inner shell) electrons of an atom, together with dots placed around the symbol to represent the *valence* (outer shell) electrons.

Examples

Lewis structure for: Na electron configuration [Ne]3s¹ is $\text{Na}^\bullet$

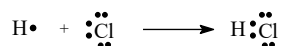
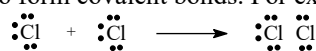
O electron configuration [He]2s²2p⁴ is $\text{O}^{\bullet\bullet}$

A **Lewis structure** is a combination of Lewis symbols that represent either the transfer or sharing of electrons in a chemical bond. For example, ionic bonding (transfer of electrons) can be represented as follows:



The shared pair is usually represented by a line e.g. H H

Atoms other than hydrogen also form covalent bonds. For example,



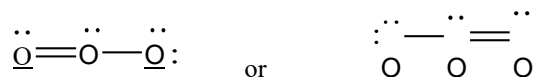
Exercise

- 1 Write Lewis structures of the following ionic compounds: a) BaO; b) MgCl₂; c) Aluminium oxide
- 2 Write Lewis structures of the following covalent compounds:
a) F₂; b) H₂O; c) NH₃ d) CH₄ e) O₂ f) N₂



6.3 Lewis Structures and Resonance

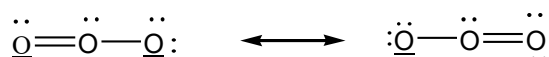
The Lewis structure of ozone O₃ are:



Which of the two Lewis structure for O₃ is correct?

In fact, neither is correct by itself. Whenever it is possible to write more than one Lewis structure for a molecule the actual electronic structure is an average of the various possibilities called a **resonance hybrid**. Ozone does not have one O=O double bond and one O-O single bond as individual structures imply, rather ozone has two equivalent O-O bonds that one can think of as having a *bond order* of 1.5- midway between pure single bond and pure double bond. Both have an identical length of 1.28 Å.

The idea of resonance between two or more Lewis structures is indicated by drawing the individual Lewis structure and using double headed “resonance arrow” to show that both contribute to the resonance hybrid.



Note: Lewis structures say nothing about bond angles or shapes of molecules in general.



Study question

Draw the resonance structure of carbonate ion, CO₃²⁻.

6.4 Valence Shell Electron Pair Repulsion Theory (VSEPR)

The main ideas of this theory put forward by Gillespie and Nyholm in 1957 are:

- The shape of the molecule is determined by repulsions between all of the electron pairs present in the valence shell. The order of repulsive energy is: Lone pair-Lone pair > Lone pair-Bond pair > Bond pair-Bond pair
- The magnitude of repulsions between bonding pairs of electrons depends on the electronegativity difference between the central atom and the other atoms.
- Double bonds cause more repulsion than single bonds, and triple bonds cause more repulsion than a double bond.

Thus, according to the VSEPR theory, bonding pairs and lone pairs take up positions around an atom that maximize their separation in order to reduce repulsions between them. The shape of the molecule is determined by the locations of the atoms attached to the central atom. Electron pairs in multiple bonds are treated as a single unit equivalent to one region of high electron concentration. Lone pairs on the central atom contribute to the shape of the molecule but are ignored when we name the shape. The molecule adjusts its shape to reduce lone pair-lone pair and lone pair – bonding pair repulsion.



6.5 Valence Bond Theory (VBT)

This was proposed by Heitler and London 1927 and developed further by Linus Pauling (See left portrait of Linus Pauling, 1930.) and widely between 1940 and 1960. It has over the time become obsolete although it is still in much use by organic chemists and provides a basis for simple description of small inorganic molecules. It was developed to use the methods of quantum mechanics to explain chemical bonding. It focuses on how the atomic orbitals of the individual atoms combine on molecular formation to give individual chemical bonds. It uses three principles to explain bonding:

- Overlap of atomic orbitals and
- Hybridization of atomic orbitals
- Resonance of molecular structures

In his 1931 paper (The Nature of the chemical bond), building on the works of Lewis, and the valence bond theory (VB) of Heitler and London, and his own earlier work, he presented six rules for the shared electron bond, the first three of which were already generally known from the work of Heitler and London:

- The electron-pair bond forms through the interaction of an unpaired electron on each of two atoms.
- The spins of the electrons have to be opposed.
- Once paired, the two electrons cannot take part in additional bonds.

His last three rules were new:

- The electron-exchange terms for the bond involves only one wavefunction from each atom.
- The available electrons in the lowest energy level form the strongest bonds.
- Of two orbitals in an atom, the one that can overlap the most with an orbital from another atom will form the strongest bond, and this bond will tend to lie in the direction of the concentrated orbital.

In other words

- Covalent bond occurs when atoms share electrons.
- The build up of electron density between two nuclei occurs when a valence atomic orbital of one atom overlaps with an atomic orbital of another.
- The orbitals share a region of space i.e. they overlap.
- The overlap of orbitals allows two electrons of opposite spin to share the common space between the nuclei forming a covalent bond.



In the hydrogen molecule for instance, the H-H bond results from overlap of two singly occupied H 1s orbitals.

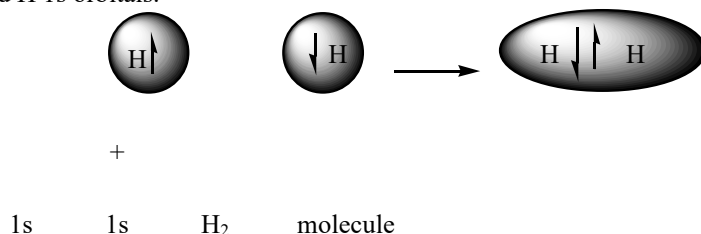


Figure 6.0.1: Formation of the H-H bond

In the valence bond model, the strength of covalent bond depends on amount of overlap. The greater the overlap, the stronger the bond. This in turn, means that bond formed by overlap other than s- orbitals have directionalities to them.

In the fluorine molecule, F_2 , for instance each atom has electron configuration: $[He]2s^2 2p_z^2 2p_y^2 2p_x^1$. This means that the fluorine bond results from overlap of two singly occupied two 2p orbitals. The 2p orbitals must point directly to one another for optimum overlap to occur, and the F-F form bond along the orbitals.

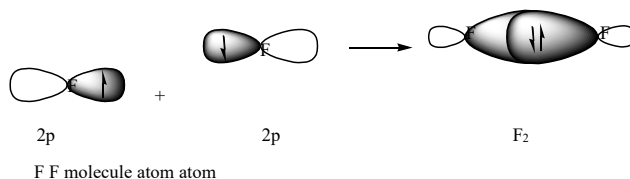


Figure 6.0.2: Formation of the F-F bond

In HCl, the covalent bond involves overlap of 1s orbital (non-directional) with chlorine 3p orbital and formed along *p*-axis.

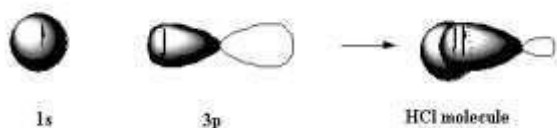


Figure 6.0.3: Overlap of atomic orbitals in the formation of HCl

The key ideas of valence bond theory can be summarized by few statements:

- 1 Covalent bonds are formed by overlap of atomic orbitals, each of which contains 1 electron of opposite spin.
- 2 Each of the bonded atoms maintains its own atomic orbitals, but the electron pair in the overlapping pair is shared by both atoms.

- 3 The greater the amount of orbital overlap the stronger the bond. This leads to a directional character to the bond when other than s – orbitals are formed.



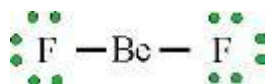
6.5.1 Molecular Geometry and Hybrid Orbitals

Hybridization (Hybrid Orbitals)

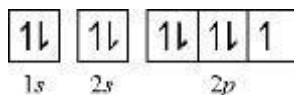
For polyatomic molecules we would like to be able to explain the number of bonds formed and the geometries of the molecules. The number of bonds formed by an atom is usually the same as the number of unpaired electrons in the ground state. However in some cases the atom may form more bonds than this. This occurs by excitation of the atom when electrons which were paired in the ground state are unpaired and promoted into suitable empty orbitals. This increases the number of unpaired electrons available for bonding.

sp Hybrid Orbitals

Consider the Lewis structure of gaseous molecules of BeF_2 :

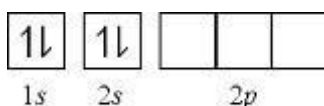


- The VSEPR model predicts this structure will be linear
- What would **valence bond theory** predict about the structure/geometry? The fluorine atom electron configuration is $1s^2 2s^2 2p^5$

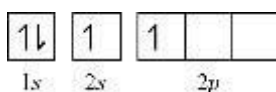


- There is an unpaired electron in a $2p$ orbital
- This unpaired $2p$ electron can be paired with an unpaired electron in the Be atom to form a covalent bond

The Be atom electron configuration is $1s^2 2s^2$



- In the ground state, there are no unpaired electrons (the Be atom is incapable of forming a covalent bond with a fluorine atom)
- However, the Be atom could obtain an unpaired electron by promoting an electron from the $2s$ orbital to the $2p$ orbital:

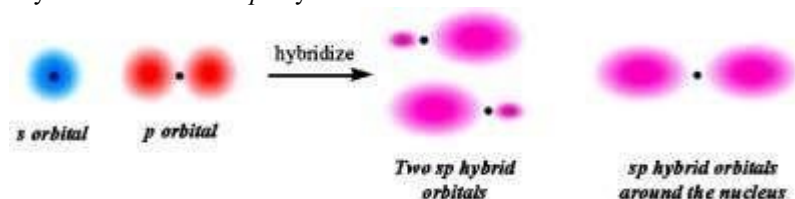


This would actually result in two unpaired electrons, one in a $2s$ orbital and another in a $2p$ orbital

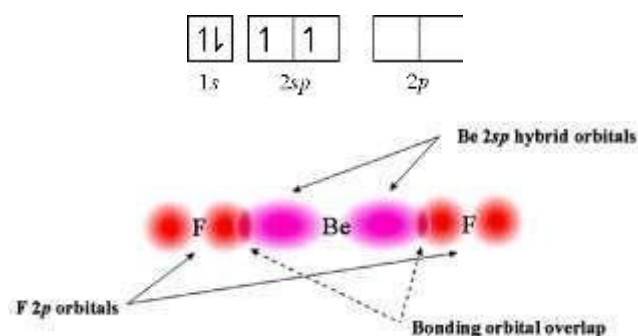
- The Be atom can now form two covalent bonds with fluorine atoms
- We wouldn't expect these bonds to be identical (one is with a $2s$ electron orbital, the other is with a $2p$ electron orbital)

However, the structure of BeF_2 is linear and the bond lengths are identical. How can this be explained?

- We can combine wavefunctions for the $2s$ and $2p$ electrons to produce a "hybrid" orbital for both electrons
- This hybrid orbital is an " sp " hybrid orbital



The orbital diagram for this hybridization would be represented as:



Note:

- The $\text{Be } 2sp$ orbitals are identical and oriented 180° from one another (i.e. bond lengths will be identical and the molecule linear)
- The promotion of a $\text{Be } 2s$ electron to a $2p$ orbital to allow sp hybrid orbital formation **requires energy**.
 - The elongated sp hybrid orbitals have one large lobe which can overlap (bond) with another atom more effectively
 - This produces a stronger bond (higher bond energy) which offsets the energy required to promote the $2s$ electron

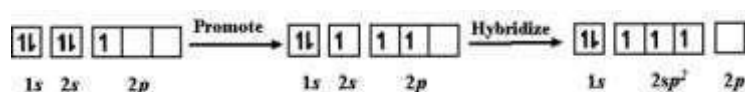
sp^2 and sp^3 Hybrid Orbitals

Whenever orbitals are mixed (hybridized):

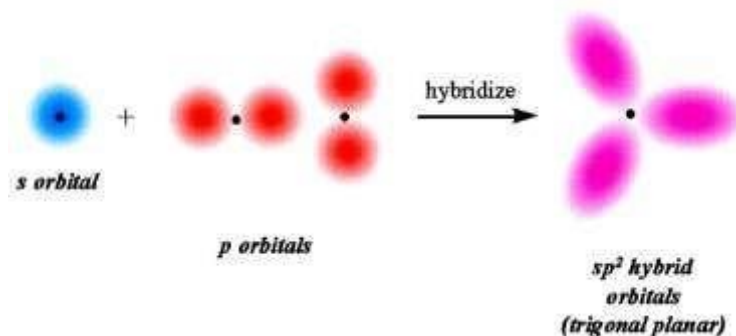
- The number of hybrid orbitals produced is equal to the sum of the orbitals being hybridized
- Each hybrid orbital is identical except that they are oriented in different directions

BF_3

Hybridization of Boron

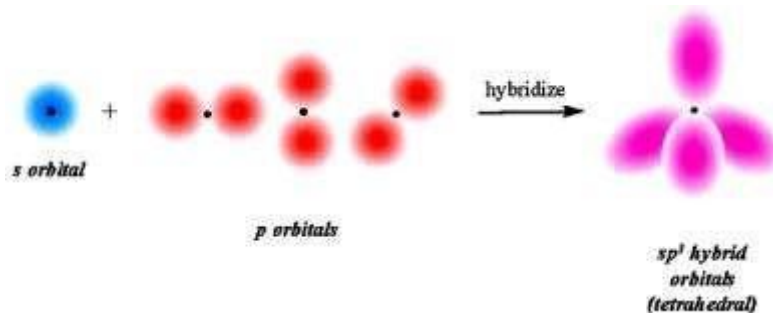
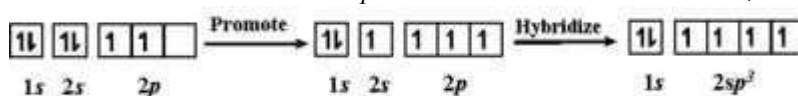


- The three sp^2 hybrid orbitals have a trigonal planar arrangement to minimize electron repulsion



NOTE: sp^2 refers to a hybrid orbital being constructed from ones orbital and twop orbitals. Although it looks like an electron configuration notation, the superscript '2' DOES NOT refer to the number of electrons in an orbital.

- An *s* orbital can also mix with all 3 *p* orbitals in the same sub-shell CH_4



Thus, using **valence bond theory**, we would describe the bonds in methane as follows: each of the carbon sp^3 hybrid orbitals can overlap with the 1s orbitals of a hydrogen atom to form a bonding pair of electrons

NOTE:

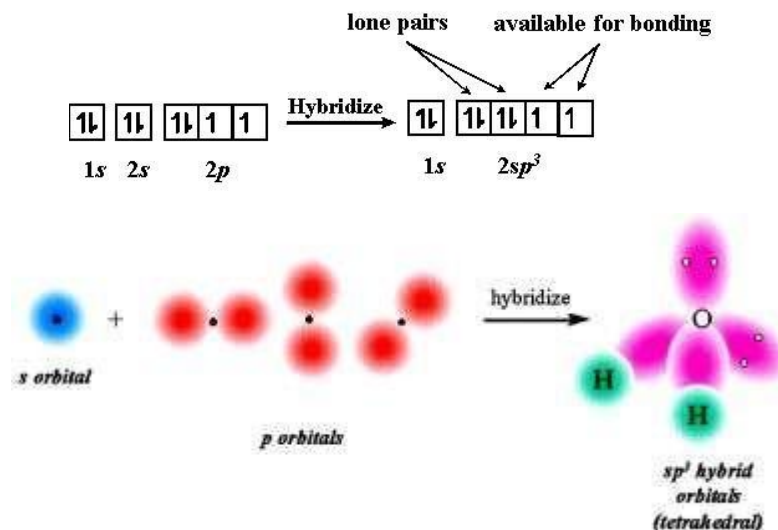
1. sp^3 refers to a hybrid orbital being constructed from ones orbital and threep orbitals. Although it looks like an electron configuration notation, the superscript '3' DOES NOT refer to the number of electrons in an orbital.

2. The two steps often observed when constructing hybrid orbitals is to

- promote a valence electron from the ground state configuration to a higher energy orbital, and then
- hybridize the appropriate valence electron orbitals to achieve the desired valence electron geometry (i.e. the correct number of hybrid orbitals for the appropriate valence electron geometry)

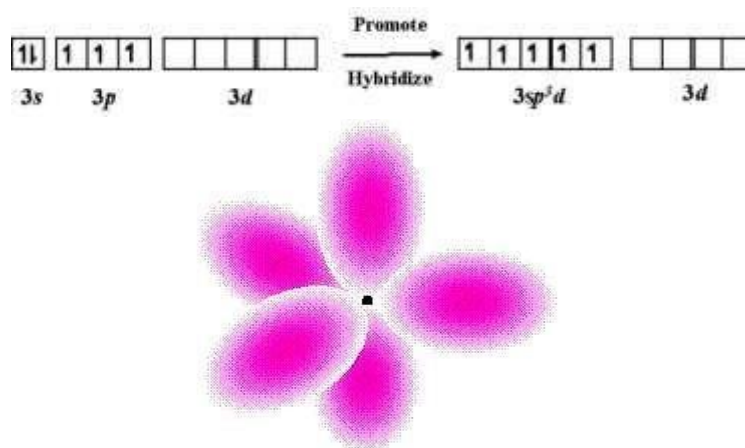
H_2O

Hybridization of Oxygen



6.6 Hybridization Involving d Orbitals

Atoms in the third period and higher can utilize d orbitals to form hybrid orbitals PF_5



Trigonal Bipyramidal

Similarly hybridizing one s , three p and two d orbitals yields six identical hybrid sp^3d^2 orbitals. These would be oriented in an *octahedral geometry*.

- Hybrid orbitals allows us to use **valence bond theory** to describe covalent bonds (sharing of electrons in overlapping orbitals of two atoms)
- When we know the molecular geometry, we can use the concept of hybridization to describe the electronic orbitals used by the central atom in bonding

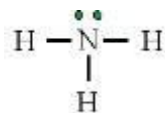
Steps in predicting the hybrid orbitals used by an atom in bonding:

- Draw the Lewis structure
- Determine the electron pair geometry using the VSEPR model
- Specify the hybrid orbitals needed to accommodate the electron pairs in the geometric arrangement



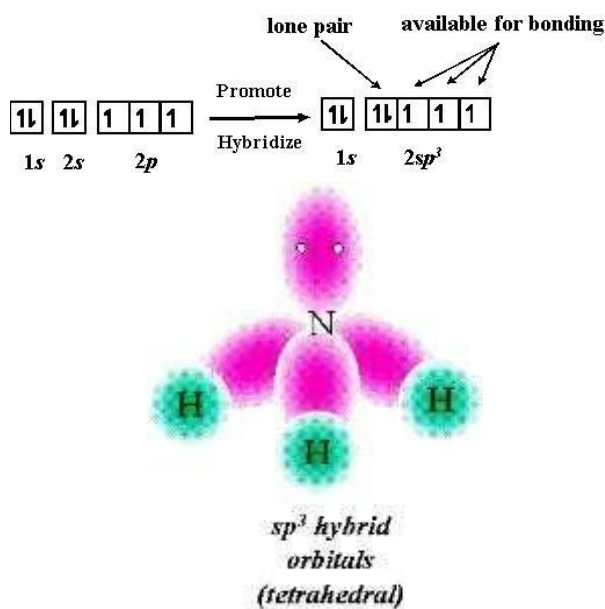
Ammonia, NH_3

1. Lewis structure



2. VSEPR indicates *tetrahedral geometry* with one non-bonding pair of electrons (structure itself will be *trigonal pyramidal*)

3. Tetrahedral arrangement indicates four equivalent electron orbitals



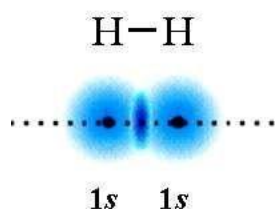
Valence Electron Pair Geometry	Number of Orbitals	Hybrid Orbitals
Linear	2	Sp
Trigonal Planar	3	Sp^2
Tetrahedral	4	Sp^3
TrigonalBipyramidal	5	sp^3d
Octahedral	6	sp^3d^2



6.7 Multiple Bonds and Orbital Overlaps

Two types of bonds: sigma (σ) and pi (π) bonds

The "internuclear axis" is the imaginary axis that passes through the two nuclei in a bond:

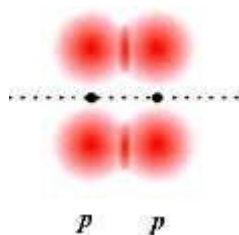


The covalent bonds we have been considering so far exhibit bonding orbitals which are *symmetrical about the internuclear axis* (either an *s* orbital - which is symmetric in all directions, or a *p* orbital that is pointing along the bond towards the other atom, or a hybrid orbital that is pointing along the axis towards the other atom)

Bonds in which the electron density is symmetrical about the internuclear axis are termed "sigma" or " σ " bonds

In multiple bonds, the bonding orbitals arise from a different type arrangement:

- Multiple bonds involve the overlap between two *p* orbitals
- These *p* orbitals are oriented perpendicular to the internuclear (bond) axis



*This type of overlap of two *p* orbitals is called a "pi" or " π " bond. Note that this is a single bond (which is made up of the overlap of two *p* orbitals)*

In π bonds

- The overlapping regions of the bonding orbitals lie *above* and *below* the internuclear axis (there is no probability of finding the electron in that region)
- The size of the overlap is smaller than a σ bond, and thus the bond strength is typically less than that of a σ bond

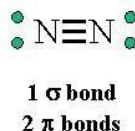
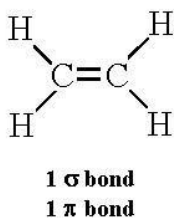
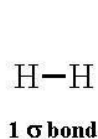
In Generally,

-



A single bond is always a σ bond.

A double bond is composed of *one* σ bond and *one* π bond. π composed of *one* σ bond and *two* π bonds.



π

A triple bond is

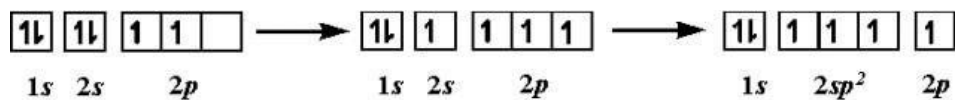
C_2H_4 (ethylene; see structure above)

The arrangement of π bonds suggests that the geometry of the bonds around each carbon is **trigonal planar**

Trigonal planar suggests sp^2 hybrid orbitals are being used (these would be σ bonds)

What about the electron configuration?

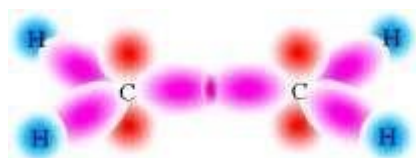
Carbon: $1s^2 2s^2 2p^2$



•

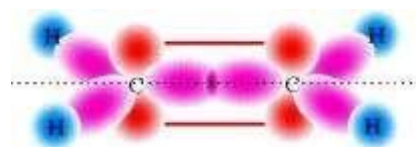
•

Thus, we have an extra unpaired electron in a p orbital available for bonding. This extra p electron orbital is oriented perpendicular to the plane of the three sp orbitals (to minimize repulsion):



•

The unpaired electrons in the p orbitals can overlap one another above and below the internuclear axis to form a covalent bond.



between the two p orbitals

□

the atoms from rotating around the C-C bond.

bond.

□

- This interaction above and below the internuclear axis represents the π bond

Experimentally,

- We know that the 6 atoms of ethylene lie in the same plane.
- If there was a single σ bond between the two carbons, there would be nothing stopping
- But, the atoms are held rigid in a planar orientation.
- This orientation allows the overlap of the two p orbitals, with formation of a π bond.
- In addition to this rigidity, the C-C bond length is shorter than that expected for a single
- Thus, extra electrons (from the π bond) must be situated between the two C-C nuclei.

The linear bond arrangement suggests that the carbon atoms are utilizing sp hybrid orbitals for bonding

- This leaves two unpaired electrons in p orbitals
- To minimize electron repulsion, these p orbitals are at right angles to each other, and



- These p orbitals can overlap to form two π bonds in addition to the single σ bond

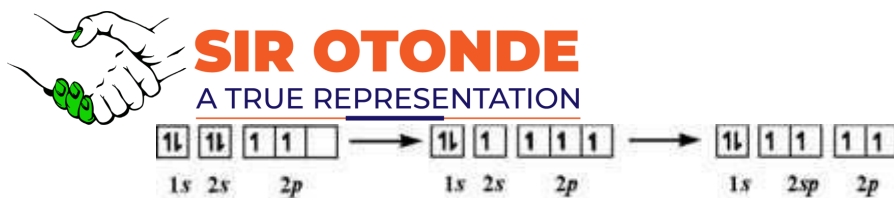
6.7.1 Delocalized Bonding

Localized electrons are electrons which are associated completely with the atoms forming the bond in question

In some molecules, particularly with resonance structures, we cannot associate bonding electrons with specific atoms

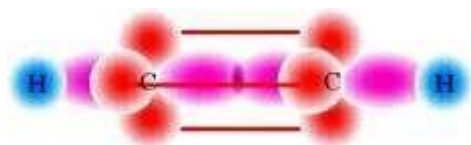
Bonding in acetylene(C_2H_2)



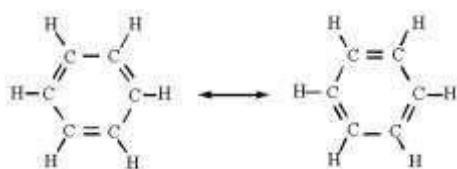


to the internuclear axis:

(forming a triple bond)



C_6H_6 (Benzene)

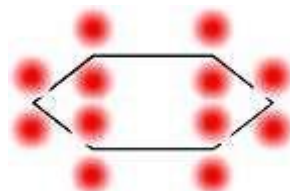


Benzene has two resonance forms

- The six carbon - carbon bonds are of equal length, intermediate between a single bond and double bond
- The molecule is planar
- The bond angle around each carbon is approximately 120°

The apparent hybridization orbital consistent with the geometry would be sp^2 (trigonal planar arrangement)

- This would leave a single p orbital associated with each carbon (perpendicular to the plane of the ring)

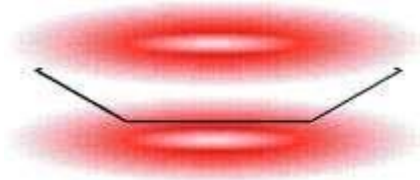


With six p electrons we could ☐ form three discrete bonds

- However, this would result in three double bonds in the ring, and three single bonds
- This would cause the bond lengths to be different around the ring (which they are not)
- This would also result in one resonance structure being the only possible structure

The best model is one in which the ☐ electrons are "smeared" around the ring, and not localized to a particular atom

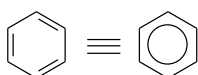
- Because we electrons in the particular atom being six carbon atoms



cannot say that the bonds are localized to a they are described as **delocalized** among the

□

Benzene is typically drawn in two different ways:

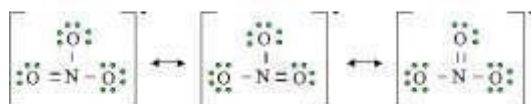


The circle in the right hand structure indicates the delocalization of the p bonds



6.8 Resonance Structures and Hybrid Orbitals

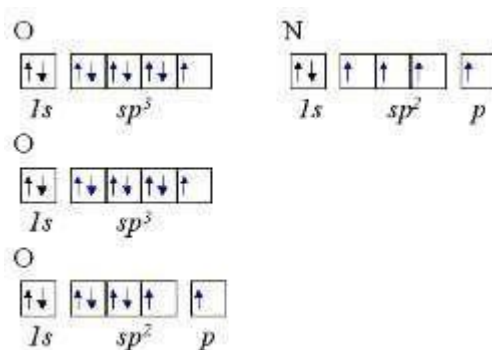
Structure of NO_3^-



The Lewis structure of NO_3^- ion suggests that three resonance structures describe the molecular structure

- For any individual Lewis resonance structure the electronic structure for the central N atom is predicted to be sp^2 hybrid orbitals participating in σ bonds with each of the O atoms, and an electron in a p orbital participating in a π bond with one oxygen (forming a double bond)
- Two of the O atoms are predicted to have sp^3 hybrid orbitals, with one orbital participating in a σ bond with the central N atom and the other orbitals filled with nonbonding electron pairs. The other O atom is predicted to have sp^2 hybrid orbitals, with one orbital participating in a σ bond with the central N and two orbitals filled with non-bonding pairs of electrons. Furthermore, this last O atom is participating in a double bond with the central N atom and therefore should have an electron in a p orbital to participate in a π bond with the central N

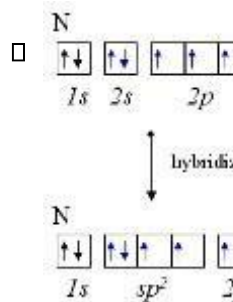
How will this arrangement look as far as the orbital diagrams?



- There are 24 valence electrons in the expected valence orbitals above
- Summing the valence electrons from the formula gives: (3×6) for O, plus 5 for N, plus 1 for ionic charge = 24

What might we expect for the electron configuration if we just started with the N atom?

□



- We would predict that the N can only make two bonds, it would have one pair of non-bonding electrons, and a p electron left over to participate in a bond with one of the bonds
- This is different from what the Lewis structure shows, and from our prediction of hybrid orbitals from the expected geometry
- If we look at the sp^3 O atoms above we see that they actually have 7 electrons (1 more than expected), while the sp^2 O atom has the expected 6. Furthermore, the N atom (in the correct sp^2 configuration) has 4 electrons (1 less than expected)
- The "extra" electron from the ionic charge is correctly accounted for in the summation of electrons

Thus, the correct way to determine electron configurations appears to be:

- begin by predicting the hybridization orbitals
- then determine lone pair arrangements and s and p bonding electrons for each atom
- confirm that all bonding electrons are correct and that the total of electrons is correct



6.9 The nature of the chemical bond

Consider a diatomic molecule such as H_2 . We shall start with 2H atoms far apart so that there is no appreciable interaction between them. Let them be labeled H_A and H_B for convenience and the electrons e_1 and e_2 respectively, just for convenience. The orbital wavefunctions for the separate atoms $H_A(e_1)$ and $H_B(e_2)$ can be written $\psi_A(1)$ and $\psi_B(2)$, respectively. The total orbital wavefunction for the separated atoms can be written as a product of the two:

$$\psi = \psi_A(1) \psi_B(2) \quad 7.1$$

Using this wave equation, the energy of the system comprising the two hydrogen atoms can be calculated as a function of the internuclear distance r_{AB} . The following graphs are obtained:

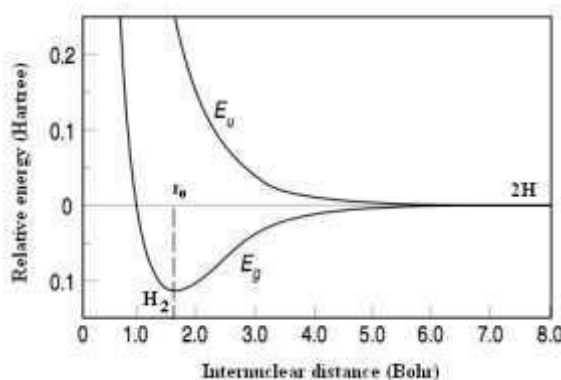


Figure 6.0.4



From the picture, the total energy of the isolated H atoms at infinite separation has been taken as zero. The curve shows how much the energy of the system of 2H atoms is below or above that of the isolated atoms. The energy value at the point of minimum on the energy curve represents the bonding energy at the equilibrium internuclear distance for the molecule described by the corresponding wavefunction.

The curve labeled E_g shows a minimum suggesting a molecule is formed. However the bonding energy is only about 6kcal/mol, which is far too small compared with the observed value of 109 kcal. Thus equation 7.1 is an oversimplification because its derivation assumed that the electrons in the two atoms are distinguishable. We know that when the two atoms are very close together so that the atomic orbitals overlap, the electrons are indistinguishable and can associate with any of the two nuclei. Thus the system of two hydrogen atoms can be separated into two different states I, $H_A(e_1)H_B(e_2)$ and II, $H_A(e_2)H_B(e_1)$.

Now let ψ_I and ψ_{II} be the wavefunctions for the states I and II. It follows from eqn 7.1 that

$$\left. \begin{aligned} \psi_I &= \psi_A(1)\psi_B(2) \\ \psi_{II} &= \psi_A(2)\psi_B(1) \end{aligned} \right\} \quad 7.2$$

The true wavefunction ψ for the system is likely to be some combination of these two. This may be written as follows:

$$\begin{aligned} \psi &= c_1\psi_I + c_2\psi_{II} \\ &= c_1\psi_A(1)\psi_B(2) + c_2\psi_A(2)\psi_B(1) \end{aligned} \quad 7.3$$

In the case of H_2 , symmetry requirements demand that the two component wavefunctions ψ_I and ψ_{II} must contribute with equal weight. Because weight is proportional to the square of the coefficients, we can write,

$$c_1^2 = c_2^2 \text{ and hence } c_1 = \pm c_2$$

Since coefficients are relative, we can put $c_1 = 1$ which means $c_2 = \pm 1$. This leads to two possible functions:

$$\psi_s = \psi_A(1)\psi_B(2) + \psi_A(2)\psi_B(1) \quad 7.4 \quad \psi_a$$

$$= \psi_A(1)\psi_B(2) - \psi_A(2)\psi_B(1) \quad 7.5$$

ψ_s remains unchanged by interchange of electrons 1 and 2. It is therefore called the symmetric combination. It represents the attractive or bonding state shown by the curve E_g . It leads to formation of the H_2 molecule with equilibrium internuclear distance $r_o = 0.8 \text{ \AA}$ at which the energy is minimum (c.f. exptal value = $0.74 \text{ \AA}$).

ψ_a changes sign upon exchange of electrons between nuclei. It is known as the asymmetric combination. It represents the repulsive or anti-bonding state shown by curve E_u . No bond is formed and the 2H stay apart.

The bonding energy corresponding to internuclear distance of $0.8 \text{ \AA} = 72 \text{ kcal/mol}$. Thus by allowing electron interchange the bonding energy has increased by about 66 kcal/mol. This is now known as exchange energy.

Other improvements that can be done on the wavefunction for the hydrogen molecule are:

- Modifying the ψ of atomic orbitals of the electrons to allow for the screening of nucleus A from nucleus B by the increased probability of the electrons being found directly between the nuclei and also to allow for the fact that the atomic orbitals will no longer be spherically symmetrical when the atoms are close together.
- Allowing for the possibility that both electrons may be simultaneously near one of the nuclei giving rise to the following two equally likely configurations:

III, $H_A(e_1)(e_2)H_B$, where all electrons are associated with nucleus A and IV, $H_A H_B(e_1)(e_2)$, where all electrons are associated with nucleus B.

The wavefunctions for the ionic forms of H_2 may therefore be represented equally well by $\psi_A(1)\psi_A(2)$ or $\psi_B(1)\psi_B(2)$. Taking into account the possibility of ionic configurations, the combined wavefunction for the H_2 molecule can be written as:

$$\psi_s = [\psi_A(1)\psi_B(2) + \psi_A(2)\psi_B(1)] + \lambda[\psi_A(1)\psi_A(2) + \psi_B(2)\psi_B(1)]$$

7.6 The first term represents the covalent component while the second term represents ionic component of the chemical bond. This equation indicates that the nature of the chemical bond in H_2 is neither purely covalent nor purely ionic but rather a resonance hybrid of the two. The coefficient λ is a measure of the degree to which the ionic forms $H_A^+H_B^-$ and $H_A^-H_B^+$ contribute to the bonding in H_2 . Since λ can be adjusted to give the best value of bonding energy, it affords a means of judging the extent of the ionic character of the bond.

Thus equation 7.6 is the essential principle of the VB theory and the result is the pairing of electrons which is equivalent to the formation of a single covalent bond.

Note:

- Since bonding involves pairing of electrons, atomic orbitals containing two electrons will not take part in bond formation since the electrons are already paired.
- As a result of resonance, the total energy of the system will seek a minimum value lying below that for any of the resonating structures.
- None of the individual structures has any physical existence



6.9.1 The total wavefunction

The total wavefunction is the product of the orbital wavefunction and the spin wavefunction. Only two types of spins are possible for each electron. If we let the corresponding spin wavefunctions be denoted by α and β , then many possibilities arise:

- If both electrons under consideration have the same spin, either α or β , the combined spin wavefunction may be written as $\alpha\alpha$ or $\beta\beta$.
- If however the spins are different, and $\alpha\beta$ and $\beta\alpha$ are acceptable wavefunctions since the electrons are indistinguishable and the allowed linear combinations are given by

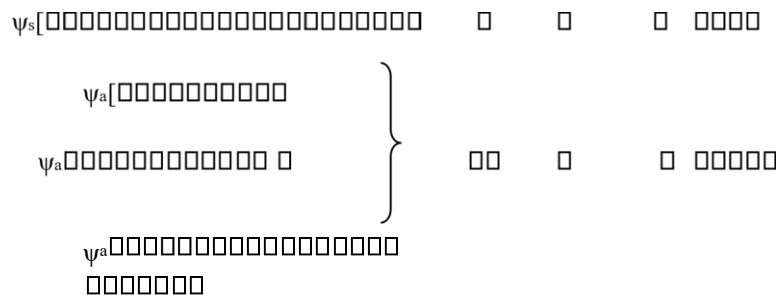
$$\frac{1}{\sqrt{2}}(\alpha\beta + \beta\alpha) \quad \text{and} \quad \frac{1}{\sqrt{2}}(\alpha\beta - \beta\alpha)$$

$$\frac{1}{\sqrt{2}}(\alpha\beta + \beta\alpha) \quad \text{and} \quad \frac{1}{\sqrt{2}}(\alpha\beta - \beta\alpha)$$

The spin wavefunction may now be combined with the orbital wavefunctions to obtain total wavefunctions.

Some combinations are forbidden by the Pauli exclusion principle which in a more general form implies that in any atomic or molecular system, the total wavefunction must be antisymmetric with respect to exchange of electrons. Therefore if the total wavefunction is symmetric, the spin wavefunction must be antisymmetric and vice versa. The symmetric orbital wave functions are given by equations 7.4 and 7.5 above. It may now be noted that of

the four possible spin wavefunctions all the except the one by equation 7.8 are symmetrical with respect to exchange of electrons. The allowed combinations of orbital and spin wavefunctions are therefore restricted to the following:



ψ_s describes the ground state of a molecule while ψ_a describes the excited state. From equations 7.9 and 7.10 above, it can be seen that the ground state can only have one spin wavefunction whereas the excited state has three spin wavefunctions. If we define multiplicity as the number of spin wavefunctions associated with a given state, then

multiplicity of the ground state = 1 (singlet state) multiplicity of

excited state = 3 (triplet state)

Note: Equation 7.9 shows that the hydrogen atoms can only combine to form a molecule when the two electrons have antiparallel spins i.e oppositely directed spins.

Summary

VBT considers the involvement of electrons in terms of atomic orbitals. The covalent bond is formed by the pairing and resultant neutralization of opposite electron spins. For an atom to form covalent bonds it must possess one or more electrons which can pair with those of another atom by canceling their spins. Thus the number of unpaired electrons determines the valence and paired electrons cannot participate in bonding. They can only do so if they can be unpaired with only a small expenditure of energy. This unpairing is generally only possible if it involves no change in the principal quantum number, n. e.g. in C and P

The strength of a bond depends on the degree of overlap of atomic orbitals. The degree of overlap also varies with type of orbitals. s orbitals are same all round while p and d overlap only at the tips of

Failures of the VBT

1. It assumes that the two electrons of the shared pair constituting the internuclear bond must come from two different atoms. It does not therefore explain the coordinate covalent bond.
2. It cannot explain the magnetic properties of some elements and compounds. For example it cannot explain why O_2 is paramagnetic because according to VBT it should be diamagnetic.

Successes of the VBT

1. It can explain the shapes of molecules in terms of hybridization.



Study Questions

1. Explain why Phosphorous is able to form pentahalides but nitrogen is not able.
2. What are the shapes of the following molecules: $BeCl_2$, BCl_3



7.1 Introduction

In this lecture you will learn how atoms bond according to the Molecular Orbital Theory (MOT) theory. You will start with Linear Combination of Atomic Orbitals (LCAO), rules for filling electrons in molecular orbitals and determination of bond orders. You will draw molecular orbital energy level diagrams for simple molecules. Molecular orbital theory can explain the paramagnetism of oxygen molecule while other previous theory could not.



7.2 Lecture 7 Objectives

At the end of this lecture, you should be able to:

- i) Describe and use molecular orbital theory to explain bonding.
- ii) Use the various bonding model to explain bonding and properties of molecules



7.3 The molecular-orbital approach

The molecular-orbital approach starts by considering the stable molecular system in which the nuclei are in their equilibrium positions, and produces molecular wavefunctions that describe the molecular energy states (molecular orbitals) into which electrons are placed.

From quantum mechanics,

1. The molecular Schrödinger equation is 'solved' to provide acceptable molecular wavefunctions that describe molecular energy states (molecular orbitals).
2. These molecular orbitals will embrace both nuclei. (In the case of polynuclear molecules the molecular orbitals will in principle involve all the nuclei.).
3. The molecular orbitals will be associated with molecular quantum numbers.
4. The molecular wavefunctions will have the same significance as do atomic wavefunctions, i.e. ψ is proportional to the probability of finding the electron in a given volume element of space $d\tau$
5. Each molecular wavefunction corresponds to a definite energy value, and the sum of the energies of individual electrons in the molecular orbitals, after correction for interaction, represents the total energy of the molecule.
6. The 'Aufbau' principle may be applied. The electrons are filled into the available molecular orbitals one at a time, the molecular orbital of lowest energy being filled first.
7. Each electron has a spin, and by the Pauli Exclusion Principle, each molecular orbital can accommodate a maximum of two electrons, provided their spins are opposed.



7.3.1 The Linear Combination of Atomic Orbitals method (LCAO)

The wavefunction for the single electron moving in the field of two nuclei A and B can be written as a linear combination of the atomic orbital wavefunctions of the constituent atoms as follows:

$$\psi = C_A \psi_A + C_B \psi_B \quad (8.1)$$

This is based on the observation that when an electron is close to a nucleus of one atom, its wavefunction closely resembles an atomic orbital of that atom. Construction of molecular orbitals may be done by adding atomic orbitals as follows:

1. All the atomic orbitals in the valence shells of the interacting atoms are combined in pairs to give molecular orbitals for the interacting nuclei.
2. Each pair consists of one atomic orbital from each of the interacting atoms. The atomic orbitals may or may not be originally occupied.
3. For combination of the atomic orbitals to occur the following conditions have to be met:
 - a) The combining orbitals must be associated with approximately the same energy
 - b) The combining orbitals must be capable of overlapping
 - c) The combining orbitals must be of the same symmetry
4. The number of molecular orbitals produced is equal to the number of combining atomic orbitals. Half will be bonding and half anti-bonding. (Constructive interference of the wave-functions leads to bonding molecular orbitals while destructive interference leads to anti-bonding molecular orbitals).
5. The molecular orbitals are arranged in ascending order of energy (which can be represented by energy level diagrams) and all the electrons originally occupying the valence shells are redistributed into these orbitals according to the *Aufbau* principle and obeying the Pauli exclusion principle.
6. If the number of electrons in the system is less than the number required to completely fill the molecular orbitals, the bonding orbitals outnumber the antibonding orbitals, producing the overall effect of bonding.

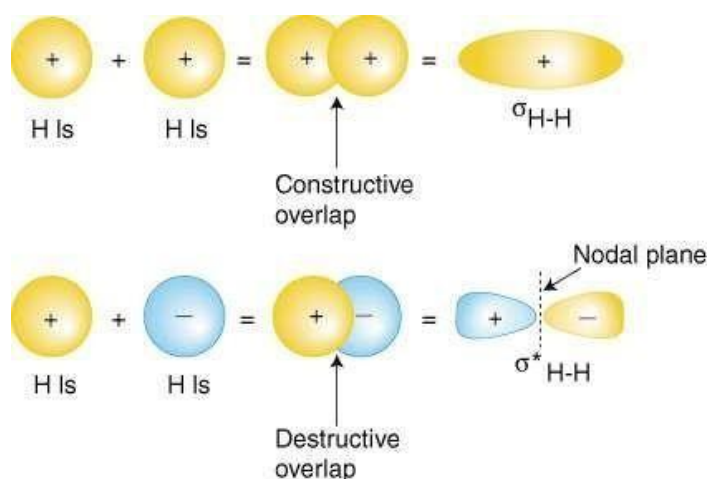


Figure 7.0.1: Two 1s orbitals combine to form a bonding and an antibonding M.O.

Examples of energy level diagrams/orbital correlation diagrams

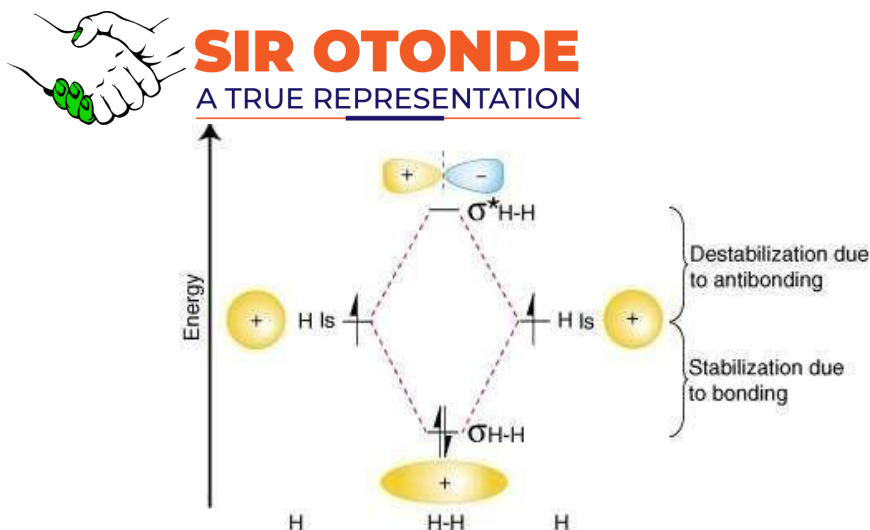


Figure 7.0.2 Orbital correlation diagram for hydrogen

Notice that the orbitals of the separated atoms are written on either side of the diagram as horizontal lines at heights denoting their relative energies. The electrons in each atomic orbital are represented by arrows. In the middle of the diagram, the molecular orbitals of the molecule of interest are written. Dashed lines connect the parent atomic orbitals with the daughter molecular orbitals. In general, bonding molecular orbitals are lower in energy than either of their parent atomic orbitals. Similarly, anti-bonding orbitals are higher in energy than either of its parent atomic orbitals. Because we must obey the law of conservation of energy, the amount of stabilization of the bonding orbital must equal the amount of destabilization of the anti-bonding orbital, as shown above.

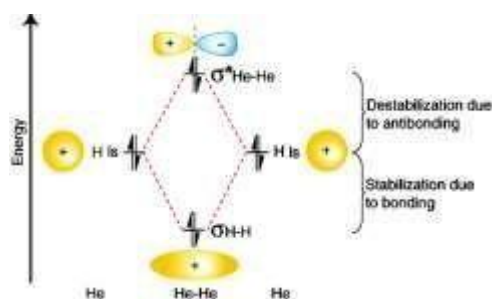


Figure 7.0.3 An orbital correlation diagram for a hypothetical He-He molecule (Source:

<http://www.sparknotes.com/chemistry/bonding/molecularorbital/section1.html>)

From the orbital correlation diagram above you should notice that the amount of stabilization due to bonding is equal to the amount of destabilization due to anti-bonding, because there are two electrons in the bonding orbital and two electrons in the anti-bonding orbital. Therefore, there is no net stabilization due to bonding so the He_2 molecule will not exist. The bond order calculation shows that there will be a bond order of zero for the He_2 molecule—exactly what we should predict given that helium is a noble gas and does not form covalent compounds.

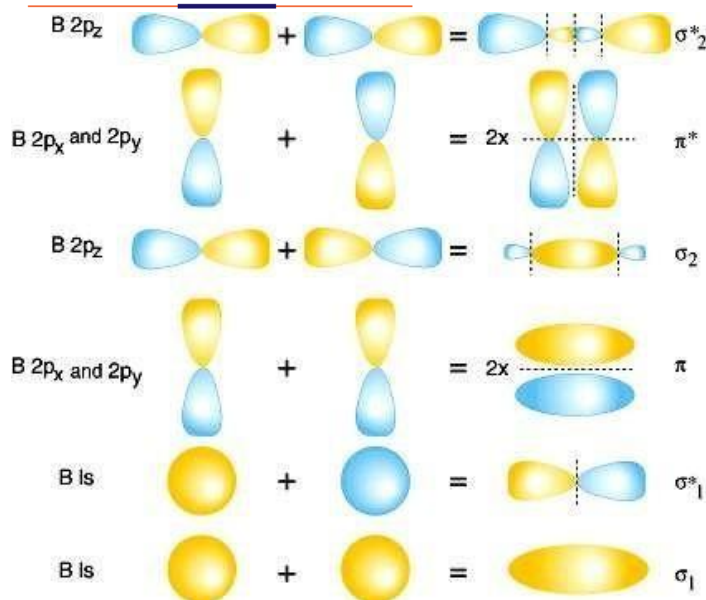


Figure 7.0.4 Molecular Orbitals of hypothetical diboron

Notice that there are two different kinds of overlap for p -orbitals - end-on and side-on types of overlap. For the p -orbitals, there is one end-on overlap possible which occurs between the two p_z . Two side-on overlaps are possible - one between the two p_x and one between the two p_y . p -orbitals overlapping end-on create σ bonds. When p -orbitals bond in a side-on fashion, they create π bonds. The difference between a π bond and a σ bond is the symmetry of the molecular orbital produced. σ bonds are cylindrically symmetric about the bonding axis, the z -direction. That means one can rotate the bond about the z -axis and the bond remains the same. In contrast, bonds lack that cylindrical symmetry and have a node passing through the bonding axis.

The orbital correlation diagram for diboron, however, is not generally applicable for all homonuclear diatomic molecules. It turns out that only when the bond lengths are relatively short (as in B_2 , C_2 , and N_2) can the two p -orbitals on the bonded atoms efficiently overlap to form a strong bond. Some textbooks explain this observation in terms of a concept called sp mixing. For any atom with an atomic number greater than seven, the bond is less stable and higher in energy than is the bond formed by the two end-on overlapping p orbitals. Therefore, the following orbital correlation diagram for fluorine is representative of all homonuclear diatomic molecules with atomic numbers greater than seven.

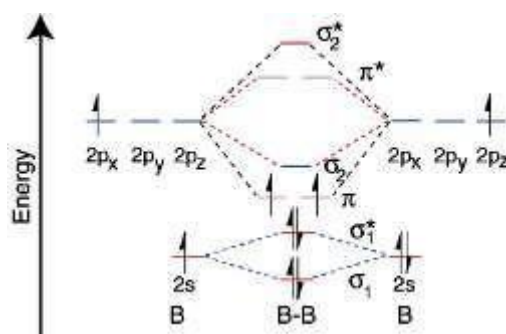


Figure 0.5 Orbital correlation diagram for diboron

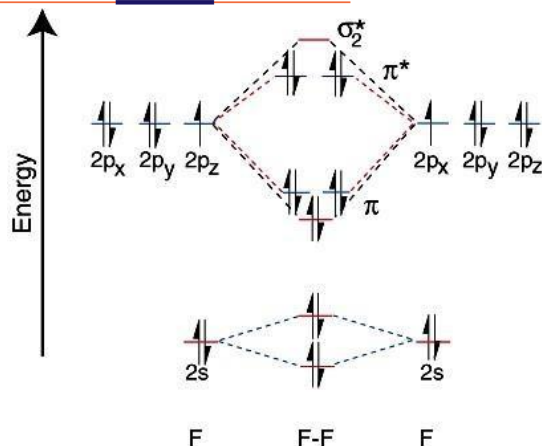


Figure 7.0.6: Orbital correlation diagram for homonuclear diatomic molecules other than B_2 , C_2 , and N

7.3.2 Molecular orbital energy level diagrams for heteronuclear diatomic molecules

To draw the correlation diagrams for heteronuclear diatomic molecules, we face a new problem: where do we place the atomic orbitals on an atom relative to atomic orbitals on other atoms? For example, how can we predict whether a fluorine 2s or a lithium 2s orbital is lower in energy? The answer comes from our understanding of electronegativity. Fluorine is more electronegative than lithium. Then electrons are more stable, i.e. lower in energy, when they are lone pairs on fluorine rather than on lithium. The more electronegative element's orbitals are placed lower on the correlation diagram than those of the more electropositive element. Fig. 7.8 illustrates this point:

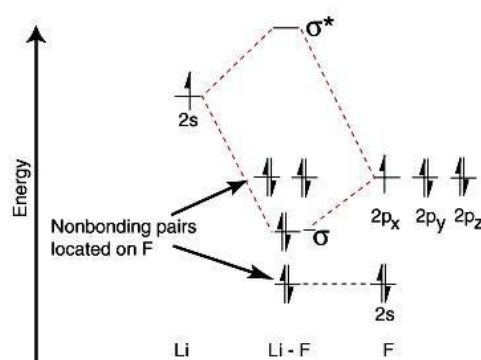


Figure 7.0.7: Orbital correlation diagram for LiF

Since lithium only has one occupied valence orbital, only one bonding and one antibonding orbital are possible. Furthermore, the electrons in orbitals on F that cannot bond with Li are left on F as lone pairs. As you can see, the electrons in the Li-F σ bond are quite close in energy to fluorine's 2p orbitals. Then the bonding orbital is primarily composed of a fluorine 2p orbital, so the M.O. diagram predicts that the bond should be polarized toward fluorine exactly what is found by measuring the bond dipole. Such an extreme polarization of electron density towards fluorine represents a transfer of an electron from lithium to fluorine and the creation of an ionic compound.

Bond Order

In any molecule bond order is given by:

$$\text{Bond order} = \frac{1}{2} (\text{Electrons in bonding MO} - \text{Electrons in antibonding MO})$$



Study Problems

Problem 1

Paramagnetic materials, those with unpaired electrons, are attracted by magnetic fields whereas diamagnetic materials, those with no unpaired electrons, are weakly repelled by such fields. By constructing a molecular orbital picture for each of the following molecules, determine whether it is paramagnetic or diamagnetic.

- a). B_2 b) C_2 c). O_2 d). NO e). CO

Solution for Problem 1: Construct energy level diagrams and then observe distribution of electrons.

- a). B_2 is paramagnetic because it has two unpaired electrons, one in each of its p orbitals.
 b). C_2 is diamagnetic because all of its electrons are paired.
 c). O_2 is paramagnetic because it has two unpaired electrons, one in each of its π orbitals.
 d). NO has an odd number of electrons and, therefore, must be paramagnetic.
 e). CO is diamagnetic because all of its electrons are paired.

8 LECTURE 8 IONIC BONDING



8.1 Introduction

In this chapter you will learn the ionic bonding and lattice energy. You will learn how lattice energy can be calculated using Born- Haber cycle and how to explain the stability of some ionic compounds from energetic view.



8.2 Lecture 8 Objectives

At the end of this lecture, you should be able to: i). Explain the nature of the ionic bond. ii). Describe the characteristics of ionic compounds. iii). Draw Born-Harber cycles for formation of ionic compounds iv). Calculate lattice energy of any crystalline material from thermochemical data.



8.3 History of Ionic bonding

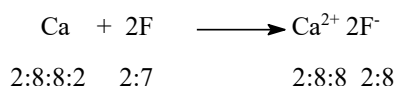
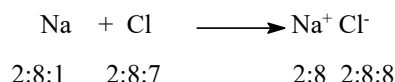
Many compounds can be thought of as a collection of ions (Mn^+ , Xn^-) held together electrostatically. This idea arose out of experiments by Arrhenius looking at the conductivity of solutions prepared by dissolving “ionic compounds” in water. He was not believed at first, but he won the 1903 Nobel Prize.



8.4 The Ionic Bond

This is also called electrovalent bond. It is usually formed between atoms that have significantly different electronegativities such that their electronegativity difference is large. In such molecules the common electron pair can be fully displaced to the more electronegative atom. This displacement leads to formation of two charged particles i.e. a positively charged ion or cation (+) and a negatively charged ion or an anion (-).

Examples:



Note that in the products each of the ions involved has a noble gas electron configuration.

Molecules with ionic bonds are rare to get unless in gaseous phase. Normally, many ions cluster together in definite arrangements forming crystal lattices or structures, such that the whole lattice is like one big molecule. In the solid lattices the ions are held together by electrostatic attraction between the oppositely charged ions.

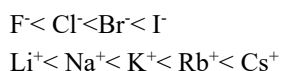


8.5 Structures of ionic solids

Generally ionic bonds are not directional like covalent bonds (electrostatic forces are isotropic). Usually one ion is surrounded by a number of oppositely charged ions in a definite manner. The number of ions surrounding any particular ion is called the coordination number. The properties of most compounds are governed by the mutual polarization of the constituent ions.

Polarization: The relative displacement of the nucleus and the electrons of the outer electron shell surrounding it under the influence of the electric field of the adjacent ion (i.e the oppositely charged ion). The extent of polarization determines the ionic character of the bond between a pair of oppositely charged ions (see Fajans' rules below).

Polarizability: The ability of an ion to be deformed under the action of an external electric field. The ease of deformation by an electric field increase with ionic size and vice versa:



Polarizing power: The ability to deform or polarize another ion. It increases with decrease in ionic, and with increasing ionic charge.

The effects of covalent vs ionic character on the physical properties of compounds can be subtle. *Normally*, ionic compounds are higher melting, higher boiling, more water soluble, less soluble in nonpolar solvent and have higher electrical conductivity in the melt than covalent substances. These rules of thumb are useful but not infallible here. Remember that quartz, SiO₂,

is a very covalent network solid with a melting point of 1710°C , higher than many salts ($\text{mpKCl} = 770^{\circ}\text{C}$). The high melting or boiling point of a covalent solid simply reflects the fact that there are more, or stronger, covalent bonds in the solid than in the liquid or gas phase for that solid. Increasing covalency will often lower the melting point of a "salt", MX, by stabilizing discrete MX molecules in the melt. In all of the examples below the more ionic compound is listed first.

Fajans' Rules

1. Covalent character increases with **decreasing cation size** or **increasing cation charge**. The high charge density of small and/or highly charged cations is able to exert a powerful polarizing effect which distorts the electron cloud around the anion. Compare NaCl (mp 800°C) with CaCl_2 (mp 772°C ; charge effect) and KI (mp 686°C) with LiI (mp 449°C ; size effect).
2. Covalent character increases with an **increase in anion charge or anion size**. The electrons of larger, more negatively charged anions are more loosely held because they are more shielded from the nuclear charge. Thus, these anions are more easily polarized by cations. Compare LiBr with Li_3P (charge effect), and LiF (mp 845°C) with LiI (mp 449°C ; size effect).
3. Covalent character is greater for cations with a **non-noble gas electron configuration** (i.e., 18-electron, pseudo-noble-gas cations) than for cations with a true noble-gas configuration (8-electron cations). This effect mostly shows up in transition-metal and post-transition metal cations in which one or more *d*-electrons ineffectively shield the outer electrons, or the electrons of an adjacent ion, from the nuclear charge. Thus, for two ions of similar size and charge, a transition metal ion with a $(n-1)d^xns^0$ configuration will be more polarizing and more polarizable than an alkali or alkaline earth metal with a noble gas configuration, $(n-1)s^2(n-1)p^6ns^0$. Compare CuCl (mp 430°C , Cu^+ is $3d^{10}4s^0$, $r = 0.91 \text{ \AA}$) with NaCl (mp 800°C , Na^+ , $2s^22p^6$, $r = 1.16 \text{ \AA}$) and HgCl_2 (mp 276°C , Hg^{2+} is $4f^{14}3d^{10}$, $r = 1.16 \text{ \AA}$) with CaCl_2 (mp 782°C , Ca^{2+} is $3s^23p^6$, $r = 1.14 \text{ \AA}$).

In an ionic crystal three forces operate and determine the stability of a lattice:

- i) The electrostatic attractive forces between the ions. These are proportional to the square of the distances between the ions.
- ii) Van der Waals forces of attraction. These are inversely proportional to the 7th power of the distance between the ions.
- iii) Interatomic repulsive forces. These are inversely proportional approximately to the 10th power of the distance between the ions.

Ultimately the reason why any particular crystal structure forms is that it gives the most favorable *lattice energy*.

It is often convenient to think about the cations in ionic compounds as lying in holes (interstices) between arrays of anions (Fig 7.10). Typically, assume ions are hard spheres. Usually, a compound will adopt a structure that maximizes the number of anions around each cation (also called coordination number) without causing the anions to touch.

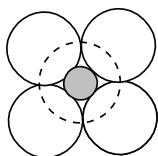


Figure 8.0.1: Coordination number in ionic solids

By applying the radius ratio rules, it is possible to predict the type of ion coordination that you will get if you know the ratio of the cation to anion size (i.e. ratio of radius of the cation to that of the anion).

Table 7.15: Preferred coordination number from r^+/r^- ratio

r^+/r^- Preferred coordination number > 0.7328

8-cubic coordination

0.414 – 0.7326

6-octahedral coordination

0.225 – 0.4144

4-tetrahedral coordination



8.5.1 Characteristics of ionic compounds

- Most simple ionic compounds tend to form hard and brittle crystals
- They usually have high melting points - several hundred or thousand Kelvin. However, salts that are liquid at room temperature have been prepared using organic cations.
- When molten they conduct electricity
- Most dissolve in high polarity solvents to form conducting solutions



8.5.2 The ionic to covalent continuum

In practice, no compound is truly ionic. Compounds of elements with very different electronegativities tend to be more ionic. Molecules with ionic bonds are rare. Normally many ions cluster together in definite arrangements forming crystal lattices or structures such that the whole lattice is like one big molecule. Only in the gaseous state can one find molecules with almost 100% ionic bond.



8.5.3 Effect of ion charge on ionic size

Isoelectronic ions get smaller as the nuclear charge increases (See Tables 8 and 9)

Table 7.24: Ionic radii of isoelectronic cations

Ion	Ionic radius/pm
Na ⁺	116
Mg ²⁺	86
Al ³⁺	68

Table 7.32: Ionic radii of isoelectronic anions

Ion	Ionic radius/pm
N ³⁻	132
O ²⁻	124
F ⁻	117

8.5.4 Trends in physical properties

Decreasing ionic size and increasing ionic charge favor better binding of the ions in the solid (higher lattice energy). This tends to give increased melting and boiling points.

Table 7.4

Compound	Melting point/ °C	Compound	Melting point/ °C
KF	857	NaF	988
KCl	772	MgF ₂	1266
KBr	735	AlF ₃	1291 sublimates
KI	685		



8.5.5 Polarization and covalency

“Ionic” compounds tend to have a considerable covalent contribution to their bonding when they contain polarizing cations. A polarizing cation is one that is capable of distorting the anion’s electron cloud towards itself. Generally;

i). Small highly charged cations are more polarizing ii).

Large highly charged anions are more polarizable iii). Polarization is favored for cations that do not have a noble gas electron configuration

e.g. Ag⁺, Cu⁺, Zn²⁺, Cd²⁺, Hg²⁺, Tl⁺ etc.



8.5.6 Physical effect of covalency

“Ionic” solids with a significant covalent contribution to bonding show “anomalous” physical properties. For example,

i). They may not be water soluble e.g. AgCl, CuI etc.

ii). AlF₃ MP 1290°C, AlI₃ MP 190°C



8.5.7 Hydration of ions

Ionic solids are usually soluble in water because the dipole on water interacts with the ion charges. The negative end of the dipole coordinates to the cation.

○ strength of interaction increases with decreasing cation size and increasing charge. Strong coordination may lead to the formation of hydrates such as [Al(OH₂)₆]³⁺.3Cl⁻. Otherwise dissolution occurs where both the cation and anion are separated by the polar water as shown below:

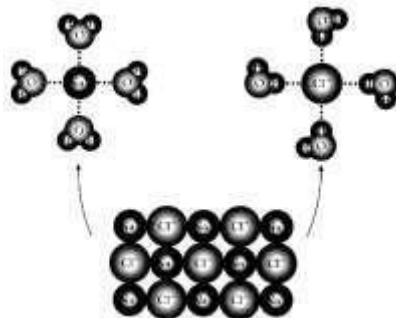


Figure 8.0.2: Ion hydration on dissolution



8.5.8 Lattice energy

Energy evolved when one gram mole of a crystal is formed from gaseous ions (-ve) or Energy required to convert one gram mole of a crystal into its constituent gaseous ions (+ve).

Lattice energies cannot be measured directly but experimental values are obtained from thermodynamic data using the Born-Haber cycle. Theoretical values can be calculated using the Born-Landé equation:

$$U = \frac{N_0 A z^+ z^- e^2}{4 \pi \epsilon_0 r_0} \left(1 - \frac{1}{n} \right)$$

where ϵ_0 = permittivity of free space = $8.854 \times 10^{-12} \text{ Fm}^{-1}$,

A = the Madelung constant which depends on the geometry of the crystal
 N_0 = the Avogadro constant = $6.023 \times 10^{23} \text{ mol}^{-1}$ Z^+ , Z^- are the charges on the cations and anions, respectively. r_0 = the equilibrium inter-ionic distance e = the charge on an electron
 n = the Born exponent

This equation allows the lattice energy of any crystal to be calculated from a knowledge of the geometry of the crystal, and hence the Madelung constant, the charges z^+ and z^- and the inter-ionic distance. Several important points arise from the Born-Landé equation:

- i) The inter-ion distance is the denominator of the equation. Thus the smaller the value of r_0 the greater the amount of energy evolved when the crystal lattice is formed, and hence the more stable the crystal will be. U is proportional to $1/r$.

$r (\text{\AA})$	$U (\text{kJ/mol})$	LIF	2.10
	-1004		
CsI	3.95	-527	

When the inter-ionic distance becomes small enough for the ions to touch, they begin to rebel each other due to the mutual repulsion of the electron clouds on the two ions.

- ii) The lattice energy depends on the product of the ionic charges, and U is proportional to $(z^+ \cdot z^-)$.

	$r (\text{\AA})$	$(z^+ \cdot z^-)$	$U (\text{kJ/mol})$
LIF	2.10	1	-1004



- iii) The experimental and theoretical values for the alkali metal halides and the oxide and halides of alkaline earth metals (except Be), all agree to within 3%. However this close agreement does not in itself prove that the equation itself or the assumptions on which it's based, are correct. There are two opposing factors in the equation. Increasing the inter-ionic distance r reduces the lattice energy. It is almost impossible to change the r without changing the structure and hence the Madelung constant A . Increasing A increases the lattice energy: hence the effects of changing r and A may largely cancel each other. iv) Crystals with a high lattice energy usually melt at high temperatures and are very hard. High lattice energy is favoured by a small inter-ionic distance and a high charge on the ions.

It has been observed that a number of salts which might be expected from radius ratio considerations to have a CsCl structure in fact adopt a NaCl structure. The Madelung constant for CsCl is larger than for NaCl and would give an increased lattice energy. However, the inter-ionic distance r will be larger in a CsCl structure than on a NaCl structure and this would decrease the lattice energy. These two factors work in opposite directions and partly cancel each other. This makes the lattice energy more favorable for a NaCl type of lattice in some cases where CsCl is geometrically possible.



8.5.9 Calculation of Lattice energy from Born-Haber cycle

The Born-Haber cycle was devised by Born and Haber in 1919 and relates the lattice energy of a crystal to the other thermochemical data.

The energy terms involved in building a crystal lattice such as NaCl or KCl may be taken in steps. The elements in their standard states are first converted to gaseous atoms, then to ions and finally packed into the crystal lattice.

The enthalpies of sublimation and dissociation and the ionization energy are positive since energy is supplied to the system. The electron affinity and lattice energy are negative since energy is evolved in these processes.



8.5.10 Hess's Law

The overall energy change in a process depends only on the energy of the initial and final states and not on the route taken.

Thus according to Hess's law, in a cycle like the one below, the enthalpy of formation ΔH_f is the algebraic sum of the terms going round the cycle (paying heed to the exothermic and endothermic nature of each step).

From Fig 7.11,

$$\begin{aligned}\Delta H_f &= \Delta H_{\text{sub}} + \Delta H_I + \Delta H_{\text{dis}} + E_A + U \\ -413 &= 109 + 119.5 + 496 + (-350.5) - U \\ U &= -(359.5 + 413) \\ U &= -772.5 \text{ kJ mol}^{-1}\end{aligned}$$

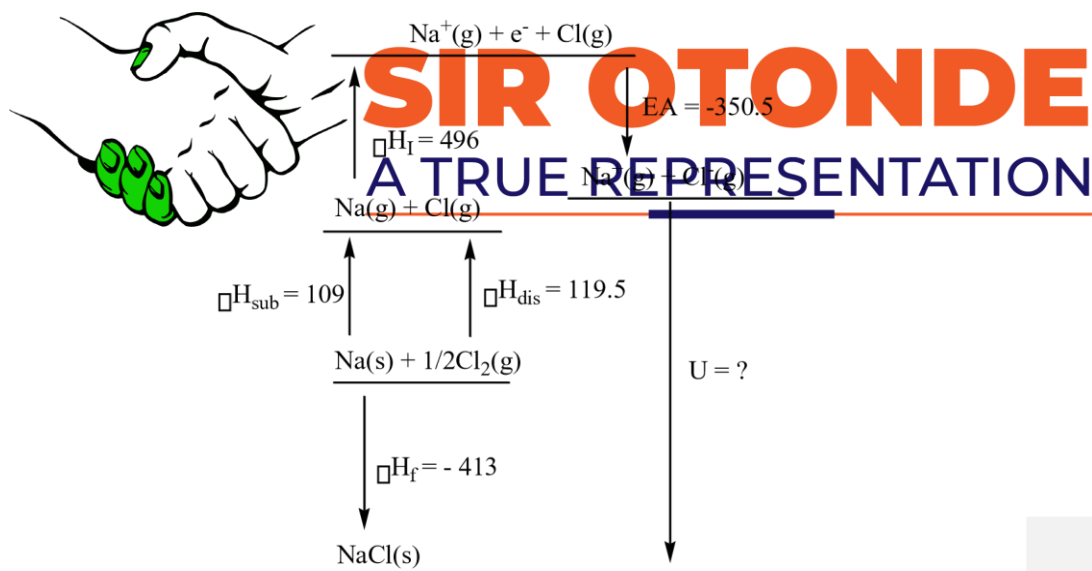


Figure 8.0.3 Bom-Haber cycle for the formation of NaCl (all enthalpies are in kJ mol^{-1})



8.5.11 Usefulness of Lattice energy

It is useful as a guide to the solubility of the crystal. When a solid dissolves, the crystal lattice must be broken up (energy must be put in). The ions so formed are solvated (with the evolution of energy). For solubility to occur, the energy of solvation must be higher than the lattice energy. Otherwise the substance will be insoluble. The noble character of many transition metals, that is their resistance to chemical attack, is related to their high heats of sublimation, ionization energy and low enthalpy of solvation of the ions.

Lattice energy may also provide some information about the ionic/covalent nature of the bonding. When calculated values of lattice energy (calculated by assuming ionic bonding) agree with the experimental values it indicates that the assumption was correct, while poor agreement may indicate that the bonding is not ionic (see Table 12).

Table 7.56: Comparison of theoretical and experimental lattice energies

	Theoretical lattice energy/ kJmol^{-1}	Born-Haber lattice energy/ kJmol^{-1}	% difference
LiCl	-825	-817	0.8
NaCl	-764	-764	0.0
KCl	-686	-679	1.0
KI	-617	-606	1.8
CaF ₂	-2584	-2611	1.0
CdI ₂	-1966	-2410	22.6



Study Questions

- The following table gives some thermochemical data:

Enthalpy of atomization, ΔH_{at} , for calcium $= +201 \text{ kJ/mole}$

). Hence or otherwise

First ionization energy ΔH_{i1} for calcium	$=+590$ kJ/mole
Second ionization energy ΔH_{i2} for calcium	$=+1100$ kJ/mole
Dissociation energy D (Cl-Cl) for chlorine	$=+242$ kJ/mole
Electron affinity, E_a for chlorine	$= -348$ kJ/mole
Lattice energy ΔH_{latt} ($\text{CaCl}_{(s)}$)	$= -719$ kJ/mole
Lattice energy ΔH_{latt} ($\text{CaCl}_{2(s)}$)	$= -2218$ kJ/mole

Calculate the enthalpy of formation for $\Delta H_f(\text{CaCl}_{(s)})$ and $\Delta H_f(\text{CaCl}_{2(s)})$ explain why calcium forms the dichloride in spite of the fact that it has a very large second ionization energy.

9 LECTURE 9 HYDROGEN BONDING



9.1 About this Lecture

This lecture covers another important type of bond known as the hydrogen bond. Hydrogen bond can either be intramolecular (i.e. within a molecule) and intermolecular (i.e. between molecules). The significance of hydrogen bond in life and chemical reactions will also be reviewed.



9.2 Lecture 9 objectives

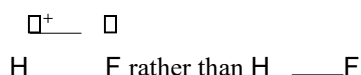
At the end of this lecture you should be able to:

- Recognize the possibility of hydrogen bond
- Discuss the importance of hydrogen
- Describe how hydrogen bonding affects the chemistry of water and other compounds



9.2.1 The concept of Hydrogen Bonding

When hydrogen is bonded to a highly electronegative element and in particular, Oxygen, Nitrogen and Fluorine, the covalent bond formed is such that, electrons are drawn mostly towards the electronegative element, and thus leaving the hydrogen rather deficient in electrons. For example, a molecule of hydrogen fluoride is correctly formulated as

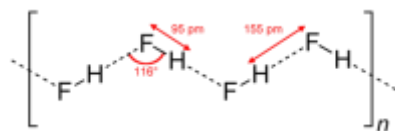


This emphasizes the fact that the electrons forming the covalent bond are not distributed evenly but are displaced more towards the more electronegative fluorine atom. Now, in a situation where there are millions of molecules as would be the case in liquid HF acid in a container the polar molecules will be existing side by side. The result is that, besides the normal covalent H-F bond, a proton, by virtue of having some net positive charge, will experience attraction by a nearby fluorine atom that belong to other molecules, as shown in Fig 9.1. The weak attraction is referred to as **Hydrogen Bonding**.



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A TRUE REPRESENTATION





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Figure 9.0.1: Hydrogen bonding in liquid hydrofluoric acid

Thus, a hydrogen bond is the attractive force between the hydrogen attached to an electronegative atom of one molecule and an electronegative atom of a different molecule. It is really a special case of dipole forces.

To recognize the possibility of hydrogen bonding, examine the Lewis structure of the molecule. The electronegative atom must have one or more unshared electron pairs as in the case of oxygen and nitrogen, and has a negative partial charge. The hydrogen, which has a partial positive charge tries to find another atom of oxygen or nitrogen with excess electrons to share and is attracted to the partial negative charge. This forms the basis for the hydrogen bond.

For example, properties of water are closely related to hydrogen bonding. The polar nature of the water molecule facilitates the hydrogen bond.

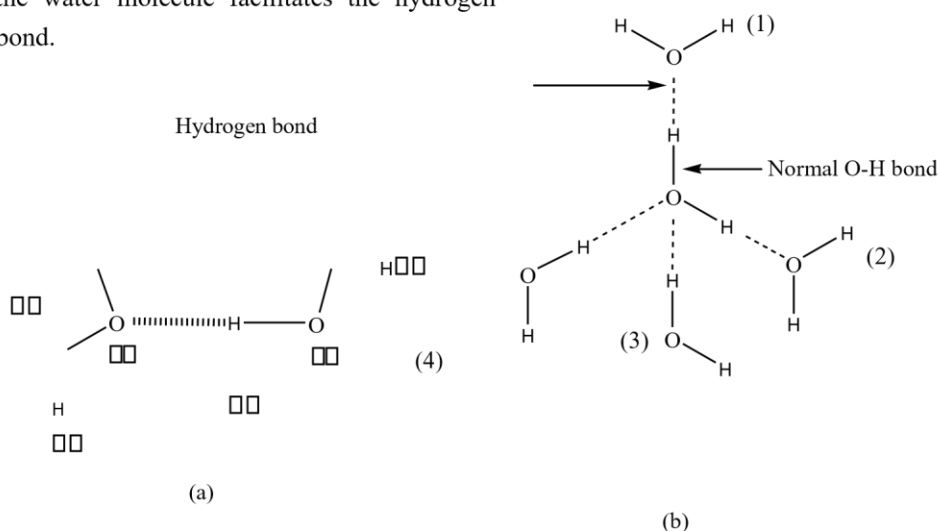


Figure 9.0.2(a) Hydrogen dipoles in water molecules leading water dimer (b) hydrogen bond network in liquid water

Two molecules of water form a hydrogen bond between them; the simplest case, when only two molecules are present, is called the water dimer and is often used as a model system (Fig 9.0.2a). When more molecules are present, as is the case of liquid water, more bonds are possible because the oxygen of one water molecule has two lone pairs of electrons, each of which can form a hydrogen bond with a hydrogen on another water molecule. This can repeat such that every water molecule is H-bonded with up to four other molecules, (two through its two lone pairs, and two through its two hydrogen atoms) (Fig 9.0.2b).

Hydrogen bonding strongly affects the crystal structure of ice, helping to create an open hexagonal lattice. The density of ice is less than water at the same temperature; thus, the solid phase of water floats on the liquid, unlike most other substances. Because of the open hexagonal structure the number of water molecules per unit volume in ice is less than the number in liquid water at the same temperature.

Hydrogen bonding also affects the properties and strength of hydrofluoric acid. Liquid HF also consists of chains of HF molecules (Fig 9.0.1), but the chains are shorter, consisting on

average of only five or six molecules. The higher boiling point of HF relative to analogous species, such as HCl, is attributed to hydrogen bonding between HF molecules, as indicated by the existence of chains even in the liquid state.

The formation of an intramolecular hydrogen bond in *o*-nitrophenol (Fig 9.3a) reduces its acidity compared with *m*-nitrophenol (Fig 9.3b) and *p*-nitrophenol (Fig 9.3c), where formation of intramolecular hydrogen bond is not possible.

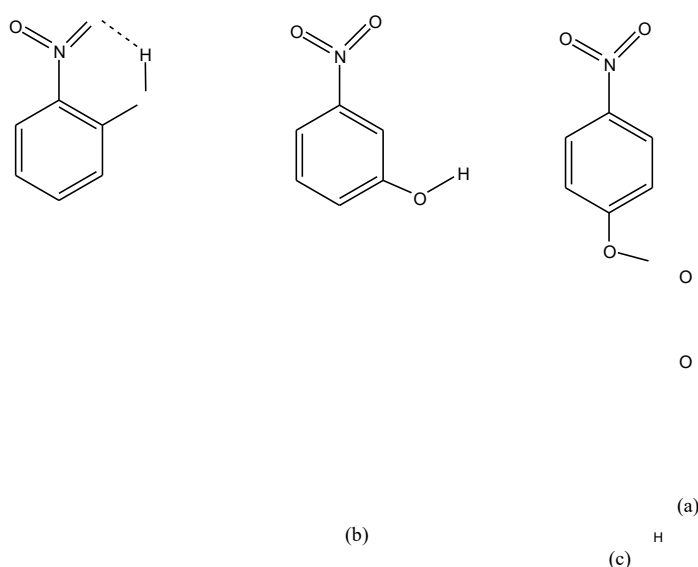


Figure 9.0.3(a)Hydrogen bonded *o*-nitrophenol (b) *m*-nitrophenol (c) *p*-nitrophenol



Among the hydrides of group 16 elements, H₂O has highest boiling point. This is attributed to the stronger and extensive intermolecular hydrogen bonding found in water and not in the rest.



9.2.2 Hydrogen bonds in DNA and proteins

Hydrogen bonding also plays an important role in determining the three-dimensional structures adopted by proteins and nucleic bases. In these macromolecules, bonding between parts of the same macromolecule causes it to fold into a specific shape, which helps determine the molecule's physiological or biochemical role. The double helical structure of DNA, for example, is due largely to hydrogen bonding between the base pairs, which link one complementary strand to the other and enable replication. In the secondary structure of proteins, hydrogen bonds form between the backbone oxygens and amide hydrogens.

In molecular biology, two nucleotides on opposite complementary DNA or RNA strands that are connected via hydrogen bonds are called a base pair (often abbreviated bp). In the canonical Watson-Crick DNA base pairing, *adenine* (A) forms a base pair with *thymine* (T) and *guanine* (G) with *cytosine* (C). In RNA, thymine is replaced by *uracil* (U). Alternate hydrogen bonding patterns, such as the wobble base pair and Hoogsteen base pair, also occur—particularly in RNA—giving rise to complex and functional tertiary structures. Importantly, pairing is the mechanism by which codons on messenger RNA molecules are recognized by anticodons on transfer RNA during protein translation. Some DNA- or RNA-binding enzymes can recognize specific base pairing patterns that identify particular regulatory regions of genes.

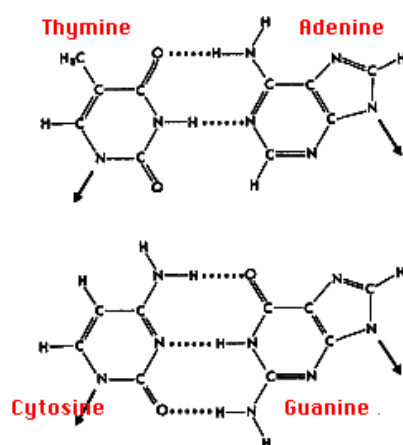


Figure 9.0.4:Hydrogen bonding causing base pairing in DNA Helix strands

A hydrogen atom attached to a relatively electronegative atom is a hydrogen bond donor. An electronegative atom such as fluorine, oxygen, or nitrogen is a hydrogen bond acceptor, regardless of whether it is bonded to a hydrogen atom or not. An example of a hydrogen bond donor is ethanol, which has a hydrogen-bonded to oxygen; an example of a hydrogen bond acceptor which *does not* have a hydrogen atom bonded to it is the oxygen atom on diethyl ether.

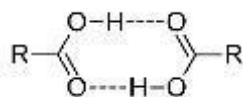


Figure 9.0.5:Hydrogen bonds in carboxylic acids dimers



9.3 Study problems

1. Consider the hydrides of group 16 elements. Explain why H₂O has higher boiling point than H₂S, H₂Se and H₂Te.



Answer: O has highest electronegativity among the group 16 elements. This results in stronger and extensive intermolecular hydrogen bonding in water than in the rest of the compounds.

2. Draw hydrogen bonded structures for:
 - a) Formic acid dimer
 - b) Ice
 - c) A layer of crystalline H_3BO_3 .
3. Explain why *para*-nitrobenzene has a higher boiling point than *o*-nitrobenzene

9.4 Revision problems and proposed solutions

Problem 1

Calculate the bond order for each of the following molecules (Hint: first draw a correlation diagram for each).

- a). B_2 b). C_2 c). O_2 d). NO e). CO

Solution for Problem 1

- a). B_2 has 2 bonding pairs and 1 antibonding pair so it has a bond order of 1.
b). C_2 has 3 bonding pairs and 1 antibonding pair so it has a bond order of 2.
c). O_2 has 4 bonding pairs and 2 antibonding pairs so it has a bond order of 2.
d). NO has 4 bonding pairs and 1.5 antibonding pairs so it has a bond order of 2.5.
e). CO has 4 bonding pairs and 1 antibonding pair so it has a bond order of 3.

Problem 2

Predict the hybridization of the central atom in each of the following molecules.

- a). CH_4 b). HCCCH c). IF_4^- d). IF_4^+ e). CH_2O

Solution for Problem 2

- a). Tetrahedral = sp^3 b). Linear = sp c). Octahedral = d^2sp^3
d). Trigonal Bipyramidal = dsp^3 e). Trigonal Planar = sp^2

Problem 3

Combining what you know about resonance and hybridization, predict the hybridization of each oxygen in the acetate ion CH_3CO_2^- .

Solution for Problem 4

Because the two oxygen atoms are equivalent, their hybridizations must also be equivalent. Therefore, we must choose between sp^2 and sp^3 hybridization. For best overlap, it makes sense to assume that the resonating lone pair is in a p -orbital so that it is properly positioned to make a π bond to the central carbon. Because we must have that p -orbital free, i.e. not hybridized, the hybridization must be sp^2 for both oxygens.

10 REFERENCES



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